Solution

From the corollary above you have

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 = 2(-2) + 3(3) = 5$$

The norm of vector **u** is $\sqrt{2^2 + 3^2} = \sqrt{13}$ and the norm of vector **v** is

$$\sqrt{(-2)^2 + 3^2} = \sqrt{13}.$$

Using the Theorem for scalar product, you have

$$\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}||\mathbf{v}| \cos\theta$$

Then, $\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|} = \frac{5}{\sqrt{13} \cdot \sqrt{13}} = \frac{5}{13}$

Therefore, $\cos\theta = \frac{5}{13}$.

Example 5

Let **u** and **v** be two vectors $|\mathbf{u}| = 3$ and $|\mathbf{v}| = 6$ and $\theta = \frac{\pi}{3}$.

a. Find $|3\mathbf{u} - 2\mathbf{v}|$.

b. Find the cosine of angle between $\mathbf{3u - 2v}$ and **u.**

Solution

a. Using the properties of dot product, you have

$$\begin{aligned}|3\mathbf{u} - 2\mathbf{v}|^2 &= (3\mathbf{u} - 2\mathbf{v}) \cdot (3\mathbf{u} - 2\mathbf{v}) \\ &= 9\mathbf{u}^2 - 12\mathbf{u} \cdot \mathbf{v} + 4\mathbf{v}^2 \\ &= 9 \cdot 3^2 - 12|\mathbf{u}||\mathbf{v}| \cos\frac{\pi}{3} + 4 \cdot 6^2, \text{ since } \mathbf{u}^2 = |\mathbf{u}|^2 \\ &= 81 - 12 \cdot 3 \cdot 6 \cdot \frac{1}{2} + 144 \\ &= 117\end{aligned}$$

Thus, $|3\mathbf{u} - 2\mathbf{v}| = \sqrt{117}$.

b. Let θ be the angle between $3\mathbf{u} - 2\mathbf{v}$ and **u**. Then

$$(3\mathbf{u} - 2\mathbf{v}) \cdot \mathbf{u} = |3\mathbf{u} - 2\mathbf{v}||\mathbf{u}| \cos\theta$$

$\Rightarrow 3\mathbf{u}^2 - 2\,\mathbf{v} \cdot \mathbf{u} = |3\mathbf{u} - 2\mathbf{v}|\,(3)\cos\theta$ but from **a** you have $|3\mathbf{u} - 2\mathbf{v}| = \sqrt{117}$

$\Rightarrow 3 \cdot 3^2 - 2|\mathbf{v}||\mathbf{u}| \cos\frac{\pi}{3} = \sqrt{117} \cdot 3\cos\theta$

$\Rightarrow 3 \cdot 3^2 - 2 \times 6 \times 3 \times \frac{1}{2} = 3\sqrt{117}\cos\theta$

$\Rightarrow 27 - 18 = 3\sqrt{117}\cos\theta$

$\Rightarrow 9 = 3\sqrt{117}\cos\theta$

$\Rightarrow \cos\theta = \frac{3}{\sqrt{117}} = \frac{3}{3\sqrt{13}} = \frac{1}{\sqrt{13}} = \frac{\sqrt{13}}{13}.$

Exercise 5.9

1. Find the dot product of the following vectors:
   a. $\mathbf{u} = 4\mathbf{i} + 2\mathbf{j}$ and $\mathbf{v} = 3\mathbf{i} - 5\mathbf{j}$ b. $\mathbf{u} = -2\mathbf{i} + 2\mathbf{j}$ and $\mathbf{v} = 7\mathbf{i} + 4\mathbf{j}$
2. Find the cosine of the angle formed by the following two vectors:
   a. $\mathbf{u} = 4\mathbf{i} + 3\mathbf{j}$ and $\mathbf{v} = 4\mathbf{i} - 3\mathbf{j}$ b. $\mathbf{u} = -2\mathbf{i} + 3\mathbf{j}$ and $\mathbf{v} = 3\mathbf{i} + 2\mathbf{j}$
3. Let $\mathbf{u} = (1, -1)$, $\mathbf{v} = (1, 1)$ and $\mathbf{w} = (-2, 3)$. Find the cosines of the angles between:
   a. $\mathbf{u}$ and $\mathbf{v}$ b. $\mathbf{v}$ and $\mathbf{w}$ c. $\mathbf{u}$ and $\mathbf{w}$
4. Say whether the vectors are orthogonal, parallel or neither
   a. $\mathbf{u} = (-5, 3)$ $\mathbf{v} = (6, -8)$ b. $\mathbf{u} = (4, 6)$ $\mathbf{v} = (-3, 2)$
   c. $\mathbf{u} = (-1, \frac{1}{2})$ $\mathbf{v} = (3, 6)$ d. $\mathbf{u} = (2, 6)$ $\mathbf{v} = (-3, -9)$
5. Find the angle between the two vectors $2\mathbf{i} + 3\mathbf{j}$ and $5\mathbf{i} - 2\mathbf{j}$.
6. Show that $(\mathbf{u} - \mathbf{v})(\mathbf{u} + \mathbf{v}) = \mathbf{u}^2 - \mathbf{v}^2$.
7. Show that $(\mathbf{u} \mp \mathbf{v})^2 = \mathbf{u}^2 \mp 2\,\mathbf{u} \cdot \mathbf{v} + \mathbf{v}^2$, where $\mathbf{u}^2 = \mathbf{u} \cdot \mathbf{u}$.
8. Vectors $\mathbf{u}$ and $\mathbf{v}$ make an angle $\theta = \frac{2\pi}{3}$ between them. If $|\mathbf{u}| = 5$ and $|\mathbf{v}| = 12$, then find
   a. $\mathbf{u} \cdot \mathbf{v}$ b. $(\mathbf{u} - \mathbf{v}) \cdot (\mathbf{u} + \mathbf{v})$ c. $(\mathbf{u} + \mathbf{v}) \cdot (\mathbf{u} + \mathbf{v})$ d. $|3\mathbf{u} + \mathbf{v}|$
9. Given a triangle with vertices A (1, 5), B (2, 2), and C (4, 6), find angle of BCA.

5.3.2 Vectors in Three-dimensional Space

Coordinate of a point in space

Activity 5.8

1. Represent each of the following points on the $xy-$ coordinate plane.
   a. A (1,1) b. B (2, 2) c. C (−4,4) d. D (0,−5) e. E(−2, −2)
2. Plot the following points, by naming the vertical axis on the plane z, and the horizontal axis y.
   a. P (2, 3) b. Q(−1, 3) c. (−2,−2)

From activity 5. 8, you observed that two numbers are necessary to locate a point in a plane. You know that any point in the plane can be represented as an ordered pair (a, b) of real numbers, where a is the x-coordinate and b is the y-coordinate. For this reason, a plane is called two-dimensional. A plane coordinate system is extended to three dimensional spaces as follows: Consider a fixed-point O in space and three lines that are mutually perpendicular at the point O. The point O is called the origin; the three lines are now called the x-axis, the y-axis and the z-axis. It is common to have the x- and the y-axes on a horizontal plane and the z-axis vertical or perpendicular to the plane containing the x- and the y-axes at the point O as shown in Figure 5.26 below.

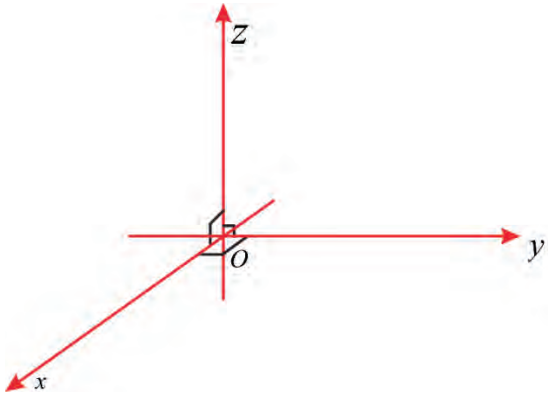


Figure 5.26

The direction of the z-axis is determined by the **right-hand rule** as illustrated in Figure 5.27:

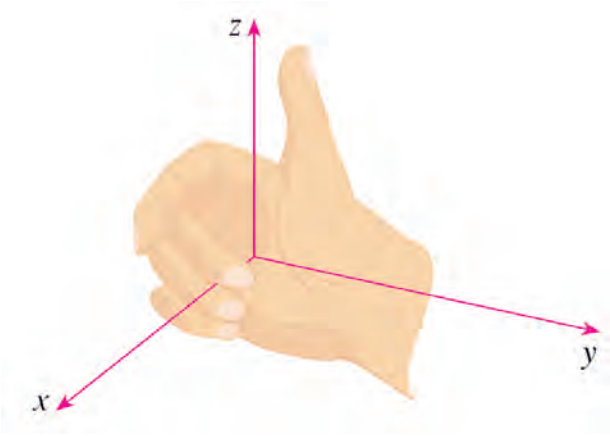


Figure 5.27

If you curl the fingers of your right hand around the z-axis in the direction of a 90° counterclockwise rotation from the positive x-axis to the positive y-axis, then your thumb points

in the positive direction of the z-axis. With these axes any point p in space can be assigned three coordinates p (x, y, z) as shown Figure 5.28.

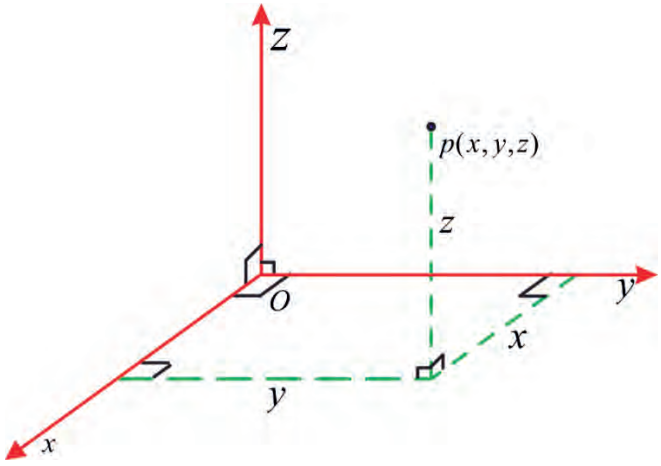


Figure 5.28

Example 1

Locate the point P (1, 3, 4) in space using the reference axes x, y and z.

Solution

The process of locating the point P may be described as follows:

The process of locating the point P may be described as follows: Start from the origin O and move 1 unit in the direction of the positive x-axis. Then move 3 units in the direction of the positive y-axis and finally move 4 units up in the direction of the positive z-axis to get point P (1, 3, 4) as shown in Figure 5.29.

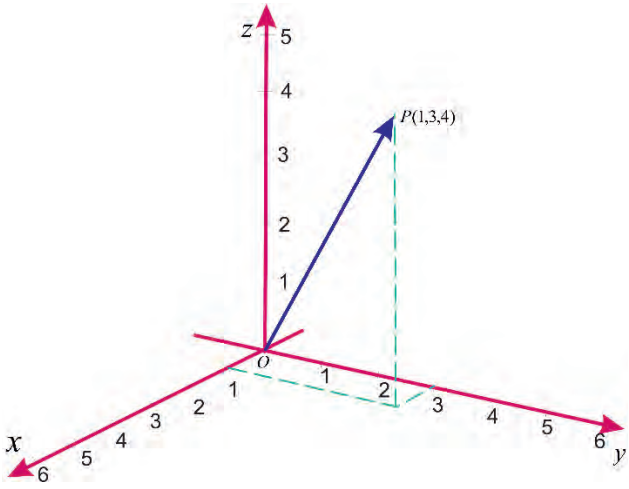


Figure 5.29

Exercise 5.10

Locate each of the following points in space using reference axes x, y and z. You may use the same or different coordinate systems in each case.

a. P (2, 1, 2)
b. Q (–3, 5, 4)
c. R (2, – 2, 3)
d. S ((2, $-\frac{1}{2}$,–4)
e. T (0, 0, –4)
f. Q (0, –4, 0)

Vectors in Space

Activity 5.9

1. How do you represent a vector on a plane?
2. How do you represent the magnitude of a vector?
3. How do you express the vector P *(x, y)* using the standard unit vectors **i** and **j**?

From activity 5.9, you have observed that the vector $\overrightarrow{OP}$ can be named using a single letter. That is, $\overrightarrow{OP}$ may be named P, or simply by P so that $\vec{P} = x\mathbf{i} + y\mathbf{j}$. Just as you worked with vectors on a plane by using the coordinates of their terminal points, you can handle vectors in a three-dimensional space with the help of the coordinates of the terminal points.

Now, let the initial point of a vector in space be the origin O of the coordinate system and let its terminal point be at P (x, y, z) as shown in Figure 5.30.

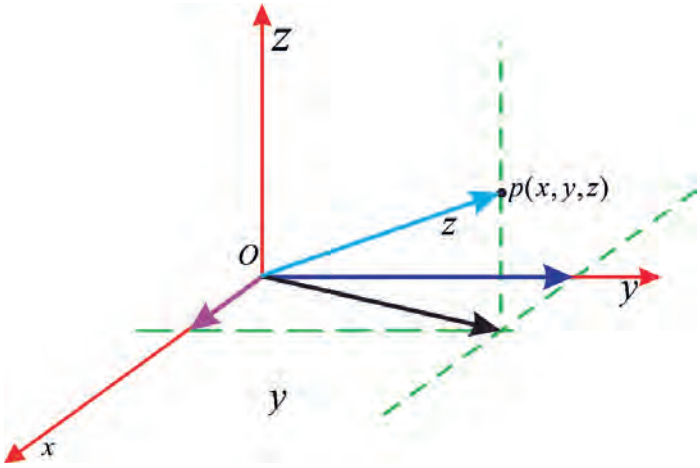


Figure 5.30

Then the vector $\overrightarrow{OP}$ can be expressed as the sum of its three components in the directions of the x, the y and the z-axis, in the form:

$\overrightarrow{OP} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ where $\mathbf{i} = (1, 0, 0)$, $\mathbf{j} = (0, 1, 0)$ and $\mathbf{k} = (0, 0, 1)$ are standard Unit vectors in the directions of the positive x, positive y and positive z-axis respectively as shown in Figure 5.31.

Do you observe that the vector $\overrightarrow{OP}$ is the sum of the three perpendicular vectors $\overrightarrow{OC}$, $\overrightarrow{OB}$ and $\overrightarrow{OD}$?

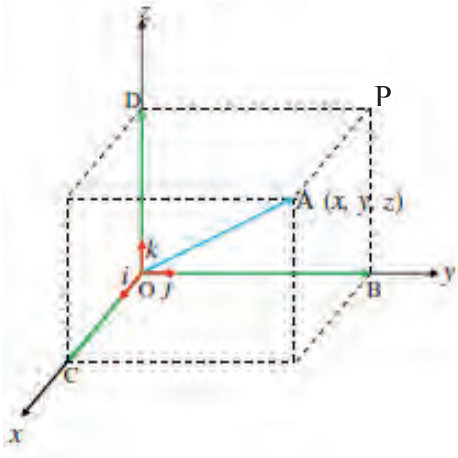


Figure 5.31

Example 2

If the initial point of a vector in space is at the origin and its terminal point or head is at P (4, 6, 5), show the vector using a coordinate system and identify its three perpendicular components in the directions of the three axes.

Solution

The three components are the vectors with common initial point O (0, 0, 0) and terminal points A (4, 0, 0) on the *x*-axis, B (0, 6, 0) on the *y*-axis and C (0, 0, 5) on the *z*-axis as shown in Figure 5.32.

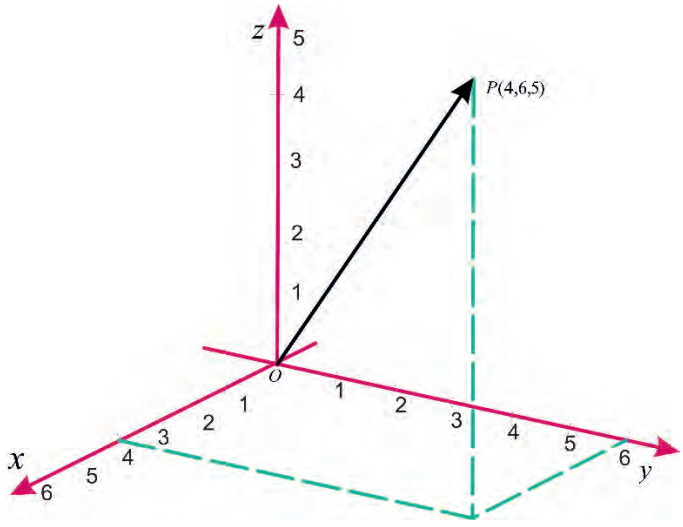


Figure 5.32

That means $\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{OD} + \overrightarrow{OP}$ or in terms of the unit vectors,

$(4, 6, 5) = 4\mathbf{i} + 6\mathbf{j} + 5\mathbf{k} = (4, 0, 0) + (0, 6, 0) + (0, 0, 5)$.

Magnitude of a vector

At the beginning of the discussion about vectors in space, it was mentioned that a vector is usually represented by an arrow, where the arrow head indicates the direction and the length of the arrow represents the magnitude of the vector. Thus, to

find the magnitude of a vector, it will be sufficient to find the distance between the initial point and the terminal point of the vector in the coordinate space.

Example 3

If the initial point of a vector is at the origin of the coordinate space and the terminal point is at P (3, 2, 2), then find the magnitude of the vector $\overrightarrow{OP}$.

Solution

The magnitude of the vector $\overrightarrow{OP}$ is the distance from O to P. This is, $\sqrt{3^2 + 2^2 + 2^2} = \sqrt{17}$.

Note

1. If the initial point of a vector **u** is at the origin and its terminal point is at a point

   Q(x, y, z) or if $\mathbf{u} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ then $|\mathbf{u}| = \sqrt{x^2 + y^2 + z^2}$.
2. If the initial point of **u** is at P(x_1, y_1, z_1) and the terminal point at Q(x_2, y_2, z_2) then $|\mathbf{u}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$.

Exercise 5.11

1. Calculate the magnitude of each of the following vectors.

   a. (–1, 2, 0) b. (2, 1, –1) c. $(\frac{1}{2}, \frac{3}{4}, \frac{4}{5})$

2. Make a three-dimensional sketch showing each of the following vectors with initial point at the origin.

   a. $\mathbf{u} = 3\mathbf{i} + \mathbf{j} + 4\mathbf{k}$ b. $\mathbf{v} = 4\mathbf{i} + 5\mathbf{j} - \mathbf{k}$

   c. $\mathbf{w} = -2\mathbf{i} + 4\mathbf{j} + 4\mathbf{k}$ d. $\mathbf{u} = 3\mathbf{j} - 5\mathbf{k}$

5.3.3 Cross Product

Activity 5.10

1. What do you know about cross product of vectors?
2. Find det(A) if A = $\begin{bmatrix} 4 & -4 \\ -5 & 2 \end{bmatrix}$
3. Find det(A) if A = $\begin{bmatrix} 5 & -3 & 1 \\ 0 & 1 & 3 \\ -1 & 4 & -1 \end{bmatrix}$

From Activity 5.10, recall 2 by 2 and 3 by 3 matrix that you have already seen in previous unit.

The determinant of 2 by 2 matrix $\begin{pmatrix} u_1 & u_2 \\ v_1 & v_2 \end{pmatrix}$ is defined by $u_1v_2 - u_2v_1$.

Again, a determinant of 3 by 3 matrix can be defined in terms of 2 by 2 determinants as follows:

$$\begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix} = u_1 \begin{vmatrix} v_2 & v_3 \\ w_2 & w_3 \end{vmatrix} - u_2 \begin{vmatrix} v_1 & v_3 \\ w_1 & w_3 \end{vmatrix} + u_3 \begin{vmatrix} v_1 & v_2 \\ w_1 & w_2 \end{vmatrix}$$

Observe that each term on the right side of the above equation involves a number u_i in the first row of the determinant, and u_i is multiplied by 2 by 2 determinant obtained from the left side by deleting the row and column in which u_i appears. This implies that 2 by 2 determinants and the standard basis vectors

$\mathbf{i} = (1, 0, 0)$, $\mathbf{j} = (0, 1, 0)$ and $\mathbf{k} = (0, 0, 1)$ define the cross product of the vectors

$\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j} + u_3\boldsymbol{k}$ and $\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\boldsymbol{k}$ as

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}$$

$$= \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} \mathbf{k}$$

$$= (u_2v_3 - u_3v_2)\,\mathbf{i} - (u_1v_3 - u_3v_1)\,\mathbf{j} + (u_1v_2 - u_2v_1)\,\mathbf{k}$$

Thus, the cross product of two vectors, say $\mathbf{u} \times \mathbf{v}$, is equal to another vector at right angles to both vectors, and it happens in the three-dimensions.

Definition 5.7

Given two non-zero vectors, $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}$ and $\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$ then the cross product of $\mathbf{u}$ and $\mathbf{v}$ is the vector

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = (u_2v_3 - u_3v_2)\,\mathbf{i} - (u_1v_3 - u_3v_1)\,\mathbf{j} + (u_1v_2 - u_2v_1)\,\mathbf{k}$$

Example 1

If $\mathbf{u} = \mathbf{i} + 3\mathbf{j} + 4\mathbf{k}$ and $\mathbf{v} = 2\mathbf{i} + 7\mathbf{j} - 5\mathbf{k}$, then find the cross product of $\mathbf{u} \times \mathbf{v}$.

Solution

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} i & j & k \\ 1 & 3 & 4 \\ 2 & 7 & -5 \end{vmatrix} = \begin{vmatrix} 3 & 4 \\ 7 & -5 \end{vmatrix}\mathbf{i} - \begin{vmatrix} 1 & 4 \\ 2 & -5 \end{vmatrix}\mathbf{j} + \begin{vmatrix} 1 & 3 \\ 2 & 7 \end{vmatrix}\mathbf{k}$$

$$= (-15 - 28)\,\mathbf{i} - (-5 - 8)\,\mathbf{j} + (7 - 6)\mathbf{k} = -43\mathbf{i} + 13\mathbf{j} + \mathbf{k}$$

Thus, the cross product of $\mathbf{u} \times \mathbf{v}$ is $-43\mathbf{i} + 13\mathbf{j} + \mathbf{k}$.

Example 2

Find $|\mathbf{u} \times \mathbf{v}|$ where $\mathbf{u} = 3\mathbf{i} + 4\mathbf{j} + 5\mathbf{k}$ and $\mathbf{v} = 7\mathbf{i} + 8\mathbf{j} + 9\mathbf{k}$.

Solution

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 4 & 5 \\ 7 & 8 & 9 \end{vmatrix} = ((4\times 9) - (5\times 8))\,\mathbf{i} - ((3\times 9) - (5\times 7))\,\mathbf{j} + ((3\times 8) - (4\times 7))\mathbf{k}$$

$$= (36 - 40)\,\mathbf{i} - (27 - 35)\,\mathbf{j} + (24 - 28)\,\mathbf{k} = -4\mathbf{i} + 8\mathbf{j} - 4\mathbf{k}$$

$$= -4\mathbf{i} + 8\mathbf{j} - 4\mathbf{k}$$

Therefore, $|\mathbf{u} \times \mathbf{v}| = \sqrt{(-4)^2 + 8^2 + (-4)^2} = 4\sqrt{6}$.

Theorem 5.6 (Properties of the cross product)

If $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ are vectors and r is a scalar

1. Anticommutativity: $\mathbf{u} \times \mathbf{v} = -(\mathbf{v} \times \mathbf{u})$
2. Multiplication by scalars: $(r\mathbf{u}) \times \mathbf{v} = r(\mathbf{u} \times \mathbf{v}) = \mathbf{u} \times (r\mathbf{v})$
3. Distributivity: $\mathbf{u} \times (\mathbf{v} + \mathbf{w}) = \mathbf{u} \times \mathbf{v} + \mathbf{u} \times \mathbf{w}$
4. Orthogonal properties: $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{u} = (\mathbf{u} \times \mathbf{v}) \cdot \mathbf{v} = 0$
5. Cross product of the zero vector: $\mathbf{u} \times 0 = 0$
6. Cross product of the vector with itself: $\mathbf{u} \times \mathbf{u} = 0$
7. The cross product of the unit vectors: $\mathbf{i} \times \mathbf{i} = \mathbf{j} \times \mathbf{j} = \mathbf{k} \times \mathbf{k} = 0$
8. Triple scalar product: $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = (\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}$

Note

$\mathbf{i} \times \mathbf{j} = \mathbf{k}$, $\mathbf{j} \times \mathbf{k} = \mathbf{i}$, $\mathbf{k} \times \mathbf{i} = \mathbf{j}$, $\mathbf{j} \times \mathbf{i} = -\mathbf{k}$, $\mathbf{k} \times \mathbf{j} = -\mathbf{i}$ and $\mathbf{i} \times \mathbf{k} = -\mathbf{j}$

Exercise 5.12

1. If $\mathbf{u} = \frac{1}{2}\mathbf{i} + 3\mathbf{j} + \mathbf{k}$ and $\mathbf{v} = 2\mathbf{i} - \frac{3}{4}\mathbf{j} - 2\mathbf{k}$, then find the following.
   a. $\mathbf{u} \times \mathbf{v}$ b. $-(\mathbf{v} \times \mathbf{u})$ c. $\mathbf{u} \times \mathbf{u}$
   d. $\mathbf{u} \cdot (\mathbf{u} \times \mathbf{v})$ e. $\mathbf{v} \cdot (\mathbf{u} \times \mathbf{v})$ f. $|\mathbf{u} \times \mathbf{v}|$
2. Let $\mathbf{u} = \mathbf{i} + 3\mathbf{j} + 5\mathbf{k}$ and $\mathbf{v} = 5\mathbf{i} + 6\mathbf{j} - 6\mathbf{k}$ be vectors. Then, find $\mathbf{u} \times \mathbf{v}$ and $|\mathbf{u} \times \mathbf{v}|$.
3. Find $\mathbf{u} \times \mathbf{v}$ if $\mathbf{u} = 2\mathbf{i} + \mathbf{k}$ and $\mathbf{v} = \mathbf{i} + \mathbf{j} + \mathbf{k}$.
4. Given a vector $\mathbf{u} = (\mathbf{i} + 3\mathbf{j} - 2\mathbf{k}) \times (\mathbf{i} + 3\mathbf{k})$ then find the magnitude of $\mathbf{u}$.

Theorem 5.7

The cross-product $\mathbf{u} \times \mathbf{v}$ is perpendicular or orthogonal to both $\mathbf{u}$ and $\mathbf{v}$.

If $\mathbf{u}$ and $\mathbf{v}$ are represented by directed line segments with the same initial point then the cross-product $\mathbf{u} \times \mathbf{v}$ points in a direction perpendicular to the plane through $\mathbf{u}$ and $\mathbf{v}$.

Its magnitude determines area of the parallelogram $A = |\mathbf{u} \times \mathbf{v}|$ as shown in Figure 5.33.

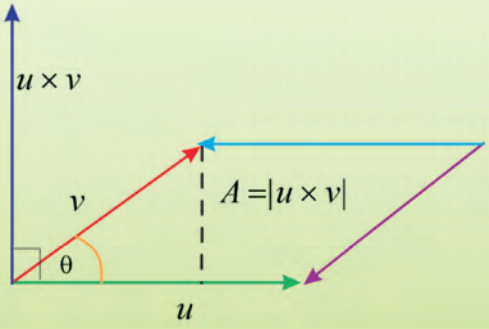


Figure 5.33

Example 3

Find the area of the parallelogram formed by the vectors $\mathbf{u} = 3\mathbf{i} + \mathbf{j}$ and $\mathbf{v} = 2\mathbf{i} + 3\mathbf{j}$, spanned between them.

Solution

Area of the parallelogram = $|\mathbf{u} \times \mathbf{v}|$

$$|\mathbf{A}| = |\mathbf{u} \times \mathbf{v}| = \left| \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 1 & 0 \\ 2 & 3 & 0 \end{vmatrix} \right|$$

$$= |(0)\,\mathbf{i} - (0)\,\mathbf{j} + (7)\,\mathbf{k}|$$

$$= 7 \text{ square units}$$

Example 4

If $\mathbf{u}$ and $\mathbf{v}$ are two vectors represented as $2\mathbf{i} + 3\mathbf{j} + 4\mathbf{k}$ and $\mathbf{i} + 3\mathbf{j} + 5\mathbf{k}$, find the area of the parallelogram spanned between them.

Solution

Area of the parallelogram A = $|\mathbf{u} \times \mathbf{v}|$

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 3 & 4 \\ 1 & 3 & 5 \end{vmatrix} = |(15 - 12)\,\mathbf{i} - (10 - 4)\,\mathbf{j} + (6 - 3)\,\mathbf{k} = 3\mathbf{i} - 6\mathbf{j} + 3\mathbf{k}$$

Then, $A = |\mathbf{u} \times \mathbf{v}| = |3\mathbf{i} - 6\mathbf{j} + 3\mathbf{k}| = \sqrt{3^2 + (-6)^2 + 3^2}$

$$= 3\sqrt{6}\,.$$

Theorem 5.8

If θ is the angle between **u** and **v** (where, $0 \leq \theta \leq \pi$) then $|\mathbf{u} \times \mathbf{v}| = |\mathbf{u}||\mathbf{v}| \sin\theta$.

Since a vector is completely determined by its magnitude and direction, we can now say that $\mathbf{u} \times \mathbf{v}$ is the vector that is perpendicular to both **u** and **v**, whose orientation is determined by the right-hand rule, and whose length is $|\mathbf{u}|\,|\mathbf{v}|\sin\theta$ as shown in Figure 5.34.

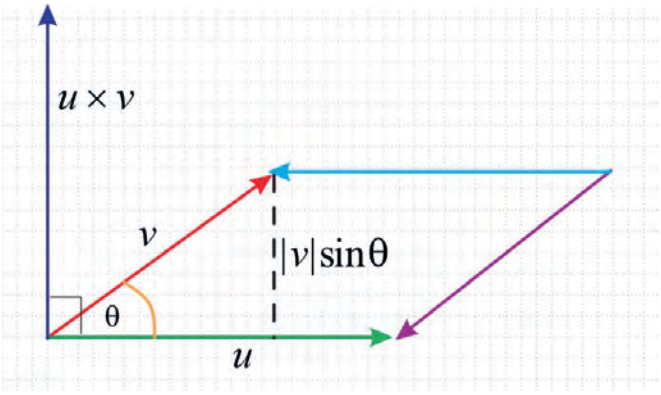


Figure 5.34

If **u** and **v** are represented by directed line segments with the same initial point, then they determine a parallelogram with base $|\mathbf{u}|$, altitude $|\mathbf{v}|\sin\theta$, and area $A = |\mathbf{u} \times \mathbf{v}| = |\mathbf{u}|\,|\mathbf{v}|\sin\theta$ as shown in the Figure 5. 34.

Exercise 5.13

1. Find the area of the parallelogram defined by the vectors $\mathbf{u} = 2\mathbf{i} + \mathbf{j}$ and $\mathbf{v} = \mathbf{i} + 3\mathbf{j}$.
2. Find the area of the triangle with vertices $(-1, 2)$, $(4, -1)$ and $(8, 7)$.
3. Find the area of the parallelogram defined by the vectors $\mathbf{u} = 2\mathbf{i} + 3\mathbf{j} + \mathbf{k}$ and $\mathbf{v} = 4\mathbf{i} + \mathbf{j} + 2\mathbf{k}$.

5.4 Application of Scalar and Cross Product

The dot product is used to find the angle between two vectors, the length of the vector as well as work done by the force, whereas the cross product is mostly used to determine the vector, which is perpendicular to the plane surface spanned by two vectors.

Example 1

Find the measure of the angle between each pair of the vectors.

a. $\mathbf{u} = 4\mathbf{i} - \mathbf{j}$ and $\mathbf{v} = 5\mathbf{i} + 3\mathbf{j}$

b. $\mathbf{u} = 2\mathbf{i} + 5\mathbf{j}$ and $\mathbf{v} = -\frac{1}{2}\mathbf{i} + \frac{3}{4}\mathbf{j}$

Solution

a. To find the cosine of the angle formed by two vectors. substitute the components of the vector into equation $\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|}$

But $\mathbf{u} \cdot \mathbf{v} = (4\mathbf{i} - \mathbf{j}) \cdot (5\mathbf{i} + 3\mathbf{j}) = 4(5) + (-1)(3) = 20 - 3 = 17$, $|\mathbf{u}| = \sqrt{4^2 + (-1)^2} = \sqrt{17}$ and $|\mathbf{v}| = \sqrt{5^2 + (3)^2} = \sqrt{34}$

$\Rightarrow \cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|} = \frac{17}{\sqrt{17}\sqrt{34}} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$

$\Rightarrow \theta = \cos^{-1}\left(\frac{\sqrt{2}}{2}\right)$

$\Rightarrow \theta = 45°$.

b. To find the cosine of the angle formed by two vectors, substitute the components of the vector into equation $\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|}$

But $\mathbf{u} \cdot \mathbf{v} = (2\mathbf{i} + 5\mathbf{j})\left(-\frac{1}{2}\mathbf{i} + \frac{3}{4}\mathbf{j}\right) = 2\left(-\frac{1}{2}\right) + 5\left(\frac{3}{4}\right) = -1 + \frac{15}{4} = \frac{11}{4}$,

$|\mathbf{u}| = \sqrt{2^2 + 5^2} = \sqrt{29}$ and

$|\mathbf{v}| = \sqrt{\left(-\frac{1}{2}\right)^2 + \left(\frac{3}{4}\right)^2} = \frac{1}{4}\sqrt{13}$

$\Rightarrow \cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|} = \frac{\frac{11}{4}}{\sqrt{29} \times \frac{1}{4}\sqrt{13}} = \frac{11}{\sqrt{377}}$

$\Rightarrow \theta = \cos^{-1}\left(\frac{11}{\sqrt{377}}\right)$

Example 2

Suppose a child is pulling a wagon with a force having a magnitude of 8 N on the handle at an angle of 60° , as shown in Figure 5.35. If the child pulls the wagon 50 m, then find the work done by the force.

Solution

According to physics formula, work done is got by dot product of force and displacement. Hence, the work done by the force F, acting at an angle $\theta = 60°$ form the line of motion which is given by

$W = |F_x| S = (|F| \cos\theta) S = |F| S \cos\theta$, where S is the displacement of the wagon.

$= 8N\ (50\ m) \cos 60° = 400\left(\frac{1}{2}\right) N\ m = 200\ J$

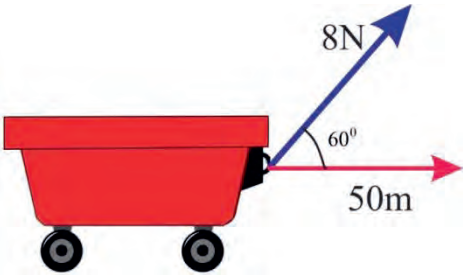


Figure 5.35

Example 3

Find the area of the triangle with vertices P (1, 4, 6), Q (–2, 5, –1), and R (1, –1, 1).

Solution

$\overrightarrow{PQ} = (-2-1)\,\mathbf{i} + (5-4)\,\mathbf{j} + (-1-6)\,\mathbf{k} = -3\mathbf{i} + \mathbf{j} - 7\mathbf{k}$

$\overrightarrow{PR} = (1-1)\,\mathbf{i} + (-1-4)\,\mathbf{j} + (1-6)\,\mathbf{k} = 0\mathbf{i} - 5\mathbf{j} - 5\mathbf{k}$

Then, $\overrightarrow{PQ} \times \overrightarrow{PR} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -3 & 1 & -7 \\ 0 & -5 & -5 \end{vmatrix} = (-5-35)\,\mathbf{i} - (15-0)\,\mathbf{j} + (15-0)\,\mathbf{k}$

$= -40\mathbf{i} - 15\mathbf{j} + 15\mathbf{k}$

Thus, $\overrightarrow{PQ} \times \overrightarrow{PR} = -40\mathbf{i} - 15\mathbf{j} + 15\mathbf{k}$. The area of the parallelogram with adjacent sides $\overrightarrow{PQ}$ and $\overrightarrow{PR}$ is the magnitude of this cross product:

$|\overrightarrow{PQ} \times \overrightarrow{PR}| = \sqrt{(-40)^2 + (-15)^2 + (15)^2}$

$= 5\sqrt{82}$

The area A of the triangle PQR is half the area of this parallelogram.
Thus, A = $\frac{5}{2}\sqrt{82}$,

Example 4

A bolt is tightened by applying a 40 N force to a 0.25 m wrench, as shown in Figure 5.36.

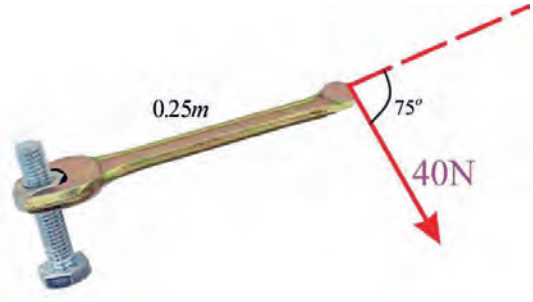


Figure 5.36

Find the magnitude of the torque about the centre of the bolt.

Solution

According to physics formula, the torque vector is calculated by cross product of the position vector and the force.

Thus, $|\tau| = |\mathbf{r} \times \mathbf{F}| = |\mathbf{r}||\mathbf{F}| \sin 75°$

$= (0.25)(40) \sin 75°$

$= 10 \sin 75° \approx 9.66$ N m,

If the bolt is right-threaded, then the torque vector itself is $\tau = |\tau|\,\mathbf{n} \approx 9.66\,\mathbf{n}$ where $\mathbf{n}$ is a unit vector directed down into the page.

Exercise 5.14

1. Find the measure of the angle between each pair of the vector.
   a. $\mathbf{u} = 3\mathbf{i} + 2\mathbf{j}$ and $\mathbf{v} = 5\mathbf{i} - \mathbf{j}$ b. $\mathbf{u} = \mathbf{i} + 7\mathbf{j}$ and $\mathbf{v} = -\mathbf{i} + \mathbf{j}$
   c. $\mathbf{u} = 6\mathbf{i} + \mathbf{j}$ and $\mathbf{v} = \mathbf{i} + 8\mathbf{j}$
2. A conveyor belt generates a force $\overrightarrow{F} = 5\mathbf{i} - 3\mathbf{j} + 3\mathbf{k}$ that moves a suitcase from point P (1, 1, 1) to Point Q (9, 4, 7) along a straight line. Find the work done by the conveyor belt. The distance is measured in meters and the force is measured in newtons.
3. If $\mathbf{u}$ and $\mathbf{v}$ are vectors represented as $2\mathbf{i} + 3\,\mathbf{j} + 4\,\mathbf{k}$ and $\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$, then find the area of the parallelogram spanned between them.
4. Find the area of the triangle whose vertices are A (3, −1, 2), B (1, − 1, −3) and C (4, − 3, 1).
5. Find the area of the quadrilateral whose vertices are A (1, 0), B (3, 5), C (7, 2) and D (5, −2).
6. A car mechanic applies a force of 800 N to a wrench to loosen a bolt. She applies the force perpendicular to the arm of the wrench. The distance from the bolt to her hand is 0.40 m. What is the magnitude of the torque applied?

5.5 Application of Vectors

In previous section, you have seen that vectors have many applications. Geometrically, any two points in the plane determine a straight line. A straight line in the plane is determined by its slope and a point through which it passes. These lines have been determined to have a certain direction. Thus, related to vectors, you will see how one can write equations of lines, circles, and tangent line to a circle using vectors.

5.5.1 Vectors and Lines

Activity 5.11

Consider two points in the line such as $R_0\ (x_0\ , y_0)$ and $R_1\ (x_1\ , y_1)$, then the vector from R_0 to R_1 is $\mathbf{v} = (x_1 - x_0,\ y_1 - y_0)$ as shown in Figure 5.36. Discuss what will happen to the magnitude and direction of vector $\mathbf{v} = \overrightarrow{r_1} - \overrightarrow{r_0} = (x_1 - x_0,\ y_1 - y_0\ )$, where $\overrightarrow{r_1}$ and $\overrightarrow{r_0}$ are position vectors corresponding to the points R_0 and R_1 respectively, if t = 1, 2, 3, 4 and so on and generalize it for $\vec{r} = \overrightarrow{r_0} + t\mathbf{v}$ where t is a scalar, $t \in \mathbb{R}$ and $\mathbf{v}$ is a direction vector of the line.

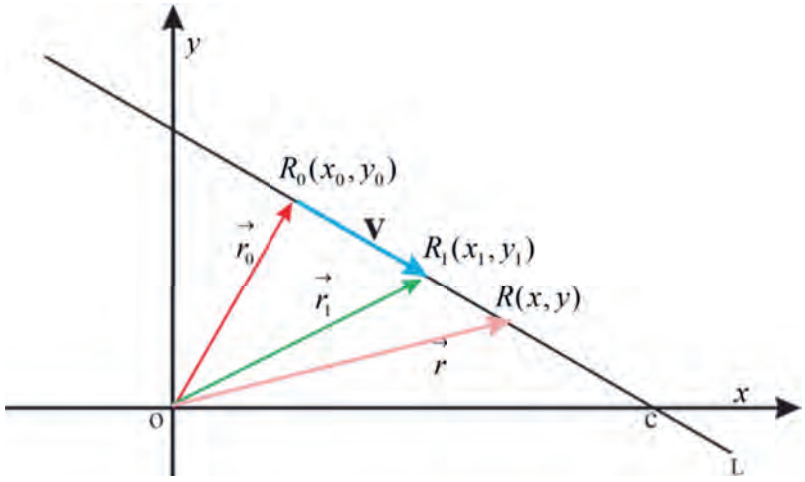


Figure 5.37

From Activity 5.11, you have observed that the line L that passes through R_0 and R_1 is parallel to the vector $\mathbf{v} = (x_1 - x_0,\ y_1 - y_0)$.

Let R (x, y) be any point on L. Then the position vector $(\vec{r})$ from the origin to any point on line L is obtained by

$\overrightarrow{OR_0} + \overrightarrow{R_0 R} = \overrightarrow{OR}$

$\Rightarrow \overrightarrow{r_0} + t\mathbf{v} = \vec{r}$

$\Rightarrow \vec{r} = \overrightarrow{r_0} + t\mathbf{v}$ where t is a scalar, $t \in \mathbb{R}$ and $\mathbf{v}$ is a direction vector of the line.

Thus,

1. If a direction vector $\mathbf{v}$ and a point $R_0(x_0, y_0)$ are given, then the vector equation of the line L determined by $\overrightarrow{r_0}$ and $\mathbf{v}$ is given by $\vec{r} = \overrightarrow{r_0} + t\mathbf{v}$, $t \in \mathbb{R}$ and $\mathbf{v} \neq \mathbf{0}$ where t is a scalar, $t \in \mathbb{R}$ and $\mathbf{v}$ is a direction vector of the line.
2. If $\mathbf{v} = (p, q)$, $R(x, y)$ and $R_0(x_0, y_0)$, then the above equation can be written as
   $(x, y) = (x_0, y_0) + t(p, q) = (x_0, y_0) + (tp, tq) = (x_0 + tp, y_0 + tq)$
   Then, from equality of vectors you have
   $x = x_0 + tp$ and $y = y_0 + tq$ where $t \in \mathbb{R}$, $(p, q) \neq (0,0)$
   This system of the equations is called the parametric equations of the line L through $R_0(x_0, y_0)$ whose direction is that of the vector $\mathbf{v} = (p, q)$, t is called a parameter.
3. If p and q are both different from 0 then from (2) above

$$t = \frac{x - x_0}{p} \text{ and } t = \frac{y - y_0}{q}$$

$$\Rightarrow \frac{x - x_0}{p} = \frac{y - y_0}{q}$$ which is called the standard equations of the line.

Example 1

Consider the line passing through the point (1, 4) and (2, 2), then

a. Find the vector equation.
b. Find the parametric equations.
c. Find the standard equations.

Solution

Let $R_0 = (1, 4)$ and $R_1 = (2, 2)$ as shown in Figure 5.38.

a. The vector equation of the line is:

$$(x, y) = (1, 4) + t\big((2, 2) - (1, 4)\big)$$

$$= (1, 4) + t(1, -2)$$

Thus, $(x, y) = (1, 4) + t(1, -2), t \in \mathbb{R}$,

b. From the vector equation of question (a), you have the parametric equations:

$x = 1 + t, y = 4 - 2t, t \in \mathbb{R}$.

c. From parametric equation of question (b), you have the standard equation

$x - 1 = \frac{y-4}{-2} \Rightarrow 2x + y - 6 = 0$.

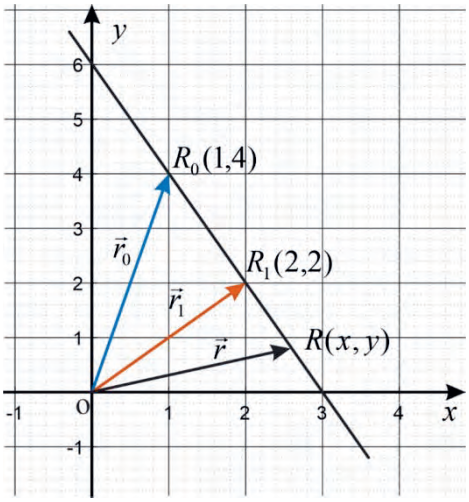


Figure 5.38

Example 2

Consider a line passing through (2, 2) and with direction vector (1, 4), then

a. Find the vector equation of the line.

b. Find the parametric equation.

Solution

Let $R_0 = (2, 2)$ and $\mathbf{v} = (1, 4)$ as shown in Figure 5.39.

a. The vector equation of the line is:

$(x, y) = (2, 2) + t(1, 4), t \in \mathbb{R}$.

b. The parametric equations are:

$x = 2 + t, y = 2 + 4t, t \in \mathbb{R}$.

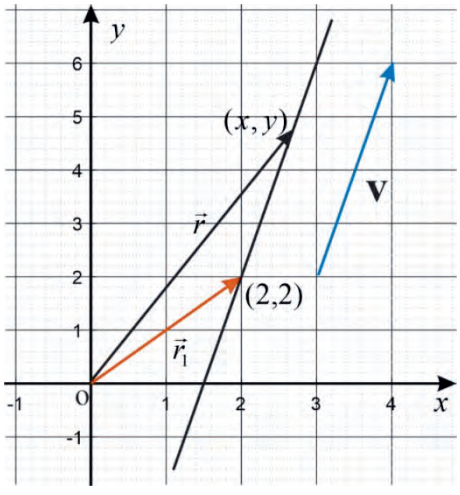


Figure 5.39

Exercise 5.15

1. Consider the line defined by the following points:
   a. (2, −9) and (8, 4) b. (7, 3) and (4, −3) c. (−1, −1) and (3, 10)
   1) Sketch the line.
   2) Find the vector equation of the line.
   3) Find the parametric equations of the line.
   4) Find the standard equation of the line.
2. Find the vector equation of the line passing through (1, −3) with direction vector (−5, 6).
3. Find the vector equation of line passing through the point having vector $5\mathbf{i} + 4\mathbf{j}$ and having direction vector (−3, 4).
4. Find the vector equation of the line passing through the point having position vector $-\mathbf{i} - \mathbf{j}$ and parallel to the line L, $\mathbf{i} + 2\mathbf{j}$.

5.5.2 Vectors and Circles

Activity 5.12

Consider a circle with centre $C\ (x_0, y_0)$ and radius r> 0 and any point $R\ (x, y)$ on a circle such that $|\vec{r} - \vec{c}| = r$. where $\vec{r}$ and $\vec{c}$ are position vectors of $R\ (x, y)$ and $C\ (x_0, y_0)$ respectively as shown in Figure 5.40. Discuss how to show that $\vec{r} \cdot \vec{r} - 2\vec{r} \cdot \vec{c} + \vec{c} \cdot \vec{c} = r^2$ represents an equation of the circle at $C(x_0, y_0)$ and radius r through a position vector of any point on the circle $R(x, y)$.

Consider a circle with centre $C\ (x_0, y_0)$ and radius r > 0 as the set of points $R\ (x, y)$ in the plane such that $|\vec{r} - \vec{c}| = r$, where $\vec{r}$ and $\vec{c}$ are position vectors of $R\ (x, y)$ and $C\ (x_0, y_0)$, respectively as shown in Figure 5.40.

By squaring both sides $|\vec{r} - \vec{c}|^2 = r^2$

$\Rightarrow (\vec{r} - \vec{c}) \cdot (\vec{r} - \vec{c}) = r^2$ 1

$\Rightarrow \vec{r} \cdot \vec{r} - 2\vec{r} \cdot \vec{c} + = r^2$ 2

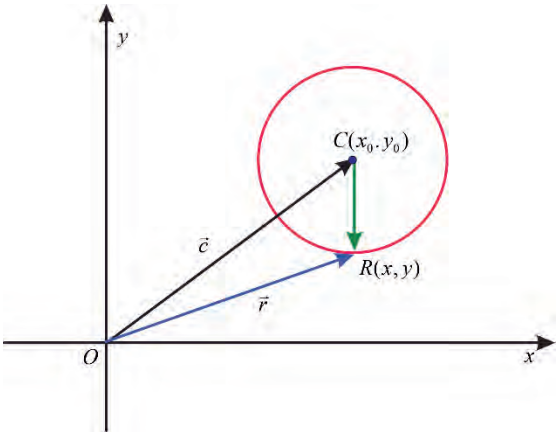


Figure 5.40

Thus, equation 2 represents the equation of the circle centred at $C(x_0, y_0)$ and radius r by using a position vector of any point on the circle $R(x, y)$.

Substituting the corresponding components of $\vec{r}$ and $\vec{c}$ in equation 1 and simplification, we obtain:

$(x - x_0)^2 + (y - y_0)^2 = r^2$ which is called **the standard equation of a circle.**

This equation can be expressed as:

$x^2 + y^2 + Ax + By + C=0$ where $A = -2x_0$, $B = -2y_0$ and $C = x_0^2 + y_0^2 - r^2$.....by expanding and rearranging the terms.

Example 1

Find an equation of the circle centred at $C(2, 3)$ and radius 4.

Solution

Let $R(x, y)$ be a point on the circle as shown in Figure 5.41.

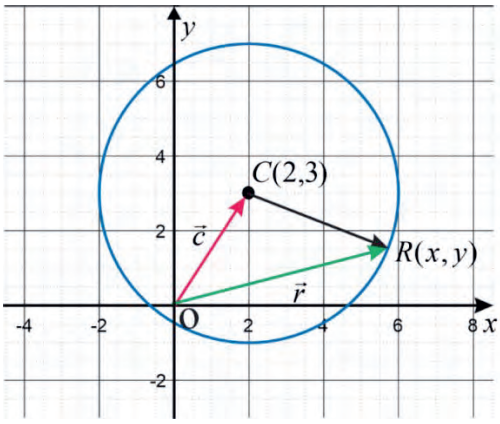


Figure 5.41

Let $\vec{r}$ and $\vec{c}$ be the position vectors of $R\ (x, y)$ and $C\ (x_0, y_0)$ respectively. Then from equation 2, you have

$\vec{r} \cdot \vec{r} - 2\vec{r} \cdot \vec{c} + \vec{c} \cdot \vec{c} = r^2$

Substituting the corresponding components of $\vec{r}$ and $\vec{c}$ in equation 2, you obtain:

$(x, y) \cdot (x, y) - 2(x, y) \cdot (2, 3) + (2,3) \cdot (2,3) = 4^2$

$\Rightarrow x^2 + y^2 - 2(2x + 3y) + 4 + 9 = 16$ -------by corollary 5.1

$\Rightarrow x^2 + y^2 - 4x - 6y + 13 = 16$

$\Rightarrow x^2 + y^2 - 4x - 6y + 13 - 16 = 16 - 16$ ---------- subtracting both sides

$\Rightarrow x^2 + y^2 - 4x - 6y - 3 = 0.$

Example 2

Find the equation of the circle whose end points of a diameter are P $(-1, -1)$ and Q = (4, 11).

Solution

The centre of the circle is $C\ (x_0, y_0) = C\ (\frac{-1+4}{2}, \frac{-1+11}{2}) = C\ (\frac{3}{2}, 5)$

The radius of the circle is given by $r = \frac{|d|}{2} = \frac{\sqrt{(4+1)^2 + (11+1)^2}}{2} = \frac{\sqrt{(5)^2 + (12)^2}}{2}$

$= \frac{\sqrt{169}}{2} = \frac{13}{2}$

Let $R(x, y)$ be a point on the circle and $\vec{r}$ and $\vec{c}$ be position vectors of R and C respectively.

Substituting the corresponding components of $\vec{r}$ and $\vec{c}$ in equation 2, you obtain:

$(x, y) \cdot (x, y) - 2(x, y) \cdot (\frac{3}{2}, 5) + (\frac{3}{2}, 5) \cdot (\frac{3}{2}, 5) = (\frac{13}{2})^2$

$\Rightarrow x^2 + y^2 - 2(\frac{3}{2}x + 5y) + \frac{9}{4} + 25 = \frac{169}{4}$, -------by corollary 5.1

$\Rightarrow x^2 + y^2 - 3x - 10y - 15 = 0.$

Exercise 5.16

1. Find an equation of the circle centred at C $(1, -3)$ and radius 6.
2. Find the equation of the circle whose end points of a diameter are P $(2, 1)$ and Q $= (6, 3)$.
3. Find an equation of the circle centred at origin and passing through $(3, 3)$.
4. Find an equation of the circle centred at C $(-4, 2)$ and passing through $(2, -5)$.

5.5.3 Equations of Tangent Lines to a Circle

Activity 5.13

Consider a circle with center R_0 (x_0, y_0) and a radius of r units as shown in the figure 5.41. Discuss how to show that the equation of the tangent line is $(x - x_o)(x_1 - x_o) + (y - y_o)(y_1 - y_o) = r^2$ where (x_1, y_1) is a point of tangency to the circle at R_1 and (x, y) is any point on line L.

Consider a circle with center C (x_0, y_0) and a radius of r units as shown in the Figure 5.42.

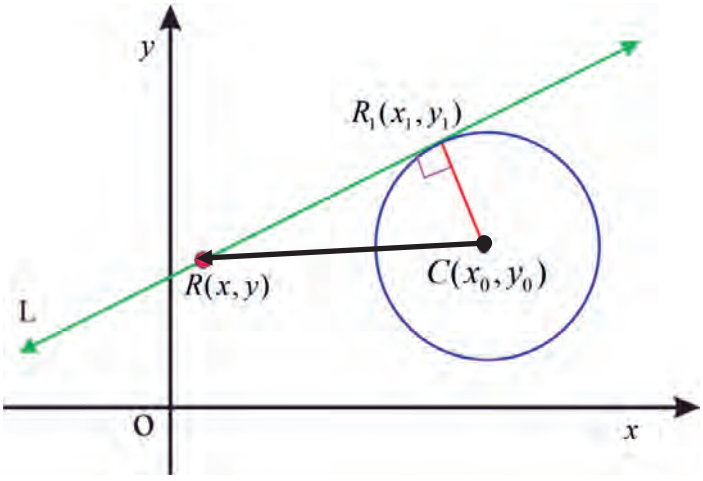


Figure 5.42

Let $R_1(x_1, y_1)$ be a point of tangency to the circle and it fulfills the following equation because it is on the point of the circle: $(x_1 - x_0)^2 + (y_1 - y_0)^2 = r^2$, $r > 0$.

A tangent line is a straight line that touches the circumference of a circle at the point of tangency.

The radius of the circle r is perpendicular (orthogonal) to the tangent L at the point of contact $R_1(x_1, y_1)$.

If $R(x, y)$ is an arbitrary point on L, $\overrightarrow{R_1R} \cdot \overrightarrow{CR_1} = 0$

Thus, the equation of the tangent line is

$$(x - x_1, y - y_1) \cdot (x_1 - x_0, y_1 - y_0) = 0$$

$\Rightarrow (x - x_1)(x_1 - x_0) + (y - y_1)(y_1 - y_0) = 0$ ..... by corollary 5.1

Adding both sides of the equation above and $(x_1 - x_0)^2 + (y_1 - y_0)^2 = r^2$ you obtain

$$(x - x_1)(x_1 - x_o) + (y - y_1)(y_1 - y_o) + (x_1 - x_o)^2 + (y_1 - y_o)^2 = r^2$$

Factorizing $(x_1 - x_o)$ and $(y_1 - y_o)$ from respective terms, you have

$$(x - x_1 + x_1 - x_o)(x_1 - x_o) + (y - y_1 + y_1 - y_o)(y_1 - y_o) = r^2$$

$\Rightarrow (x - x_o)(x_1 - x_o) + (y - y_o)(y_1 - y_o) = r^2$.

Note

If the circle is centred at the origin, then the above equation becomes:

$xx_1 + yy_1 = r^2$.

Example 1

Find the equation of the tangent line to the circle $x^2 + y^2 + 10x + 2y + 13 = 0$ at the point $(-3, 2)$

Solution

Let $R_1(x_1, y_1) = R_1(-3, 2)$ as shown in Figure 5.43

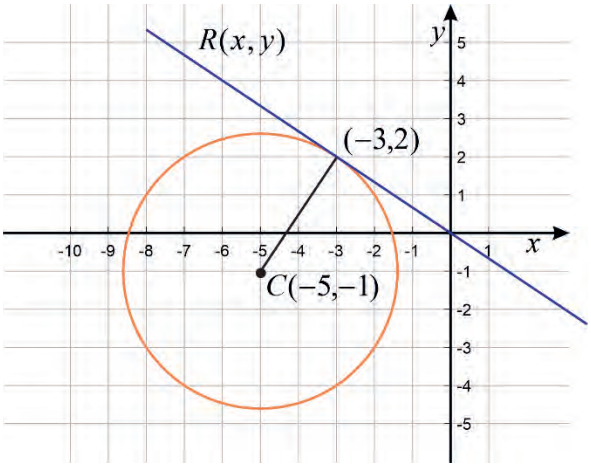


Figure 5.43

From $x^2 + y^2 + 10x + 2y + 13 = 0$

$\Rightarrow x^2 + 10x + y^2 + 2y + 13 = 0$

$\Rightarrow (x + 5)^2 - 25 + (y + 1)^2 - 1 + 13 = 0$ ..... By completing square

$\Rightarrow (x + 5)^2 + (y + 1)^2 - 26 + 13 = 0$

$\Rightarrow (x + 5)^2 + (y + 1)^2 = 13$

Therefore, the equation of the circle is

$(x + 5)^2 + (y + 1)^2 = 13$ with centre C $(-5, -1)$ and radius r = $\sqrt{13}$.

The equation of the tangent line is given by $(x - x_o)(x_1 - x_o) + (y - y_o)(y_1 - y_o) = r^2$

Substituting the corresponding components of $R_1(-3, 2)$ and $R_0(-5, -1)$ in equation above, you obtain:

$(x + 5)(-3 + 5) + (y + 1)(2 + 1) = \sqrt{13}^2$

$\Rightarrow (x + 5)(2) + (y + 1)(3) = 13$

$\Rightarrow 2x + 10 + 3y + 3 = 13$

$\Rightarrow 2x + 3y = 0$

Thus, the tangent line to the graph of the circle at the point $(-3, 2)$ is $2x + 3y = 0$

Example 2

Find equation of tangent line through R_1 (3, 4) a point on the circle $x^2 + y^2 = 25$ as shown in Figure 5.44

Solution

Recall that the equation of tangent line if the circle is centred at the origin is given by $xx_1 + yy_1 = r^2$

Since, the circle is centred at the origin with radius 5, the equation of the tangent line to the circle is

$x \times 3 + y \times 4 = 5^2 \Rightarrow 3x + 4y = 25.$

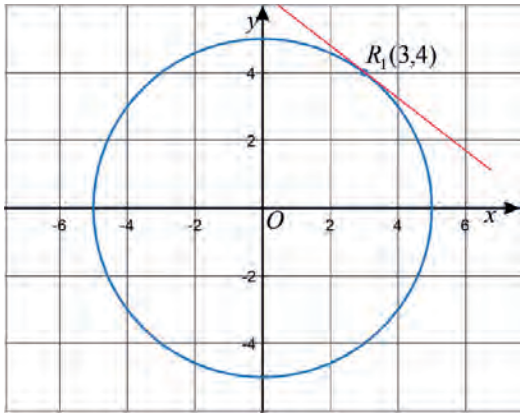


Figure 5.44

Exercise 5.17

1. Find the equation of the tangent line to the circle $x^2 + y^2 + 4x - 8y = 0$ at the point $R_1(-4, 8)$.
2. Find the equation of the line tangent to the circle $x^2 + y^2 - 8x - 10y - 128 = 0$ at the point $R_1(-8, 10)$.
3. Find equation of the tangent line through $R_1(-4, -3)$ point on the circle $x^2 + y^2 = 25$.
4. Find equation of tangent line to the circle $x^2 + y^2 = 29$ at the point R $(-2, 5)$.

5.6 Applications

In this unit, you discussed definition and properties of scalar as well as cross product of vectors with their application, application of vectors such as determining the equations of a line, and the equations of a tangent line to a circle. Now, you will consider further applications involving vectors.

Example 1

Prove the diagonals of a parallelogram meet at right angles if and only if it is a rhombus.

Proof

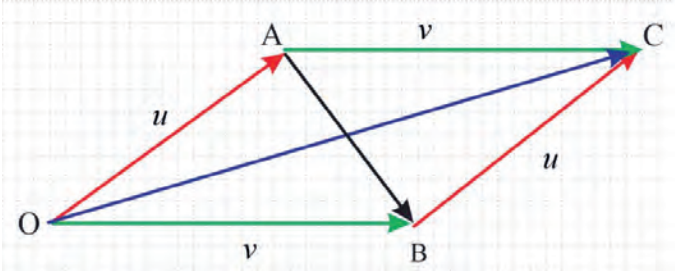


Figure 5.45

Let $\overrightarrow{OA} = \mathbf{u}$ and $\overrightarrow{OB} = \mathbf{v}$, as shown in Figure 5.45.

Therefore, $\overrightarrow{BC} = \overrightarrow{OA} = \mathbf{u}$

$\overrightarrow{AC} = \overrightarrow{OB} = \mathbf{v}$

$\overrightarrow{OC} = \mathbf{u} + \mathbf{v}$

$\overrightarrow{AB} = \mathbf{v} - \mathbf{u}$

$\overrightarrow{OC} \cdot \overrightarrow{AB} = (\mathbf{u} + \mathbf{v}) \cdot (\mathbf{v} - \mathbf{u})$

$= \mathbf{u} \cdot \mathbf{v} - \mathbf{u} \cdot \mathbf{u} + \mathbf{v} \cdot \mathbf{v} - \mathbf{u} \cdot \mathbf{v}$ ----- distributive property

$= \mathbf{v} \cdot \mathbf{v} - \mathbf{u} \cdot \mathbf{u} = |\mathbf{v}|^2 - |\mathbf{u}|^2$

Now for vectors to be perpendicular, the scalar product must be 0.

Therefore, $0 = |\mathbf{v}|^2 - |\mathbf{u}|^2$

$\Rightarrow |\mathbf{v}|^2 = |\mathbf{u}|^2 \Rightarrow |\mathbf{v}| = |\mathbf{u}|$

Hence, when the length of $\mathbf{u}$ is the same as $\mathbf{v}$ (i.e., a rhombus) the angle is 90°.

Example 2

A ball is thrown with an initial velocity of 70 m/s at an angle of 35° horizontally. Find the vertical and horizontal components of the velocity.

Solution

Let **v** represent the velocity and use the given information to write **v** in unit vector form:

$\mathbf{v} = 70\cos 35°\boldsymbol{i} + 70\sin 35°\boldsymbol{j}$

Simplify the scalars, you get:

$\mathbf{v} \approx 57.34\boldsymbol{i} + 40.15\boldsymbol{j}$ since the scalars are the horizontal and vertical components of **v**.

Therefore, the horizontal component is 57.34 m/s and the vertical component is 40.15 m/s.

Exercise 5.18

1. Prove the diagonals of a rectangle meet at right angles if and only if it is a square.
2. Use vectors to prove the sum of the squares of the lengths of a parallelogram's diagonals is equal to the sum of the squares of the lengths of the sides.
3. Ahmed is pulling her toy duck (mass 1kg) at a constant speed of 2.5 m/s. The string she uses to pull the duck makes an angle 45° above the horizontal and Ahmed keeps a constant tension in the string of 2 N. What is the work done by the tension force when the duck is pulled forward a distance of 2.8 m?
4. An airplane is flying in the direction of 43° east of north (also abbreviated as N43E at a speed of 550 m/h. A wind with speed 25 m/h comes from the southwest at a bearing of N15E. What is the ground speed and new direction of the airplane?

Problem Solving

1. A plane leaves the airport on the bearing of 45° traveling at 400 km/h if the bearing angles are measured clockwise from north. The wind is blowing at a bearing of 135° at a speed of 40 km/h. What is the actual velocity of the plane?
2. Two children are playing with a ball. The girl throws the ball to the boy. The ball travels in the air, curves 3 m to the right, and falls 5 m away from the girl, as shown in Figure 5.46. If the plane that contains the trajectory of the ball is perpendicular to the ground, find its equation.

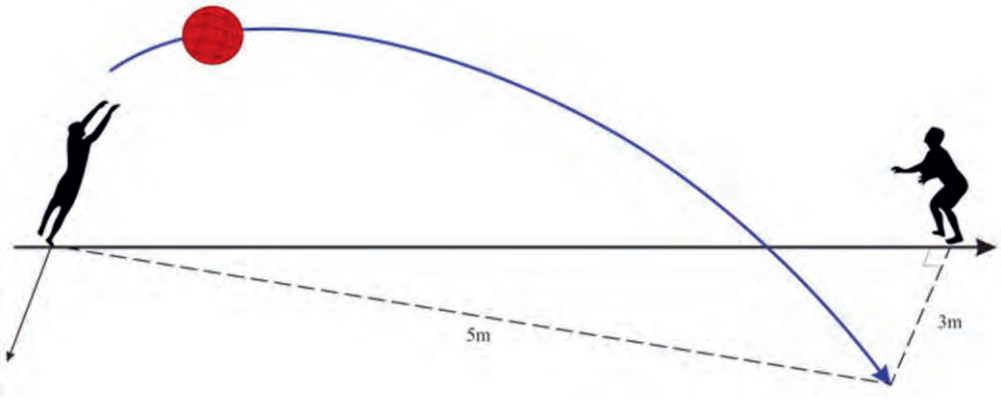


Figure 5.46

3. In square DEFG, side DE has column vector $\begin{pmatrix}2\\1\end{pmatrix}$. Find two possible column vectors for $\overrightarrow{EF}$.

Summary

1. **Vector**

   a. A scalar quantity is a physical quantity which can be expressed completely by its magnitude and particular units alone.

   b. A vector quantity is a physical quantity which can be expressed completely by stating both its magnitude with particular unit and directions.

   c. Two vectors are said to be equal, if they have the same magnitude and direction.

   d. A zero vector or null vector is a vector whose magnitude is zero and whose direction is indeterminate.

   e. A unit vector is a vector whose magnitude is one.

2. **Addition and subtraction of vectors**

Let $\mathbf{u}$ and $\mathbf{v}$ be vectors, then the sum $\mathbf{u} + \mathbf{v}$ is a vector given by the parallelogram law or triangle law satisfying the following properties.

   a. Vector addition is commutative. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$.

   b. Vector addition is associative. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$.

   c. $\mathbf{u} + \mathbf{0} = \mathbf{u}$.

   d. $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$.

   e. For any three vectors $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$, if $\mathbf{u} = \mathbf{v}$ and $\mathbf{v} = \mathbf{w}$, then $\mathbf{u} = \mathbf{w}$.

3. **Multiplication of a vector by a scalar**

Let $\mathbf{u}$ and $\mathbf{v}$ be vectors, let k_1 and k_2 be scalars hen the following properties are true.

   a. $|k_1\mathbf{u}| = |k_1|\mathbf{u}$.

   b. Associative Property: $k_1(k_2\,\mathbf{u}) = (k_1k_2)\,\mathbf{u}$.

   c. Distributive Property: $(k_1 + k_2)\,\mathbf{u} = k_1\mathbf{u} + k_2\mathbf{u}$ or $k_1(\mathbf{u} + \mathbf{v}) = k_1\mathbf{u} + k_2\mathbf{v}$.

   d. Identity Property: $1 \cdot \mathbf{u} = \mathbf{u}$.

   e. Multiplicative Property of zero: $0 \cdot \mathbf{u} = 0$.

4. **Scalar product or dot product**

If $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ be vectors, k is a scalar and θ is an angle between two vectors then the dot product of two vectors is defined as $\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}||\mathbf{v}| \cos\theta$ satisfying the following properties.

a. The scalar product of vectors is commutative. $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$.

b. $k(\mathbf{u} \cdot \mathbf{v}) = (k\mathbf{u}) \cdot \mathbf{v} = \mathbf{u} \cdot (k\mathbf{v})$.

c. The scalar product of vectors is distributive: $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$.

d. If $\mathbf{u} = \mathbf{0}$ or $\mathbf{v} = \mathbf{0}$, then $\mathbf{u} \cdot \mathbf{v} = 0$.

e. Two vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal if $\mathbf{u} \cdot \mathbf{v} = 0$.

f. If $\mathbf{u} = (u_1, u_2)$ and $\mathbf{v} = (v_1, v_2)$ are vectors, then $\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2$.

5. **Cross product**

If $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ are vectors and r is a scalar the cross product of vectors has the following properties

a. Anticommutativity: $\mathbf{u} \times \mathbf{v} = -\mathbf{v} \times \mathbf{u}$.

b. Multiplication by scalars: $(r\mathbf{u}) \times \mathbf{v} = r(u \times \mathbf{v}) = \mathbf{u} \times (r\,\mathbf{v})$.

c. Distributivity: $\mathbf{u} \times (\mathbf{u} + \mathbf{w}) = \mathbf{u} \times \mathbf{v} + \mathbf{u} \times \mathbf{w}$.

d. Orthogonal properties: $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{u} = (\mathbf{u} \times \mathbf{v}) \cdot \mathbf{v} = 0$.

e. Cross product of the zero vector: $\mathbf{u} \times 0 = 0$.

f. Cross product of the vector with itself: $\mathbf{u} \times \mathbf{u} = 0$.

g. The cross product of the unit vectors: $\mathbf{i} \times \mathbf{i} = \mathbf{j} \times \mathbf{j} = \mathbf{k} \times \mathbf{k} = 0$.

h. Triple scalar product: $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = (\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}$.

Review Exercise

1. If vector $\mathbf{u} = (3, -2)$, $\mathbf{v} = (-2, -1)$ and $\mathbf{w} = (5, 1)$, then find

   a. $2\mathbf{u} - 3\mathbf{v}$ b. $|3\mathbf{u} - 2\mathbf{v} + \mathbf{w}|$ c. $\mathbf{u} + 4\mathbf{v} - \mathbf{w}$ d. $|2\mathbf{u} + 2\mathbf{w}|$

   e. Find unit vector in the direction of $\mathbf{u} + \mathbf{w}$ f. Find $\mathbf{a}$ if $\mathbf{a} + \mathbf{v} = 2\mathbf{u} - \mathbf{w}$

2. If $\mathbf{u} = \mathbf{i} + 2\mathbf{j}$, $v = 2\mathbf{i} + 4\mathbf{j}$ and $w = 5\mathbf{i} - \mathbf{j}$, then find each of the following product

   a. $(\mathbf{u} \cdot \mathbf{v})\,\mathbf{w}$ b. $\mathbf{u} \cdot (3\mathbf{w})$ c. $|\mathbf{v}|^2$

3. If $\mathbf{u} = 3\mathbf{i} + 2\mathbf{j} + 5\mathbf{k}$ and $\mathbf{v} = \mathbf{i} + 3\mathbf{j} + 6\mathbf{k}$, then find

   a. $\mathbf{u} \times \mathbf{v}$ b. $\mathbf{v} \times \mathbf{u}$

4. If vector $\mathbf{v} = 2\mathbf{i} + 3\mathbf{j}$, then find the angle formed by the following vectors:

   a. $\mathbf{v}$ and $\mathbf{i}$ b. $\mathbf{v}$ and $\mathbf{j}$

5. Determine whether $\mathbf{u} = \mathbf{i} + 5\mathbf{k}$ and $\mathbf{v} = 10\mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$ are orthogonal vectors or not.

6. Let $\mathbf{u} = (-5, 3, 7)$ and $\mathbf{v} = (6, -8, 2)$. Determine if the vectors are parallel, perpendicular or neither.

7. Given the vectors $\mathbf{u} = 6\mathbf{i} + \mathbf{j} - 3\mathbf{k}$ and $\mathbf{v} = 2\mathbf{i} + 3\mathbf{j} + \mathbf{k}$, then find

   a. the area of the parallelogram formed by $\mathbf{u}$ and $\mathbf{v}$.

   b. the area of the triangle formed by $\mathbf{u}$ and $\mathbf{v}$.

8. A constant force of 60N is applied at angle of 30° to pull a handcart of 10 m across the ground, as shown in Figure 5.47. What is the work done by this force?

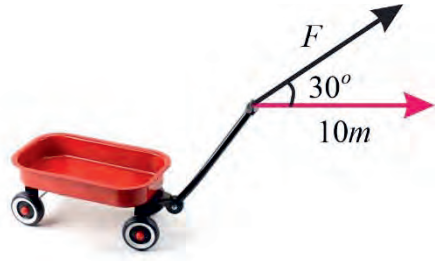


Figure 5.47

9. If the line L passing through the point P $(-2, 1)$ and parallel to the vector $2\mathbf{i} + 3\mathbf{j}$, then find a. vector equation of the line.

   b. parametric equation of the line.

10. Find the equation of the line tangent to the circle $x^2 + y^2 + x - y = 7$ at point

    a. A $(2\ -5)$ b. B $(-1, 4)$

11. If three forces $F_1 = 3\mathbf{i} - 4\mathbf{j}$, $F_2 = 4\mathbf{i} + 5\mathbf{j}$ and $F_3 = 5\mathbf{j}$ are acted on the body, then find the resultant force.

12. A balloon is rising vertically at 4 m/s. Wind is blowing from east to west at 3 m/s. Find the resultant velocity of the balloon.

13. A 50-N weight is hung by a cable so that the two portions of the cable make angles of 40° and 53°, respectively, with the horizontal, as shown in Figure 5.48. Find the magnitudes of the forces of tension T_1 and T_2 in the cables if the resultant force acting on the object is zero.

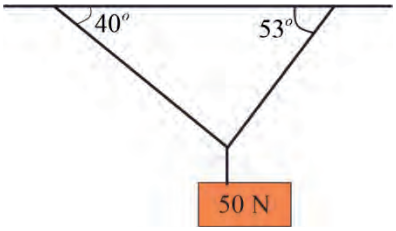


Figure 5.48

UNIT 6

TRANSFORMATIONS OF THE PLANE

Unit Outcomes

By the end of this unit, you will be able to

- Know basic concepts about transforming of the plane.
- Apply methods and procedures of transformation to transform plane figures.

Unit Contents

6.1 Introduction

6.2 Translation

6.3 Reflection

6.4 Rotation

6.5 Applications

Summary

Review Exercise

- identity transformation
- initial point
- non-rigid motion
- reflection
- rigid motion
- rotation
- standard position
- terminal point
- transformation
- vector translation

6.1 Introduction

In unit 5, you have discussed that various geometric and algebraic aspects of vector representation and vector algebra. In this unit, you focus on transformations and rigid motions which will be helpful to discuss translations, reflections and rotations.

Activity 6.1

Consider the following conditions and discuss what will happen to the shape or size or both of the object:

a. when a spring is compressed or stretched.

b. when the earth rotates about its axis.

c. when you see your image in a plane mirror.

d. when you draw your home's door.

From Activity 6.1, you have observed that some mappings of objects preserve shape, size or distance between any two points and some other mapping of objects are not preserve shape, size or distance between any two points. Due to this reason, transformations are classified as either rigid motion or non-rigid motion. But in this section, you will focus only on rigid motions.

Definition 6.1

A rigid motion is a motion which preserves distance. That means, for A and B, AB = A'B' where A' and B' are the images of A and B respectively as shown in Figure 6.1

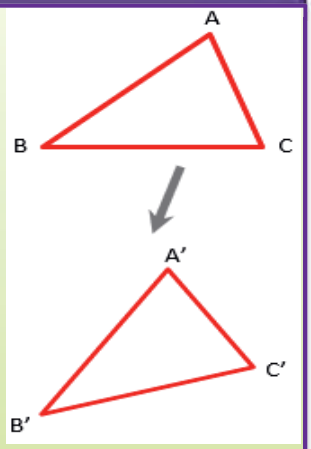


Figure 6.1

Note

Rigid motion carries any plane figures to a congruent plane figure. That means rigid motion carries triangles to congruent triangles, parallelograms to congruent parallelograms, etc.

Within the rigid motion, there are three main types of transformations that you will learn in this unit. These are translation, reflection and rotations.

6.2 Translation

Activity 6.2

Consider a triangle PQR and discuss what will happen to the shape, size and orientation of a triangle when you slide it as shown in Figure 6.2.

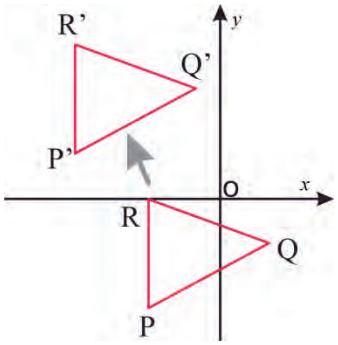


Figure 6.2

From Activity 6.2, you have observed that Δ PQR and Δ $P'Q'R'$ have the same size, shape and orientation.

Definition 6.2

A translation is a transformation that occurs when every point of a figure is moved from one location to another location along the same direction through the same distance.

If point P is translated to point P', then the vector $\overrightarrow{PP'}$ is called **the translation vector**.

You can state a translation formula in terms of coordinates as follows:

Let T = ***(a, b)*** be a translation vector, then

a. The origin is translated to ***(a, b)*** That means (0, 0) $\xrightarrow{T(a,b)}$ ***(a, b)***

b. The image of the point $P\ (x, y)$ under the translation vector T will be the point $P'\ (x + a, y + b)$.

Example 1

1. Let T = (2, 3) be the translation vector. Find the image of the points A (2, 1), B (−1, 2) and C (−5, 1) through translation vector T.
2. If translation T takes the origin to (−2, 2) then find the image of the points P (3, 5) and Q (−1, 4).
3. The image of point (1, 2) under translation T is (2, 4). What is the image of the point (-2, 3) under the same translation?

Solution

1. To translate A (2, 1) through translation vector T = (2, 3), it has to move 2 units right and then 3 units up which is the point (4, 4) as shown in the Figure 6.3.

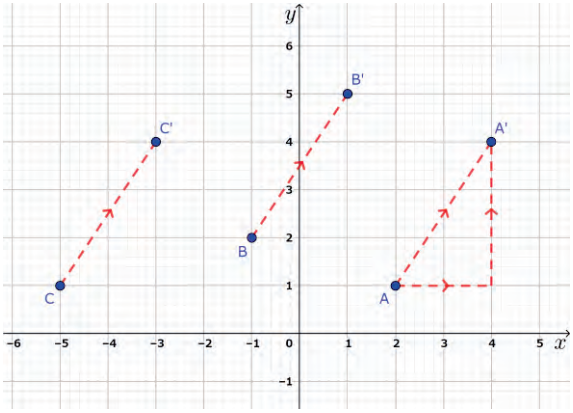


Figure 6.3

Hence, A' (4, 4) is the image of the point A (2, 1) under the translation vector T, $\overrightarrow{AA'} = (2, 3)$;

i.e., A (2,1) $\xrightarrow{T(a,\ b)}$ A' (4, 4)

Let us see the following table for some other points on the same graph and their corresponding images under the translation through translation vector T = (2, 3).

No.	Points in the Plane	Corresponding images
1	A (2, 1)	A' (4, 4)
2	B (-1, 2)	B' (1, 5)
3	C (-5, 1)	C' (-3, 4)
4	R (x, y)	R' ($x + 2, y + 3$)

From the above table, you can see that the image of any point under the translation T through translation vector (2, 3) is obtained by adding 2 to x-coordinate and 3 to y-coordinate of the given point; i.e., the image of R (x, y) is $R' = (x + 2, y + 3)$

Similarly, if the translation vector is $\overrightarrow{PQ}$ where P = (a, b) and Q = (c, d) then

a. the origin is translated to $(c - a, d - b)$

b. the point R (x, y) is translated to $(x + c - a, y + d - b)$

2. The image of point P (3, 5) is P': $(3 + (-2), 5 + 2) = (1, 7)$.

The image of point Q $(-1, 4)$ is Q': $((-1) + (-2), 4 + 2) = (-3, 6)$.

3. The translation vector is $T = (2-1,\ 4-2) = (1,\ 2)$.

The image of point $(-2,\ 3)$ is $(-2+1,\ 3+2) = (-1,\ 5)$

Example 2

If A $(-4, -4)$, B $(-2, -1)$ and C $(-1, -5)$ are the vertices of a triangle ABC, find the coordinates of the image of ΔABC under the translation T = (6, 5). Draw ΔABC and its image on the same plane.

Solution

As A $(-4, -4)$, B $(-2, -1)$ and C $(-1, -5)$ are the vertices of ΔABC,
the coordinates of the vertices of image of ΔABC can be obtained by using the formula below:

you have, $P(x, y)=(x + a, y + b) =(x + 6, y + 5)$.

Hence, A': $(-4 + 6, -4 + 5) = (2, 1)$.

B': $(-2 + 6, -1 + 5) = (4, 4)$.

C': $(-1 + 6, -5 + 5) = (5, 0)$.

Graphically, you can show these points as shown in Figure 6.4.

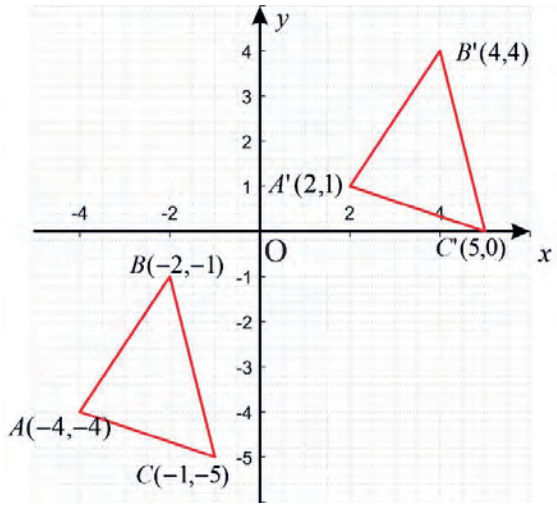


Figure 6.4

Exercise 6.1

1. Let T = (2, 3) be the translation vector. Find the image of the pint A (−1, 4) and (3, 5) through the translation vector T.
2. The image of point (−2, 3) under translation T is (3, −1). What is the image of the point (4, 2) under the same translation?
3. A (−4, 2), B (0, 1) and C (−2, −3) are the vertices of ΔABC. If A' (−1, 6) be the image under the translation of vertex A, find the images B' and C' of B and C under the same translation.
4. Draw a square S (1, 2), Q (4, 1), R (5, 4) and E (2, 5). Find the image after the translation $(x, y) \rightarrow (x - 2, y + 3)$. Then, draw and label the image.
5. If a translation T takes the origin to (3, 4), then find the image of the points P (5, 7) and Q (−3, 6).

Line Translation

Example 3

If a translation T takes the point (−2, 3) to the point (1, 2), then what are the images of the following line

a. L: $3x - y + 4 = 0$

b. L: $4y + 2x + 1 = 0$

Solution

Let T be the translation vector, A $= (-2, 3)$ and $B = (1, 2)$. Then,

T $= \overrightarrow{AB}$ =B − A = (1−(−2), 2−3) = (3, −1). Thus, T:$(3, -1)$.

The point P (x, y) is translated to the point P′ $= (x', y')$.

Here, $x' = x + 3, y' = y - 1$.

Therefore, $x = x' - 3, y = y' + 1$.

A translation map lines onto parallel lines. Let L' be the image of L under T. Then,

a. $L: 3x - y + 4 = 0$

$\Rightarrow \quad 3(x' - 3) - (y' + 1) + 4 = 0$

$\Rightarrow \quad 3x' - 9 - y' - 1 + 4 = 0$

$\Rightarrow \quad 3x' - y' - 6 = 0$

$\Rightarrow L': 3x - y - 6 = 0.$

b. $L: 4y + 2x + 1 = 0$

$\Rightarrow \quad 4(y' + 1) + 2(x' - 3) + 1 = 0$

$\Rightarrow \quad 4y' + 4 + 2x' - 6 + 1 = 0$

$\Rightarrow \quad 4y' + 2x' - 1 = 0$

$\Rightarrow L': 4y + 2x - 1 = 0.$

Note

Generally, a translation vector T: (a, b) translates point P (x, y) to point $P'(x', y')$.

The equation of $P'(x', y')$ can be got by P $\Rightarrow$ P', $x' = x - a$, $y' = y - b$.

Exercise 6.2

1. If a translation T takes (1, –4) to (–3, 2), then find the image of the line L: $5x + 4y + 9 = 0$.
2. If a translation T takes the origin to (0, 2), then find the image of each of the following lines.
   a. $y = 3x - 1$ b. $3y + x = 12$
3. If the image of the line $2x - 3y = 7$ under a translation is $2x - 3y = 0$, then what is the translation vector of the translation line which is parallel to the $x-$axis or $y-$axis?

Circle Translation

Example 4

If a translation T takes the origin to $(-2, 2)$, then find the equation of the image for the circle whose equation is $x^2 + y^2 = 4$.

Solution

Let T be the translation vector. The image of (x, y) under T is T $(x, y) = (x - 2, y + 2)$ as shown in Figure 6.5.

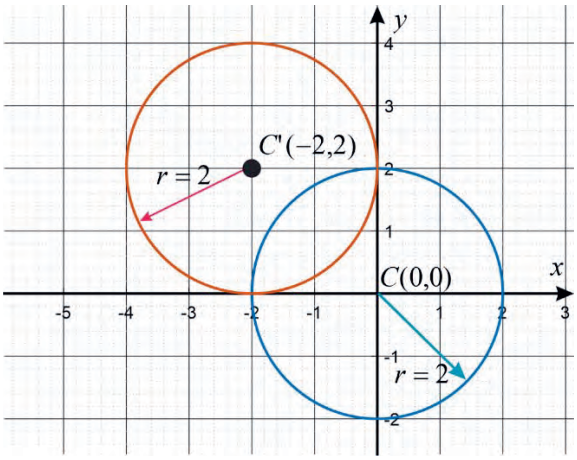


Figure 6.5

The centre of the circle $(0, 0)$ is translated to $(-2, 2)$

Therefore, the image of the circle given by $x^2 + y^2 = 4$ is $(x + 2)^2 + (y - 2)^2 = 4$.

Note

As a translation is a rigid motion, to translate a circle it suffices to translate the center of the circle by the translation vector T (a, b) and keep its radius for the image too.

Exercise 6.3

1. If a translation T takes the origin to $(-3, 3)$, then find the equation of the image for the circle whose equation is $x^2 + y^2 = 16$.
2. If a translation T takes the origin to $(-6, 8)$, then find the equation of the image for the circle whose equation is $x^2 + y^2 = 9$.
3. Find the equation of the image of the circle $(x + 4)^2 + (y + 2)^2 = 7$ when translated by the vector $\overrightarrow{AB}$ where A = (2, 3) and B = (–5, 4).
4. Translate a circle $x^2 + y^2 - 4y = 21$ by the translation vector T $(3, -5)$. Find the image circle.

6.3 Reflection

Activity 6.3

Consider a triangle ABC and discuss what will happen to the shape, size and orientation of a triangle when you fold or flip over the fixed line as shown in Figure 6.6.

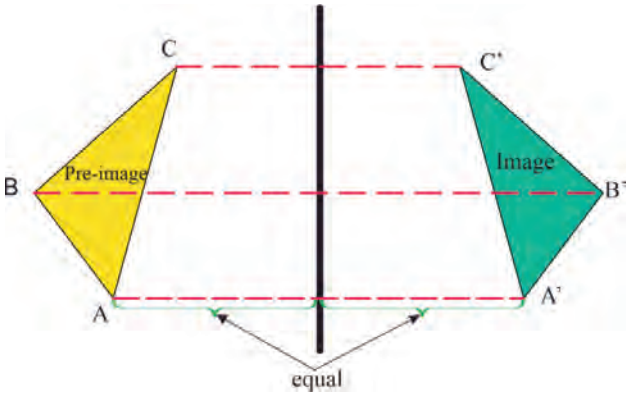


Figure 6.6

From Activity 6.3, you have observed that Δ ABC and its reflection (Δ A'B'C') have the same shape and size, but the figures face in opposite directions. Having these concepts, let us define reflection as follows.

Definition 6.3

Let L be a fixed line in the plane. A reflection **A** about a line **L** is a transformation of the plane onto itself which carries point **A** of the plane into the point **A'** of the plane such that L is the perpendicular bisector of $\overline{AA'}$.

The line L is called the **line of reflection or the axis of reflection**

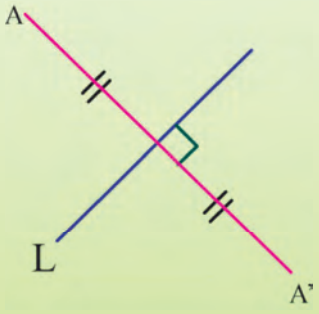


Figure 6.7

Notation: The reflection of a point **A** about the line **L**, is denoted by M(**A**);
i.e., A' = M(**A**).

Basic Properties of Reflection

1. A reflection about a line L has the property that, for two points **A** and **B** in the plane, if **A** = **B**, then M(**A**) = M(**B**). Hence, **reflection** is a function mapping the set of points in the plane into the set of points in the plane except reflection of points about the axis of reflection.
2. A reflection about a line L maps distinct points to distinct points, i.e., if **A** ≠ **B**, then M(**A**) ≠ M(**B**). Equivalently, it has the property that, for points **A** and **B** in the plane, M(A) = M(B), then **A** = **B**. Thus, reflection is **a one-to-one mapping.**
3. For every point **A'** in the plane, there exists a point **A** such that M(**A**) = **A'**. If the point **A'** is on L, then there exists **A** = **A'** such that M(**A**) = **A'**. Thus, reflection is **an onto mapping.**

Now, consider reflections with respect to the line $y = mx$ and the line $y = mx + b$ for $b \neq 0$.

A. Reflection in the line y = mx

Let L be a line passing through the origin and making an angle θ with the positive x-axis.

You will now find the image of a point P (x, y) when it is reflected about the line L as shown in Figure 6.8.

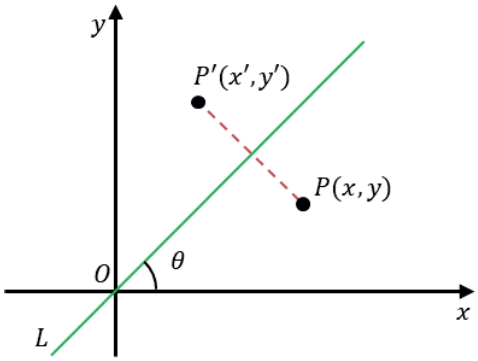


Figure 6.8

You will have the following four special cases based on the value of θ:

1. When θ = 0, you will have reflection in the x- axis. Thus, (x, y) is mapped to $(x, -y)$

Example 1

Find a reflection of a point P (2, 3) over the x-axis.

Solution

As shown in the Figure 6.9, the reflection is P' (2, −3).

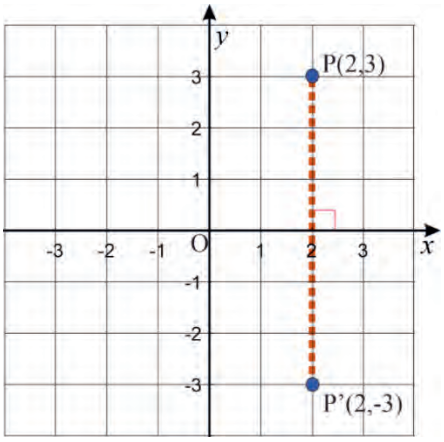


Figure 6.9

2. When $\theta = \frac{\pi}{2}$, you will have reflection in the y - axis and (x, y) is mapped to $(-x, y)$.

Example 2

Find a reflection of a point P (2, 3) over the y – axis.

Solution

As shown in the Figure 6.10, the reflection is p'(−2, 3).

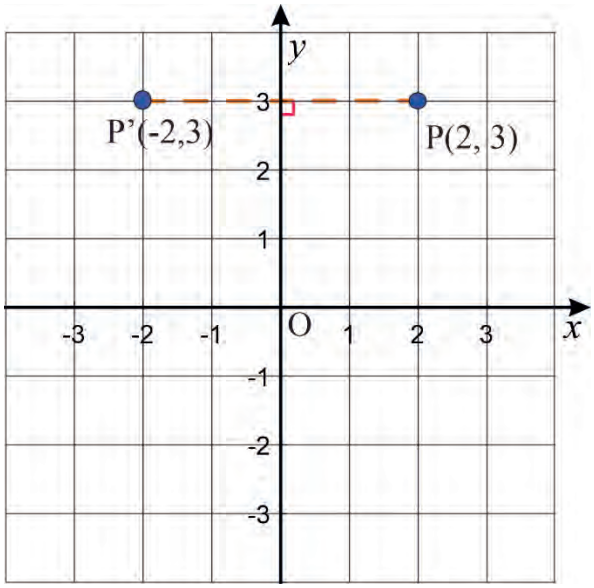


Figure 6.10

3. When $\theta = \frac{\pi}{4}$, you will have reflection about the line $y = x$ and hence (x, y) is mapped to (y, x).

Example 3

Find a reflection of a point P (−3, 2) over the line $y = x$.

Solution

As shown in the Figure 6.11, the reflection *is* P' (2, −3).

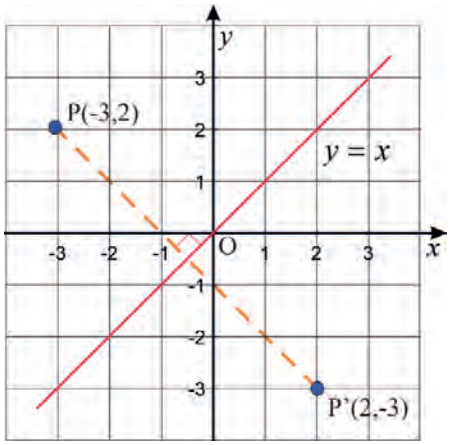


Figure 6.11

4. When $\theta = \frac{3\pi}{4}$, you will have reflection about the line $y = -x$ and (x, y) is mapped to $(-y, -x)$.

Example 4

Find a reflection of a point P (2, 3) over the line $y = -x$.

Solution

As shown in the Figure 6.12, the reflection is P'(−3, −2)

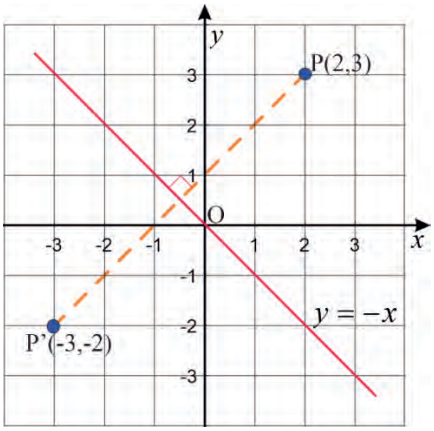


Figure 6.12

In general, reflection of points by the lines $y = 0$, $x = 0$, $y = x$ *and* $y = -x$ using coordinate rules for reflection are:

1. If (x, y) is reflected in the x-axis $(y = 0)$, then its image is the point $(x, -y)$.
2. If (x, y) is reflected in the y-axis $(x = 0)$, then its image is the point $(-x, y)$.
3. If (x, y) is reflected in the line $y = x$, then its image is the point (y, x).
4. If (x, y) is reflected in the line $y = -x$, then its image is the point $(-y, -x)$.

Exercise 6.4

1. Find a reflection of a point P (4, 5) over the following.
   a. x-axis. b. y-axis c. the line $y = x$ d. the line $y = -x$
2. The vertices of △DEF are D (1, 4), E (5, 5), and F (4, 1). Find a reflection Graph △ DEF over the following.
   a. x-axis. b. y-axis c. the line $y = x$ d. the line $y = -x$

Let L be a line passing through the origin and making an angle θ with the positive x-axis.

Then, the slope of L is given by $m = \tan\theta$ and its equation is $y = mx$ as shown in Figure 6.13.

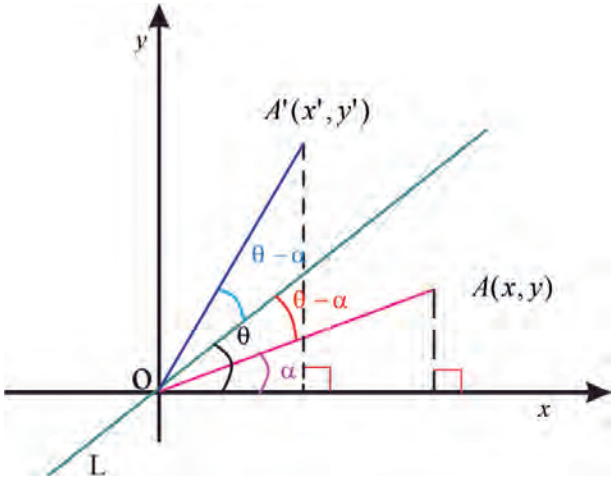


Figure 6.13

You will now find the image of a point A (x, y) when it is reflected about the line L as shown in Figure 6.13.

Suppose α is the angle measured from positive x-axis counter clockwise to the vector $\overrightarrow{OA}$ and A'(x', y') be the image of A (x, y), and let OA = r, then $OA' = r$ (why?).

The coordinates of A are:

$$x = r\cos\alpha \text{ and } y = r\sin\alpha$$

The angle measured from L to vector $\overrightarrow{OA'}$ is $\theta - \alpha$. Thus, the angle measured from positive x-axis to the vector $\overrightarrow{OA'}$ is

$$\theta + (\theta - \alpha) = 2\theta - \alpha$$

Therefore, the coordinates of A' are:

$$x' = r\cos(2\theta - \alpha) \text{ and } y' = r\sin(2\theta - \alpha)$$

Now, use the following trigonometric identities

1. Sine of the sum and the difference

$$\sin(\alpha + \beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

$$\sin(\alpha - \beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$$

2. Cosine of the sum and difference

$$\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$$

$$\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$$

Using these trigonometric identities, you obtain:

$x' = r[\cos 2\theta\cos\alpha + \sin 2\theta\sin\alpha] = (r\cos\alpha)\cos 2\theta + (r\sin\alpha)\sin 2\theta.$

$= x\cos 2\theta + y\sin 2\theta$ and

$y' = r[\sin 2\theta\cos\alpha - \cos 2\theta\sin\alpha] = (r\cos\alpha)\sin 2\theta - (r\sin\alpha)\cos 2\theta,$

$= x\sin 2\theta - y\cos 2\theta$

Therefore, the coordinates of P'(x', y') is the image of the point P (x, y) when reflected about the line $y = mx$ is:

$x' = x\cos 2\theta + y\sin 2\theta$

$y' = x\sin 2\theta - y\cos 2\theta$ where θ is the angle of inclination of the line L: $y = mx$.

Example 5

Find the image of the points A (2, 1), B (0, −3) and C (−6, 8) when reflected about the line $y = \frac{1}{\sqrt{3}}x$.

Solution

Since $\tan\theta = \frac{1}{\sqrt{3}} \Rightarrow \theta = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) \Rightarrow \theta = \frac{\pi}{6}$

Thus, if A' (x', y') is the image of A (2,1), then

$$x' = x\cos 2\theta + y\sin 2\theta = 2\cos 2\left(\frac{\pi}{6}\right) + 1\sin 2\left(\frac{\pi}{6}\right) = 2\cos\frac{\pi}{3} + 1\sin\frac{\pi}{3}$$

$$= 2 \times \frac{1}{2} + 1 \times \frac{\sqrt{3}}{2} = \frac{2+\sqrt{3}}{2}$$

$$y' = x\sin 2\theta - y\cos 2\theta = 2\sin 2\left(\frac{\pi}{6}\right) - 1\cos 2\left(\frac{\pi}{6}\right) = 2\sin\frac{\pi}{3} - 1\cos\frac{\pi}{3} = 2 \times \frac{\sqrt{3}}{2} - 1 \times \frac{1}{2}$$

$$= \frac{2\sqrt{3}-1}{2}$$

Therefore, the images of A (2, 1) is A'$\left(\frac{2+\sqrt{3}}{2}, \frac{2\sqrt{3}-1}{2}\right)$

Again, if B' (x', y') is the image of B (0, -3), then

$$x' = x\cos 2\theta + y\sin 2\theta = 0 \times \cos\frac{\pi}{3} + (-3)\sin\frac{\pi}{3} = 0 \times \frac{1}{2} - 3 \times \frac{\sqrt{3}}{2} = \frac{-3\sqrt{3}}{2}$$

$$y' = x\sin 2\theta - y\cos 2\theta = 0 \times \sin\frac{\pi}{3} - (-3)\cos\frac{\pi}{3} = 0 \times \frac{\sqrt{3}}{2} + 3 \times \frac{1}{2} = \frac{3}{2}$$

Thus, the image of B (0, –3) is B' $\left(\frac{-3\sqrt{3}}{2}, \frac{3}{2}\right)$.

Similarly, if C' (x', y') is the image of C (−6, 8), then

$$x' = x\cos 2\theta + y\sin 2\theta = -6 \times \cos\frac{\pi}{3} + 8\sin\frac{\pi}{3} = -6 \times \frac{1}{2} + 8 \times \frac{\sqrt{3}}{2} = -3 + 4\sqrt{3}$$

$$y' = x\sin 2\theta - y\cos 2\theta = -6 \times \sin\frac{\pi}{3} - 8\cos\frac{\pi}{3} = -6 \times \frac{\sqrt{3}}{2} - 8 \times \frac{1}{2} = -3\sqrt{3} - 4$$

Thus, the image of C (−6, 8) is C' $(-3 + 4\sqrt{3}, -3\sqrt{3} - 4)$.

Exercise 6.5

1. Find the image of the point (0, 1) when reflected about the line $y = \sqrt{3}x$.
2. Find the image of the point (2, 3) when reflected about the line $y = -\sqrt{3}x$.
3. Find the image of the point (3, 0) when reflected about the line $y = 2x$.

B. Reflection in the line $y = mx + b$

1. Reflection of a point in the line

Let L: y = $mx + b$ be the line of reflection, where $m \in \mathbb{R} \setminus \{0\}$.

Let A (x, y) be a point on the plane, not on L.

Let A'(x', y') be the image of A (x, y) when reflected about the line L.

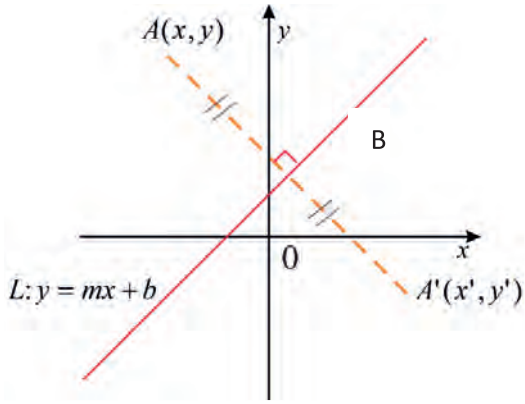


Figure 6.14

Let s be the line passing through the points A (x, y) and A' (x', y'), $\overline{AA'}$ is perpendicular to L, since L is perpendicular bisector of $\overline{AA'}$. Since the slope of L is m, the slope of $\overline{AA'}$ is $-\frac{1}{m}$.

Then, one can determine the equation of the line $\overline{AA'}$. If B is the point of intersection of L and $\overline{AA'}$, taking B as the mid-point of $\overline{AA'}$, you can find the coordinates of A'.

Thus, to find the image of a point A (x, y) when reflected about a line L, you follow the following four steps.

1. Find the slope of the line L, say m.
2. 2. Find the equation of the line $\overline{AA'}$, which passes through the point A (x, y) and has slope $-\frac{1}{m}$.
3. Find the point of intersection B of L and $\overline{AA'}$ which serves as the midpoint of $\overline{AA'}$
4. Using B as the mid-point of $\overline{AA'}$, find the coordinates of A' .

Example 6

Find the image of the point (4, 0) when reflected about the line L: $y = 2x + 1$.

Solution

Let A (4, 0) be a point on the plane, not on L.

Let A'(x', y') be the image of A (4, 0) when reflected about the line L.

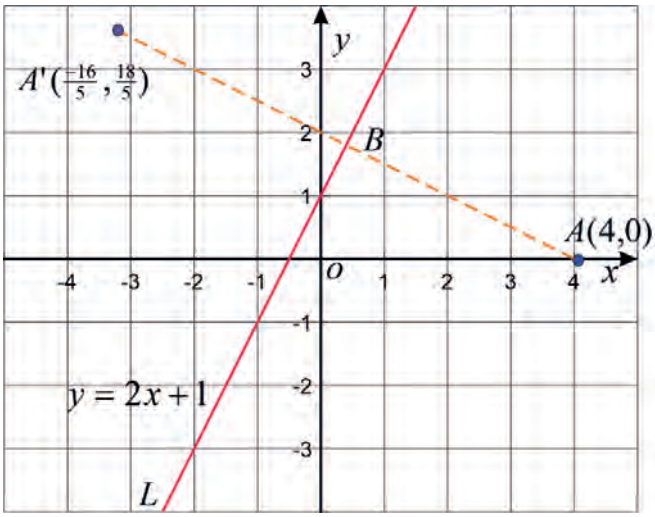


Figure 6.15

Let s be the line passing through the points A (*4, 0*) and A' (*x', y'*). $\overline{AA'}$ is perpendicular to L since L is perpendicular bisector of $\overline{AA'}$ and the slope of L is 2. Thus, slope of $\overline{AA'}$ is $-\frac{1}{2}$.

The equation of the line $\overline{AA'}$, which passes through the point A (*4, 0*) and has slope $-\frac{1}{2}$ is $\frac{y-0}{x-4} = -\frac{1}{2} \Rightarrow y = -\frac{1}{2}x + 2$

Now, let us find the point of intersection B of L and $\overline{AA'}$ which serves as the midpoint of $\overline{AA'}$

Then, B is obtained by solving the system of linear equations $\begin{cases} y = -\frac{1}{2}x + 2 \\ y = 2x + 1 \end{cases}$.

Thus, $(\frac{2}{5}, \frac{9}{5})$ is its solution.

Using B as the midpoint, the image of A (4, 0) can be obtained by using the two points B $(\frac{2}{5}, \frac{9}{5})$ and A (4, 0).

Then, the image of the point A (4, 0) is:

$\frac{x'+4}{2} = \frac{2}{5} \Rightarrow x' = \frac{-16}{5}$ and $\frac{y'+0}{2} = \frac{9}{5} \Rightarrow y' = \frac{18}{5}$

Thus, the image of A (4, 0) is A'$(\frac{-16}{5}, \frac{18}{5})$.

Example 7

Find the image of (–2, 4) when reflected about the lines

a. $y = -2$ **b.** $x = 1$

Solution

a. Line $y = -2$ is a straight line parallel to x-axis and at a distance of 2 units in negative direction of y-axis. Point $(-2,4)$ is 6 units away from given line $y = -2$. Therefore, distance of image from the line is also 6 units. Thus, distance of image from x-axis = 6 + 2= 8 units in negative direction of y-axis. Thus, the image of the point (–2, 4) when reflected about the line $y = -2$ is (–2, –8).

b. Line $x = 1$ is a straight line parallel to y-axis and at a distance of 1 unit in positive direction of x-axis. Point $(-2,4)$ is 3 unit away from given line $x = 1$. Therefore, distance of image from the line is also 3 units. Thus, distance of image from y-axis = $3 + 1 = 4$ units in positive direction of x-axis. Thus, the image of the point (–2, 4) when reflected about the line $x = 1$ is (4, 4).

Exercise 6.6

1. Find the images of the points (4, 2) when reflected about the line $y = 2x - 1$.
2. Find the images of the points $(-3, -2)$ when reflected about the line $y = -x - 2$.

3. Find the image of (2, 4) when reflected about the lines
   a. $y = 1$ b. $y = -2$
4. If the image of the point (–5, 4) under reflection is (3, 0), find the line of reflection.
5. If P is the reflection of point $(4, -4)$ in the line $2y = x + 1$, find the coordinates of point P.

1. Reflection of a line in the line

Note

To find the images of a line s when reflected about the line L, take the following way.

1. If a line $\overline{AA'}$ is perpendicular to the axis of reflection L, then $\overline{AA'}$ is its own image.
2. If $\overline{AA'}$ is a line parallel to the line of reflection **L**, to find the image of $\overline{AA'}$ when reflected about L, we follow the following steps.
   a. Choose any point B on $\overline{AA'}$.
   b. Find the image of B, M(B) = B'
   c. Find the equation of $\overline{AA'}$ which is the line passing through B' with slope equal to the slope of L.

Example 8

Find the image of the line $s: -3x + y = 2$ after reflection in the line L: $y = x + 4$.

Solution

Let s: $y = 3x + 2$ and s' be its image after reflecting about the line L: $y = x+4$.

Let A (a, b) be any point on s, say A (0, 2), so that its image $A'(a', b')$ lies on s' as shown in figure 6.16.

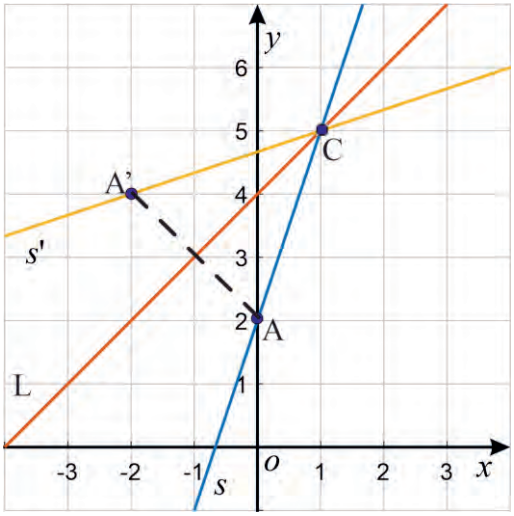


Figure 6.16

By using the following reflecting procedure:

a. Choose any point A on s.

b. Find the image of A, $\mathrm{M}(A) = A'$

c. Find the intersection of L and s as C.

d. Find the line passing through both A' and C. This line is the image s'

$\mathrm{M}(0, 2) = (a', b') \Rightarrow \frac{b' - 2}{a' - 0} = -1$

$\Rightarrow a' = -b' + 2.$

The midpoint of (0, 2) and (a', b') which is $\left(\frac{a'}{2}, \frac{b' + 2}{2}\right)$ lies on the reflecting axis L.

Then, from

L: $y = x + 4$, it follows

$\Rightarrow \frac{b' + 2}{2} = \frac{a'}{2} + 4.$

$\Rightarrow$ So, $a' = b' - 6.$

Since $a' = -b' + 2$ and $a' = b' - 6$, then

$-b' + 2 = b' - 6$

It implies $b' = 4$ and $a' = -b' + 2 = -4 + 2 = -2.$

Therefore, A' (a', b') = A' (-2, 4).

It follows that $(-2, 4)$ lies on s'.

Now, let us find a point that belongs to both L and s, say C. Then C is obtained by solving the system of equations $\begin{cases} y = x + 4 \\ y = 3x + 2 \end{cases}$.

Thus, (1, 5) is its solution.

Finally, the equation of the image line can be obtained by using the two points of

$A'(-2, 4)$ and C (1, 5).

Then, the equation of the imaged line is:

s': $\frac{y-4}{x-(-2)} = \frac{5-4}{1-(-2)}$.

or s': $\frac{y-4}{x+2} = \frac{1}{3}$.

or s': $x - 3y + 14 = 0$.

Thus, the image of line $-3x + y = 2$ after reflecting about the line $y = x + 4$ is s':

$x - 3y + 14 = 0$.

Exercise 6.7

1. Find the image of the line s: $y = 2x + 3$ after it has been reflected about the line L: $y = x - 2$.
2. What's the equation of the image of a line s: $y = 3x + 4$ when it is reflected along the line $y = 2x - 3$?

2. Reflection of a circle in the line

Note

1. If the centre of a circle C is on the line of reflection L, then the image of C is itself.
2. If the centre O of a circle C has image O' when reflected about a line L, then the image circle has centre O' and radius the same as C.

Example 9

Find image of the circle $x^2 + y^2 - 4x - 2y + 4 = 0$ after reflection in the line L: $y = x - 1$.

Solution

By applying completing square method on $x^2 + y^2 - 4x - 2y + 4 = 0$,
you find $(x - 2)^2 + (y - 1)^2 = 1$.
Clearly, the centre of the circle is (2, 1) and it lies on the line of reflection $y = x - 1$
Therefore, the circle is its own image.

Example 10

Given the equation of the circle $x^2 + (y - 3)^2 = 1$, find the equation of the circle after a reflection about the line $y = x$.

Solution

The centre of the circle is (0, 3). The reflection of (0, 3) about the line $y = x$ is (3, 0), which is the centre of the image circle. Therefore, the equation of the image circle is $(x - 3)^2 + y^2 = 1$.

Exercise 6.8

1. Find the image of the circle $x^2 + y^2 = 4$ after reflection in the line $L: y = x$.
2. Find the image of the circle $(x - 4)^2 + (y - 1)^2 = 9$ after reflection in the line $L: y = x - 1$.
3. Given an equation of a circle $(x - 1)^2 + (y - 2)^2 = 36$, find the equation of the image circle after a reflection about the line $y = x + 2$.

4. The image of the circle $x^2 + y^2 - x + 2y = 0$ when it is reflected about the line L is $x^2 + y^2 - 2x + y = 0$. Find the equation of L.

6.4 Rotation

Activity 6.4

1. Discuss rotation of points through 90 ° and 180° about the origin.
2. Consider a triangle ABC, discuss what happened the shape, size and orientation of a triangle when you turn counter clockwise it around a fixed point as shown in Figure 6.17 and try to define what does rotation of figures mean.

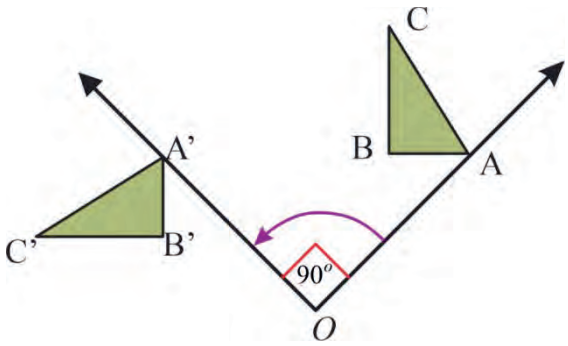


Figure 6.17

As you have observed form Figure 6.17, when Δ ABC is rotated about a point O or the centre of rotation to be the origin, Δ ABC and Δ $A'B'C'$ have the same shape and size.

Definition 6.4

A rotation R about a point O through an angle θ is a transformation of the plane onto itself which carries every point A of the plane into the point A' of the plane such that $OA = OA'$ and m $(\angle AOA') = \theta$. O is called the centre of rotation and θ is called the angle of rotation.

A. Rotation when the center of rotation is about the origin

Note

1. The rotation is in the counter clockwise direction, if $\theta > 0$ and in the clockwise direction if $\theta < 0$.
2. Rotation is a rigid motion but the figures may be turned in different directions.

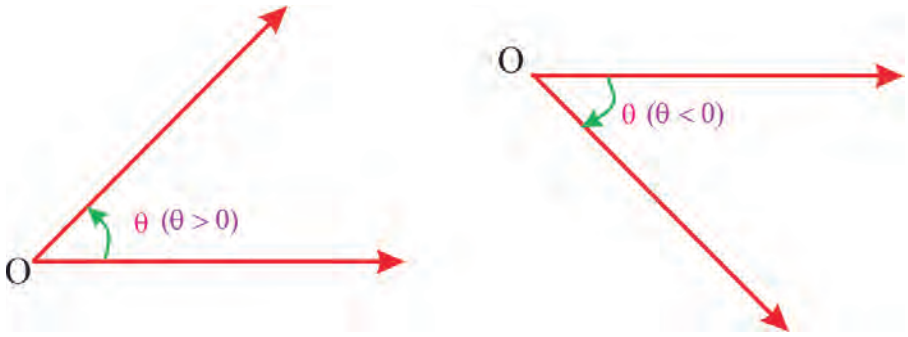


Figure 6.18

Example 1

Find the image of point A (2, 0) when it is rotated through 30° about the origin.

Solution

Let the image of the point A (2, 0) be A'(a, b) as shown in the Figure 6.19.

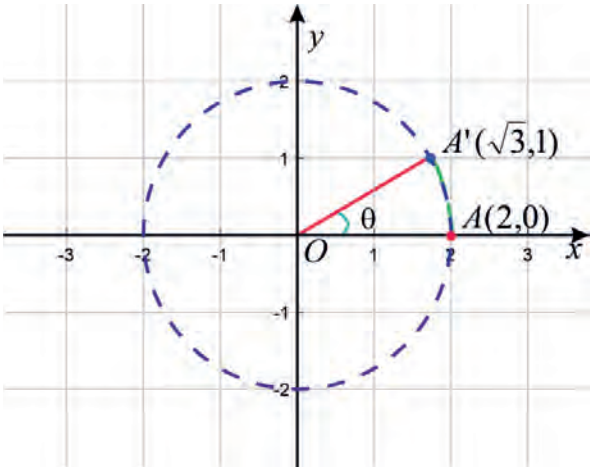


Figure 6.19

From trigonometry, you have A'*(a, b)* = (r cos θ, r sin θ) where r = 2 and θ = 30°

Therefore, A'*(a, b)* = (2cos30°, 2sin30°).

Hence, A' *(a, b)* = $(2\,(\frac{\sqrt{3}}{2}), 2\,(\frac{1}{2})) = (\sqrt{3}, 1)$.

Thus, the image of A (2, 0) is A'$(\sqrt{3},\ 1)$.

Notation

If R is rotation through an angle θ, then the image of A *(x, y)* is denoted by $R_\theta\ (x, y)$. For example, in the above example, $R_{30°}\ (2, 0) = (\sqrt{3}, 1)$

Exercise 6.9

Find the image of point P (3, 0) when it is rotated through the following angle about the origin:

a. $\theta = 45°$ b. $\theta = 60°$ c. $\theta = 120°$ d. $\theta = 135°$

e. $\theta = 150°$ f. $\theta = -45°$ g. $\theta = -60°$ h. $\theta = -120°$

i. $\theta = -135°$ j. $\theta = -150°$

Rotation when the center of rotation is about the origin (1)

Theorem 6.1

Let R be a rotation through angle θ about the origin. If $R_\theta\ (x, y) = (x', y')$, then

$x' = x\cos\theta - y\sin\theta$ $\qquad$ $y' = x\sin\theta + y\cos\theta$

Proof

From trigonometry, you have,

$(x, y) = (r\cos\alpha, r\sin\alpha)$ and $(x', y') = (r\cos(\alpha + \theta), r\sin(\alpha + \theta))$,

where $r\cos(\alpha + \theta) = r\cos\alpha\cos\theta - r\sin\alpha\sin\theta$… (cosine of the sum formula)

$= x\cos\theta - y\sin\theta$……because $x = r\cos\alpha$ and $y = r\sin\alpha$

and $r\sin(\alpha + \theta) = r\sin\alpha\cos\theta + r\cos\alpha\sin\theta$…… (sine of the sum formula)

$= y\cos\theta + x\sin\theta$……because $x = r\cos\alpha$ and $y = r\sin\alpha$

$\therefore R_\theta\ (x, y) = (x\cos\theta - y\sin\theta, y\cos\theta + x\sin\theta)$

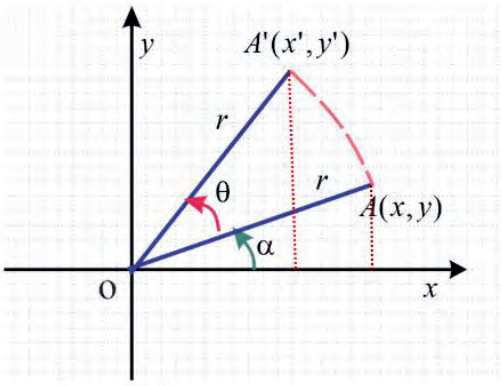


Figure 6.20

Example 2

Find the images of the following points under a rotation by the indicated angle about the origin.

a. $(1, 3), 30°$ b. $(3, -4), 810°$

Solution

From theorem 6.1, you have

a. $x' = x\cos\theta - y\sin\theta;\ x = 1, y = 3;\ \theta = 30°$

$= 1\cos 30° - 3 \times \sin 30°$

$= 1 \times \frac{\sqrt{3}}{2} - 3 \times \frac{1}{2} = \frac{\sqrt{3} - 3}{2}$

$y' = x\sin\theta + y\cos\theta$

$= 1\sin 30° + 3\times\cos 30°$

$= \frac{1}{2} + 3\times\frac{\sqrt{3}}{2} = \frac{1+3\sqrt{3}}{2}$

Thus, $R_{30°}(1, 3) = (\frac{\sqrt{3} - 3}{2}, \frac{1+3\sqrt{3}}{2})$.

b. A rotation of 810 ° is the same as two consecutive rotations by 360 ° followed by a rotation by 90° (because 810 ° $= 2\times360$ ° $+ 90°$).

Hence, a rotation by $810°$ is the same as a rotation by $90°$. Thus, we can simply use theorem 6.1 for $\theta = 90°$

$$x' = x\cos\theta - y\sin\theta;\ x = 3, y = -4;\ \theta = 90°$$
$$= 3 \times \cos(90°) - (-4) \times \sin(90°)$$
$$= 3\times 0 + 4 \times 1 = 4$$
$$y' = x\sin\theta + y\cos\theta$$
$$= 3 \times \sin(90°) + (-4) \times \cos(90°)$$
$$= 3 \times 1 - 4 \times 0 = 3$$

Thus, $R_{810°}(3, -4) = R_{90°}(3, 4) = (4, 3)$.

Exercise 6.10

1. Find the images of the following points under a rotation by the indicated angle about the origin:

   a. $(2, 4), 30°$ b. $(4, -5), \frac{\pi}{2}$ c. $(5, -6), \frac{\pi}{2}$ d. $(2, 1), -45°$

2. Determine the images of the following points after a rotation of the indicated angle about the origin:

   a. $(-2, 4), -60°$ b. $(5, -6), -180°$ c. $(2, -2), -\frac{3\pi}{2}$ d. $(3, 4), -540°$

Rotation when the center of rotation is about the origin (2)

Note

1. $\theta = \frac{\pi}{2}$

Let R be a counter-clockwise rotation through an angle θ about the origin. Then $\theta = \frac{\pi}{2} \Rightarrow (x', y') = (-y, x)$. This means when $\theta = \frac{\pi}{2}$,

$$x' = x\cos\theta - y\sin\theta = -y$$
$$y' = x\sin\theta + y\cos\theta = x$$

Example 3

The image of a triangle ABC when it is rotated through $\theta = \frac{\pi}{2}$ or 90° about the origin is shown in Figure 6.21.

To see that this is a rotation of $\theta = \frac{\pi}{2}$, imagine point B attached to the red arrow. The red arrow is then moved $\theta = \frac{\pi}{2}$ (Notice that the $\frac{\pi}{2}$ or 90 ° is formed by the two red arrows).

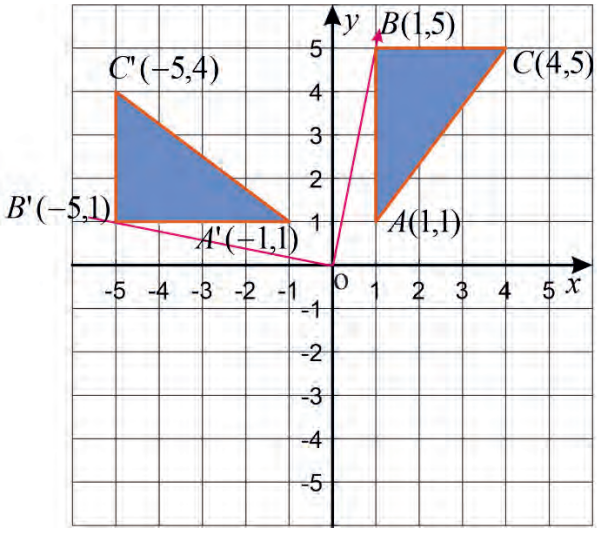


Figure 6.21

Look at the new position of B, labelled $B' = (-5, 1)$. This same approach can be used for all three vertices.

2. $\theta = \pi \Rightarrow R(x, y) = (-x, -y)$.

Example 4

The image of a triangle ABC when it is rotated through $\theta = \pi$ or 180 ° about the origin is shown in Figure 6.22.

Starting with ΔABC, draw the rotation of 180° centred at the origin. As you did in the previous example, imagine point B attached to the red arrow from the center (0, 0).

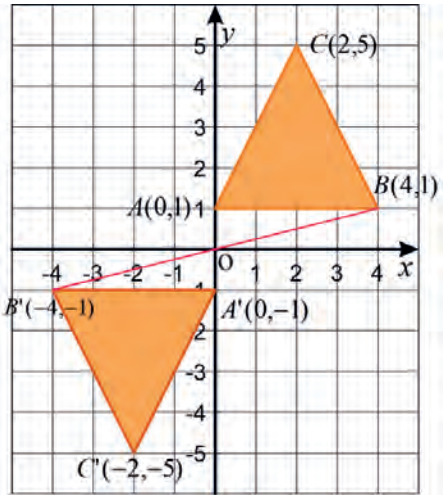


Figure 6.22

The arrow is then moved $\theta = \pi$ or 180º (which forms a straight line). Notice the new position of B, labeled B' = (−4, −1).

3. $\theta = \frac{3\pi}{2} \Rightarrow R(x, y) = (y, -x)$.

Example 5

The image of a quadrilateral ABCD when it is rotated through $\theta = \frac{3\pi}{2}$ or 270 ° about the origin is shown in Figure 6.23.

Starting with quadrilateral *ABCD*, draw the rotation of 270º centred at the origin. As you did in the previous examples, imagine point *A* attached to the red arrow from the centre (0, 0). The arrow is then moved 270º. Notice the new position of *A*, labelled *A′* = (*1, 0*).

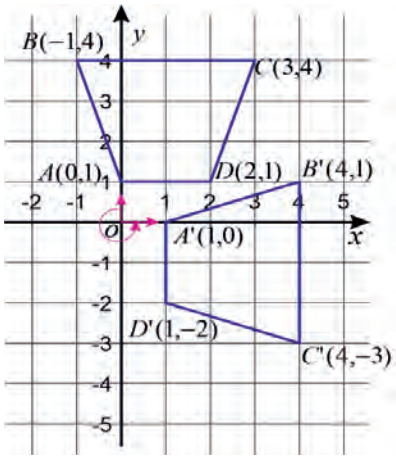


Figure 6.23

4. $\theta = 2n\pi$ for $n \in Z \Rightarrow$ R is the identity transformation.

5. Every circle with centre at the centre of rotation is fixed.

In general, you can summarize rotation of point centred at the origin as follows

Rotation of 90°	(x, y) becomes $(-y, x)$
Rotation of 180°	(x, y) becomes $(-x, -y)$
Rotation of 270°	(x, y) becomes $(y, -x)$

Exercise 6.11

1. A triangle CDE has vertices C = (2, 1), D= (5, 3) and E = (3, 4). Find the image of the triangle when rotated about the origin through an angle $\theta = \frac{\pi}{2}$.

2. Determine the image of the straight-line AB under an anticlockwise rotation of 90° about O in the Figure 6.24.
3. If A (2, −2), B (3, −1) and C (−6, 1) are the vertices of a triangle ABC as shown in Figure 6.25, determine the image of the triangle ABC under a clockwise rotation of 180° about the origin.

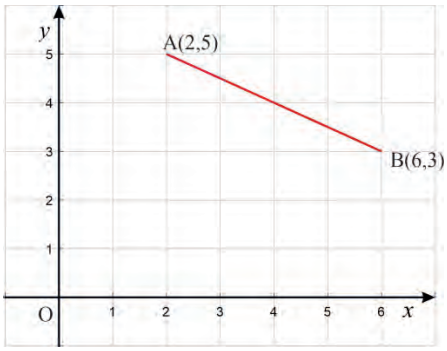


Figure 6.24

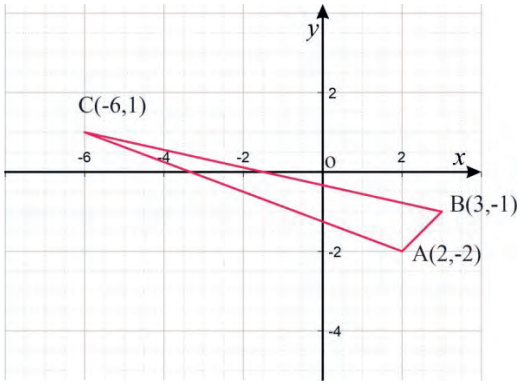


Figure 6.25

4. Given an equation of a circle $(x+1)^2 + (y+3)^2 = 4$, find the image circle under an anticlockwise rotation of 180° about the origin.

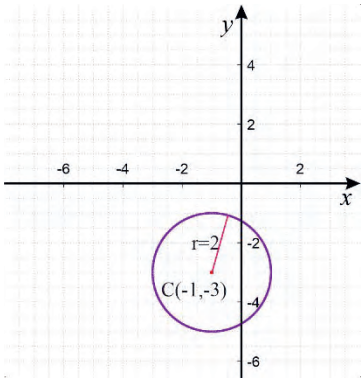


Figure 6.26

B. Rotation when the center of rotation is the point (a, b)

In previous section, you saw rotation about the origin. In this section you will discuss rotation about an arbitrary point (a, b).

Consider the Figure 6.27 as shown below.

It indicates how the original triangle ABC that has vertices A (4,7), B (2,7) and C (4, 2) is rotated through 180° about the point (4, 1) to give triangle $A'B'C'$.

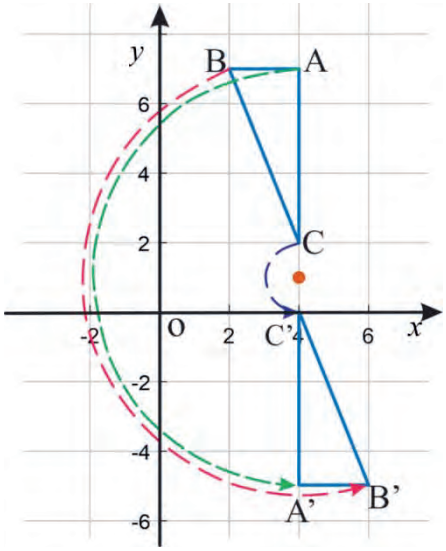


Figure 6.27

As you can observed from Figure 6.27, you can generalize the following formula:

Corollary 6.1

Rotation about the point (a, b) can be divided the following three transformation.

1. Translation by $T = (-a, \ -b)$

$$(x_1', \ y_1') = (x - a, \quad y - b)$$

→ The center of rotation is moved to the origin

2. Rotation through an angle θ about the origin

$$x_2' = x_1' \cos\theta - y_1' \sin\theta$$
$$y_2' = x_1' \sin\theta + y_1' \cos\theta$$

3. Translation by $T = (a, \ b)$

$$x_3' = a + x_2' = a + x_1' \cos\theta - y_1' \sin\theta$$
$$= a + (x - a)\cos\theta - (y - b)\sin\theta$$
$$y_3' = b + y_2' = b + x_1' \sin\theta + y_1' \cos\theta$$
$$= b + (x - a)\sin\theta + (y - b)\cos\theta$$

Example 6

Find the image of the point A (3, 4) when it is rotated through $\theta = \pi$ about (6, 5).

Solution

From the corollary 6.1, you have,

$$x' = a + (x - a)\cos\theta - (y - b)\sin\theta$$

$$y' = b + (x - a)\sin\theta + (y - b)\cos\theta$$

where $(x, y) = (3, 4)$, $(a, b) = (6, 5)$, $\theta = \pi$

$$\begin{aligned} x' &= 6 + (3 - 6)\cos\pi - (4 - 5)\sin\pi \\ &= 6 + (-3)(-1) - (-1)(0) \\ &= 9 \end{aligned}$$

$$\begin{aligned} y' &= 5 + (3 - 6)\sin\pi + (4 - 5)\cos\pi \\ &= 5 + (-3)(0) + (-1)(-1) \\ &= 6 \end{aligned}$$

Thus, the image of the point A (3, 4) after rotating 180° about (6, 5) is (9, 6).

Note

One can also obtain the image of a line under a given rotation as follows:

1. Choose two points on the line.
2. Find the images of the two points under the given rotation.

Thus, the image line will be the line passing through the two image points.

Example 7

Determine the image of the line $L: y = 3x + 1$ after a rotation of $\theta = \frac{\pi}{2}$ about the point (2, 2).

Solution

From the above note, you choose any two arbitrary points, say $(x_1, y_1) = (1, 4)$ and $(x_2, y_2) = (-2, -5)$ as shown in figure 6.28

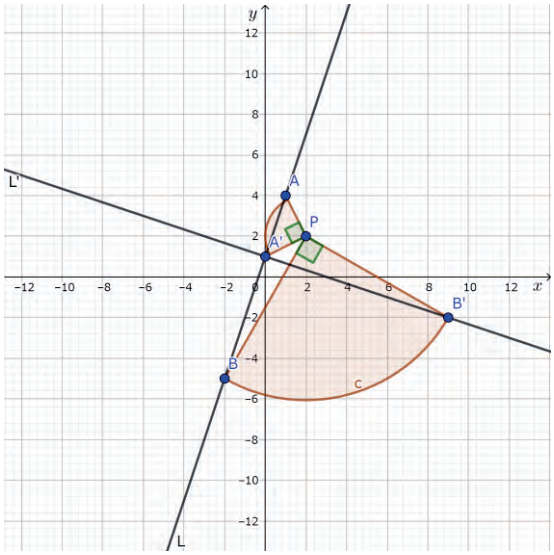


Figure 6.28

But from corollary 6.1, you have

$$x' = a + (x-a)\cos\theta - (y-b)\sin\theta$$

$$y' = b + (x-a)\sin\theta + (y-b)\cos\theta$$

where $(x, y) = (x_1, y_1) = (1, 4)$ or $(x_2, y_2) = (-2, -5)$, $(a, b) = (2, 2)$ and $\theta = \frac{\pi}{2}$

Thus, $P(x_1, y_1) = (a + (x_1 - a)\cos\theta - (y_1 - b)\sin\theta, b + (x_1 - a)\sin\theta + (y_1 - b)\cos\theta)$

$$P(1, 4) = (2 + (1-2)\cos\tfrac{\pi}{2} - (4-2)\sin\tfrac{\pi}{2},\ 2 + (1-2)\sin\tfrac{\pi}{2} + (4-2)\cos\tfrac{\pi}{2})$$

$$= (2 - 0 - 2(1), 2 - (1) + 2(0))$$

$$= (0, 1)$$

$$P(x_2, y_2) = (a + (x_2 - a)\cos\theta - (y_2 - b)\sin\theta, b + (x_2 - a)\sin\theta + (y_2 - b)\cos\theta)$$

$$P(-2, -5) = (2 + (-2-2)\cos\tfrac{\pi}{2} - (-5-2)\sin\tfrac{\pi}{2}, 2 + (-2-2)\sin\tfrac{\pi}{2} + (-5-2)\cos\tfrac{\pi}{2})$$

$$= (2 + ((-2-2)0) - ((-5-2)1), 2 + ((-2-2)1) + ((-5-2)0)$$

$$= (9, -2)$$

Now, let L' be the imaged line containing the points (0,1) and (9, −2).

Thus, slope of L' $= \frac{-2-1}{9-0} = -\frac{3}{9} = -\frac{1}{3}$

L': $\frac{y-1}{x-0} = \frac{-1}{3}$

$\Rightarrow$ L': $y = -\frac{1}{3}x + 1$.

Note

As in the case of translation and reflection, to find the image of a circle under a given rotation we follow the following steps:

1. Find the centre and radius of the given circle.
2. Find the image of the centre of the circle under the given rotation.
3. Equation of the image circle will be an equation of the circle centred at the image of the centre of the given circle with radius the same as the radius of the given circle.

Example 8

Find the image of the circle $(x + 4)^2 + (y - 6)^2 = 9$ when it is rotated through $\theta = \pi$ about (−7, −3).

Solution

From the above note, you compute only the image of the centre of the circle.

The centre is (−4, 6) and its radius is 3

But from corollary 6.1, you have

$$x' = a + (x - a)\cos\theta - (y - b)\sin\theta$$

$$y' = b + (x - a)\sin\theta + (y - b)\cos\theta$$

where $(x, y) = (-4, 6)$, $(a, b) = (-7, -3)$, $\theta = \pi$

$$x' = -7 + (-4 - (-7))\cos\pi - (6 - (-3))\sin\pi$$

$$= -7 + 3(-1) - 9(0)$$

$$= -10$$

$$y' = -3 + (-4 - (-7))\sin\pi + (6 - (-3))\cos\pi$$
$$= -3 + 3(0) + 9(-1)$$
$$= -12$$

Thus, the equation of the image of the circle is $(x + 10)^2 + (y + 12)^2 = 3^2$.

Exercise 6.12

1. Find the image of (0, 1) after it has been rotated 45^o about (5, 3).
2. Find an equation of the line into which the line with the given equation is transformed under a rotation through the indicated angle about (0, 1):
   a. $2x - 3y = 5; \theta = \frac{8\pi}{3}$ b. $4x + 5y = 4; \theta = \frac{\pi}{4}$
3. Find an equation of the circle into which the circle with the given equation is transformed under a rotation through the indicated angle, about the (1, 0):
   a. $x^2 + y^2 = 3, \theta = \frac{3\pi}{2}$ b. $(x + 1)^2 + (y - 2)^2 = 6^2, \theta = \frac{\pi}{3}$

6.5 Applications

In this unit, you discussed definition and properties of transformation (translation, reflection and rotation) with their application in determining the images of a point, a line, a circle and plane figures. Now, you will consider further applications involving transformations.

Example 1

Figure 6.29 shows a reflection in line L.

a. Which two points are invariant?
b. If the length of FG is 14cm, what is the length of GH?
c. If the length of FH is 16 cm, what is the length of EH?

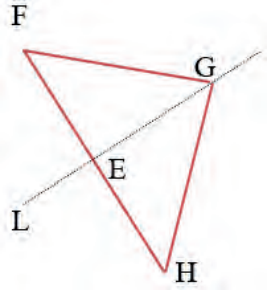


Figure 6.29

Solution

a. The invariant points are E and G.

b. Let $\overrightarrow{FG} = 14$cm then $\overrightarrow{GH} = 14$cm. Since, $\overrightarrow{GH} = \overrightarrow{FG} = 14$cm.

c. If the length of FH = 16cm then the length of

$EH = \frac{1}{2}$(length of FH) $= \frac{1}{2} \times 16\text{cm} = 8\text{cm}$.

Example 2

Given the translation $(x + 6, y - 2)$, find the image of the point (2, 6).

Solution

The given translation is, $(x, y) \rightarrow (x + 6, y - 2)$.

It is given that $(x, y) = (2, 6)$. Hence, the translated point is,

$(x + 6, y - 2) = (2 + 6, 6 - 2) = (8, 4)$.

Exercise 6.13

1. Figure 6.30 below shows a reflection in a horizontal axis. Some length in this Figure are AB = 4cm, HI = 3cm and FH = 3cm.

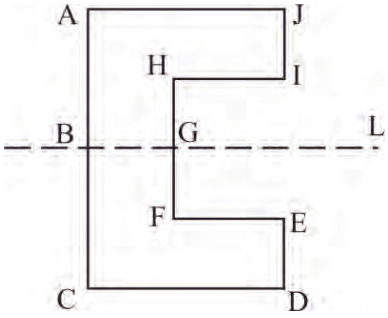


Figure 6.30

a. What point is the image of I?
b. What line segment is the image of CD?
c. Write down any invariant points.
d. What is the length of EF?
e. What is the length of GH?
f. What is the length of AC?

2. A shape is formed with vertices (1, 6), (−3, −3), (−4, 5), and (−6, −4). Plot the image of this shape with respect to the translation $(x, y) \rightarrow (x + 6, y + 1)$.

Problem Solving

1. Let D (−5, 3), E (−3, 0) and F(−5, −2) be the three vertices of a triangle. If this triangle is reflected about the line $x = -2$, what will be the new vertices D', E' and F'?
2. ΔABC with A (1,1), B (7,−2) and C (1,−2) is rotated 90° about point R (−2,−4). What are the coordinates of the image of point B?

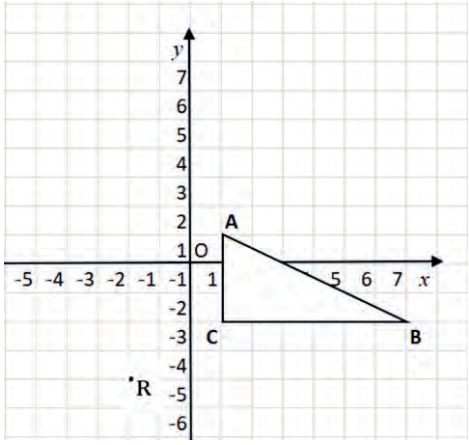


Figure 6.31

3. Triangle ABC has vertices at A (2,3), B (5,7), and C (5,3) in the coordinate plane. The triangle will be reflected over the x-axis and then rotated 180° about the origin to form $\Delta A'B'C'$. What are the vertices of $\Delta A'B'C'$?

Summary

1. Transformation of the plane

a. Transformation can be classified as rigid motion and non-rigid motion.

b. Rigid motion is a motion that preserves distance. Otherwise, it is non-rigid.

c. Identity transformation is a transformation in which image of every point is itself.

2. Translation

A translation is a transformation that occurs when every point of a figure is moved from one location to another location along the same direction through the same distance:

1. If point A is translated to point A', then the vector $\overrightarrow{AA'}$ is called the translation vector.

2. If T = *(a, b)* is a translation vector, then

i. The origin is translated to (a, b). That means $(0, 0) \longrightarrow (a, b)$.

ii. The image of the point A (x, y) under the translation T = A' $(x + a, y + b)$.

3. Reflection

Let L be a fixed line in the plane. A reflection M about a line L is a transformation of the plane onto itself which carries each point A of the plane into the point A' of the plane such that L is the perpendicular bisector of AA'.

a. Reflection in the *x-axis*, M $(x, y) = (x, -y)$.

b. Reflection in the *y-axis*, M $(x, y) = (-x, y)$.

c. Reflection in the line $y = x$, M $(x, y) = (y, x)$.

d. Reflection in the line $y = -x$, M $(x, y) = (-y, -x)$.

e. Reflection in the line $y = mx$, M $(x, y) = (x', y')$ where

$x' = x \cos 2\theta + y \sin 2\theta$; $y' = x \sin 2\theta - y \cos 2\theta$ and $m = \tan \theta$

4. Rotation

A rotation R about a point O through an angle θ is a transformation of the plane onto itself which carries every point A of the plane into the point A' of the plane such that OA = OA' and m ($<AOA'$) = θ. O is called the centre of rotation and θ is called the angle of rotation.

1. If R is a rotation through angle θ about the origin), then

   $R_\theta\ (x, y) = (x', y') = (x' = x \cos\theta - y \sin\theta, y' = x \sin\theta + y \cos\theta)$

2. Let R be a counter-clockwise rotation through an angle θ about the origin. Then

   a. If $\theta = \frac{\pi}{2}$ then $R_\theta(x, y) = (-y, x)$.

   b. If $\theta = \pi$ then $R_\theta\ (x, y) = (-x, y)$.

   c. If $\theta = \frac{3\pi}{2}$ then $R_\theta\ (x, y) = (y, x)$.

   d. If $\theta = 2n\pi$ then $R_\theta\ (x, y)$ is the identity transformation.

   e. Every circle with centre at the centre of rotation is fixed.

Review Exercise

1. If a translation T takes the point $(3, -5)$ to the point $(1, 4)$, then find the images of the following line and circle.

   a. $5x - 3y = 5$ b. $y + 3x = 6$

   c. $x + 2y = 7$ d. $x^2 + y^2 - 4x + 6y = 8$

   e. $x^2 + y^2 = 4$ f. $x^2 + y^2 - 3x + 5y = 0$

2. The image of the point A (4, 9) in a reflection is A' (5, 1). Find the line of reflection's equation.

3. Find the image of the circle $x^2 + y^2 - 6y = 3$ when it is rotated through $\frac{\pi}{3}$ about $(-1,5)$.

4. Rotate triangle ABC if A (5, 1), B (3, 2), C $(3, -2)$ 90° clockwise about the origin, then reflect the image over the line $x = 4$.

5. If the image of the point $(-3, 1)$ under reflection is (0, 2), then find the line of reflection.

6. If the image of the line $3x - 4y = 11$ under a translation is $3x - 4y = 0$. What is the translation vector of the translation line?

7. Find the equation of the image of the circle $x^2 + y^2 + 16x - 24y + 183 = 0$ when it is reflected about the line $4x + 7y + 13 = 0$?

8. If the line with equation $3x + 2y = 1$ is rotated clockwise about the point $(-1, 2)$ through $\theta = 45^\circ$, then find the equation of this new line?

9. Find the image of the circle $(x + 10)^2 + (y + 12)^2 = 9$ when it is rotated through $\theta = 30^\circ$ about (7, 3).

10. Find the image of the line s: $y = x + 5$ after it has been reflected about the line

    a. $L: y = \frac{1}{2}x + 2.$ b. $L: y = x - 1.$

    c. $L: y = \frac{1}{3}x + 1.$ d. $L: y = 2x + 1.$

UNIT 7

STATISTICS

Unit Outcomes

By the end of this unit, you will be able to:

- Know specific facts about types of data.
- Understand basic concepts about grouped data.
- Use statistical method to solve real life problems.
- Analyze statistical data.
- Compare statistical data.
- Appreciate the value of statistics in real life.

Unit Contents

7.1 Types of Data

7.2 Introduction to Grouped Data

7.3 Graphical Representation of Grouped Data

7.4 Measures of Central Tendency and Their Interpretation

7.5 Real-life Application of Statistics

Summary

Review Exercise

- class interval
- class limit
- class midpoint
- continuous data
- deciles
- discrete data
- frequency
- mean
- measures of location
- mode
- percentiles
- qualitative data
- quantitative data
- quartiles

Introduction

In grade 9, you have seen the meaning, importance and purpose of statistics. Moreover, you have seen presentation of data using different forms like a histogram, measures of central tendency, and measures of dispersion of ungrouped data.

At this grade, you will start types of data but initially through discussion.

Florence Nightingale (1820 – 1910), was a famous nurse during the Crimean War, which took place between 1853 and 1856. She is generally considered to have founded the modern nursing profession, by establishing her own nursing school at St Thomas' Hospital in London in 1860. She was also a gifted mathematician, and used statistical diagrams to illustrate the conditions that existed in the hospitals where she worked. Although she did not invent pie chart, she popularized its use, along with other diagrams such as the rose diagram, which is like a circular histogram. In 1859, she was elected the first female member of the Royal Statistical Society.

Source: Cambridge IGCSE Mathematics Fifth Edition.

7.1 Types of Data

1. Qualitative and Quantitative Data

Activity 7.1

Consider the following data: gender, height, weight, beauty, time, colour, price and temperature. Discuss which of these data can be or cannot be measured/expressed in the form of numbers.

From Activity 7.1 you have observed that there are data that can be measured in a form of number values while there are also data that cannot be expressed in the form of number values. These data can be classified as quantitative or qualitative data.

Definition 7.1

Quantitative data is the type of data whose value is measured in the form of numbers or counts, with a unique numerical value associated with each data set.

Qualitative data is non-numerical data describing the attributes or properties that an object possesses.

Example 1

Classify the following data as qualitative or quantitative: height, weight, the street a person lives on, the car a person drives, income, length, time, distance, religion, social status, number of mathematics teachers in your school and price.

Solution

The street a person lives on, the car a person drives, religion and social status are qualitative, while height, weight, income, length, time, number of mathematics teachers in your school, price and distance are quantitative.

2. Discrete Data and Continuous Data

Activity 7.2

Consider the following quantitative data: height, weight of the students, number of teachers in your school, length of a road, number of chairs in a room, temperature of a body of students and number of houses along a street. Discuss which of these data can be quantified in terms of whole numbers values. Which can be quantified in terms of fractional values?

There are quantitative data that can be quantified in terms of whole number values only or fractional values and varying data values that are measured over a specific time interval. Depending on this criterion, therefore, these data can be classified as discrete or continuous data.

Definition 7.2

A Discrete data is a numerical type of data that includes whole, concrete numbers with specific and fixed data values determined by counting. There is a gap between values.

A Continuous data is one which takes any fractional point along a specified interval of values.

Example 2

Which of the following are discrete data? Which are continuous?

The number of persons per household, the weight of each shipment of exported coffee, the units of an item in inventory, the length of time between successive landings of airplane at Bole Airport, the number of assembled components which are found to be defective.

Solution

Table 7.1

Discrete data	Continuous data
The number of persons per house hold	The weight of each shipment of exported coffee
The units of an item in inventory	The length of time between successive landings of airplane at Bole Airport
The number of assembled components which are found to be defective	

Definition 7.3

Variables are the characteristics of a unit being observed that may assume more than one of a set of values to which a numerical measure can be assigned and denoted by letters as x, y, z, ...

Example 3

Suppose in your class the height, weight, age of students, eye colour of students, dog breed, climate, electrical conductivity, customer service satisfaction and class attendance vary. Thus, these quantities are some examples of variables.

Exercise 7.1

1. Classify the following data as quantitative and qualitative:
   a. Gender b. Religion c. Amount of money you have
   d. State of birth e. Number of students f. Educational qualification
   g. Shoe size h. Hair color i. Volume of water in a barrel
   j. score of a team in a soccer match
2. Identify whether each of the following is discrete or continuous data:
   a. weight of a package ready to be shipped
   b. rank of students by examination results

c. volume of water in a glass
d. yield of wheat in quintals
e. number of red marbles in a jar
f. time it takes to get to school
g. distance travelled between classes
h. students' grade level
i. year of birth
j. winning time in a race

7.2 Introduction to Grouped Data

Having a collected and edited data, the next important step is to organize it. That is to present it in a readily comprehensible condensed form in order to draw conclusion from it. Hence, before discussing about grouped data let us define a grouped data and frequency distribution of a data as follows.

Definition 7.4

1. **Grouped data are** data formed by aggregating individual observations of a variable into groups.
2. **Frequency distribution** is a list, table or graph that displays the frequency of various outcomes in a sample.

7.2.1 Grouped Discrete Data

Activity 7.3

Consider the following data.

4	5	2	8	8	7	5	5	6	7
6	6	8	8	7	5	7	7	4	4
4	4	1	0	0	6	1	9		

It represents students' results of mathematics test out of ten. Discuss how this data can be organized into three groups of classes.

From Activity 7.3, you can make a frequency distribution table with a group of three classes that help you to define the grouped frequency distribution, grouped discrete data and other related definitions.

Definition 7.5

1. **Grouped discrete data** is a discrete data that has been grouped into categories.
2. **Grouped frequency distribution** is the organization of raw data in table form using classes and frequencies for the purpose of summarizing a large sample of data. When the range of the data is large, the data can be divided into classes with more than one unit in width.
3. **Class interval** is the numerical width of any class in a particular distribution.
4. **Class limit** is the minimum value and the maximum value the class interval may contain.
5. **Class width** is the difference between the upper- and lower-class limit of any class (category).

Note

Steps for constructing grouped frequency distribution

1. Determine the number of classes required (usually between 5 and 20).
2. Approximate the interval of each class or class width using the following formula

$$\text{Class width} = \frac{\text{Largest Value in ungrouped data} - \text{Smallest Value in ungrouped data}}{\text{number of classes required}}$$

Example 1

The following raw data represents the number of people who visited Unity Park per day for 20 days

11	29	6	33	14	31	22	27	19	20
18	17	22	38	23	21	26	34	39	27

a. Construct a grouped frequency distribution with 6 classes.

b. What is the frequency of the 3rd class?

c. What is the frequency of the 6th class?

Solution

a. The data given is raw data, or ungrouped data. Follow these steps to summarize the raw data into a grouped frequency distribution:

Step 1: Determine the number of classes.

To prepare the grouped frequency distribution, you decide the number of classes. For this case, let the number of classes be 6.

Step 2: Find class Width.

From the formula for class width, you have:

$$\text{Class width } (w) = \frac{\text{Largest Value in ungrouped data } - \text{Smallest Value in ungrouped data}}{\text{number of classes required}}$$

$$= \frac{39 - 6}{6} = 5.5$$

From the formula, the class width (w) is calculated as 5.5. For practical purposes, it will be convenient to choose the class width to be a whole number. For this case, you can take class width as 6. (This is obtained by rounding 5.5 to the nearest whole number). Therefore, $w = 6$.

The complete grouped frequency distribution is as follows:

Table 7.2

Number of people (Class interval)	Tally	Number of visiting days (f)
6 – 11	//	2
12 – 17	//	2
18 – 23	### //	7
24 – 29	////	4
30 – 35	///	3
36 – 41	//	2
Total		20

b. The frequency of 3rd class is 7.

c. The frequency of 6th class is 2.

Example 2

For the above table, give the lower- and upper-class limits for the 3rd and the 6th classes.

Solution

For the 3rd class 18 – 23, 18 is called the lower-class limit and 23 is called the upper-class limit, while the lower limit and the upper limit of the 6th class are 36 and 41, respectively.

Exercise 7.2

The following data shows the test results of 20 students in mathematics:

1	2	3	3	4	5	5	5	6	6
6	6	6	7	8	8	9	10	10	10

a. Construct a grouped frequency distribution with 5 classes.

b. What is the frequency of the 4th class?

c. What is the frequency of the 2nd class?

d. What is the upper-class limit and lower-class limit of the 3rd class?

Definition 7.6

1. **Cumulative frequency** is the total of a frequency and all frequencies in a frequency distribution until a certain defined class interval.
2. **Cumulative frequency distribution** is a form of frequency distribution that represents frequency distribution.

Example 3

Construct a cumulative frequency distribution for example 1 above.

Solution

In the above grouped frequency distribution, you are considering frequencies of each class. But in reality, you may be interested to know about other issues such as how many days fewer than 24 people visited the Unity Park. To answer such a question, the frequency distribution given above may not always be suitable. For such a purpose, you need to construct what is called a cumulative frequency distribution.

A cumulative frequency distribution is constructed by successively adding the frequencies of each class.

The cumulative frequency distribution of the above data of people that visit Unity Park per day is as follows:

Table 7.3

Number of people (Class interval)	Tally	Number of visiting days(f)	Cumulative frequency (Cf)
6 – 11	//	2	2
12 – 17	//	2	4
18 – 23	~~////~~ //	7	11
24 – 29	////	4	15
30 – 35	///	3	18
36 – 41	//	2	20

Exercise 7.3

1. The following data represents the number of people who paid TAX per day for 30 working days (hint: use excel software to present the ascending order of the data)

1	1	4	4	3	6	5	3	6	5
3	7	7	6	7	8	7	8	8	8
9	9	10	11	21	16	12	22	19	16

   a. Construct a grouped frequency distribution with 6 classes.
   b. What is the frequency of the 3rd class?
   c. What is the frequency of the 6th class?
   d. Determine the cumulative frequency distribution.

2. The following are scores of 80 students in a mathematics exam.

22	36	32	31	32	33	33	36	31	22
31	41	49	36	36	36	36	42	34	35
36	49	50	49	38	38	41	43	41	42
46	50	51	58	48	41	42	47	41	45
46	56	54	59	50	46	48	50	44	46
50	69	56	60	56	47	56	54	47	56
50	72	60	65	62	55	63	67	58	62
55	76	71	66	76	58	76	70	62	62

Construct a grouped frequency distribution, using 8 classes width. Answer the following questions.

a. Determine the number of classes.

b. Determine the lower-class limit of the fourth class.

c. Determine the upper-class limit of the fourth class.

d. Determine the frequency of the first class.

e. Determine the cumulative frequency distribution.

f. How many students got mathematics score less than 55?

g. Determine the cumulative frequency at the fifth class.

7.2.2 Grouped Continuous Data

Activity 7.4

Consider the following data.

46	47	60	63	69	49	50	60	60	70
50	50	50	63	63	48	48	49	70	70
49	49	60	61	62	70	51	51	70	70
55	60	45	46	46	60	55	57	59	60
67	50	55	55	69	47				

It represents weight of grade 11 students (in kg) in which the highest observed weight was 70 kg and the lowest was 45 kg. Discuss how this data can be organized into 5 categories using class width.

For Activity 7.4, you have to group this data in table form into 5 categories using class width that represents grouped frequency distribution that helps you to define grouped continuous data and other related definitions.

Definition 7.7

1. **Grouped continuous data** is a continuous data that has been grouped into categories.
2. **Class mark (Mid points)** is the average of the lower- and upper-class limits or the average of upper- and lower-class boundary. This means if m_i is the class mark (midpoint) of a class then it is calculated by

$$m_i = \frac{(\text{lower class boundary} + \text{upper class boundary})}{2}$$

Note

Steps for constructing grouped frequency distribution

1. Determine the number of classes required
2. Approximate the interval of each class or class width using the following formula

$$\text{Class width} = \frac{\text{Largest Value in ungrouped data} - \text{Smallest Value in ungrouped data}}{\text{number of classes required}}.$$

Example

On a certain construction site, the weekly wages (in Birr) of 100 labourers taken from a list (i.e., ungrouped data) in which the highest observed wage was Birr 258 and the lowest was Birr 142 are required to be given in 6 classes of a frequency distribution as follows. Let x be the weekly wage.

Table 7.4

Weekly wage (in Birr)	Number of labourers *(f)*
$140 \leq x < 160$	7
$160 \leq x < 180$	20
$180 \leq x < 200$	33
$200 \leq x < 220$	25
$220 \leq x < 240$	11
$240 \leq x < 260$	4
Total	**100**

Based on the above table, answer the following questions.

a. Construct a cumulative frequency distribution.

b. What is the range?

c. What is the class width?

d. What is the frequency of 4^{th} class?

e. How many people are paid Birr 220 and more?

Solution

a.

Table 7.5

Weekly wage (Birr) (Class interval)	Number of Laborers	Cumulative frequency (Cf)
$140 \leq x < 160$	7	7
$160 \leq x < 180$	20	27
$180 \leq x < 200$	33	60
$200 \leq x < 220$	25	85
$220 \leq x < 240$	11	96
$240 \leq x < 260$	4	100

b. Range = $258 - 142 = 116$.

c. Class width is 20.

d. The frequency of 4^{th} class is 25.

e. Number of people that are paid Birr 220 and more is 15.

Exercise 7.4

In a certain secondary school, the monthly wages (in Birr) of 100 teachers taken from a list (i.e. ungrouped data) in which the highest observed wage was Birr 11,000 and the lowest was Birr 4,200 are required to be given in 5 categories (classes) of a frequency distribution as shown in Table 7.6.

Table 7.6

Monthly wage (in Birr)	Number of Teachers(*f*)
$4{,}200 \leq x < 5{,}600$	23
$5{,}600 \leq x < 7{,}000$	16
$7{,}000 \leq x < 8{,}400$	30
$8{,}400 \leq x < 9{,}800$	20
$9{,}800 \leq x < 11{,}200$	11

Based on the above table, answer the following questions.

a. Construct a cumulative frequency distribution.

b. What is the range?

c. What is the frequency of 4^{th} class?

d. What is the class width?

e. Determine the lower-class limit for the third class.

f. What is the class mark of the second class?

g. How many people are paid Birr 7,000 and more?

7.3 Graphical Representation of Grouped Data

Activity 7.5

What knowledge do you have about graphical representation of grouped data such as histogram and frequency polygon?

In the previous section representation of data by tables has already been discussed. Now let us turn our attention to another representation of data, i.e., the graphical representation. Usually, comparisons among the group data are best shown by means

of graphs. The representation then becomes easier to understand than the actual data. The histogram, and frequency polygon graph are most commonly applied graphical representation for continuous data.

Note

Steps for constructing histogram, and frequency polygon

1. Draw and label the x- and y- axis.
2. Choose a suitable scale for the frequencies or cumulative frequencies and label it on the y- axis.
3. Represent the class boundaries for the histogram or the mid points for the frequency polygon on the x -axis.
4. Plot the points.
5. Draw the bars or lines to connect the points.

Definition 7.8

Frequency Polygon is a graph constructed by using lines to join the midpoints of each interval, or bin. The frequency is placed along the vertical axis and classes mid points are placed along the horizontal axis.

Example

Construct histograms, and Frequency polygons for data shown in Table 7.7. Let x be the weekly wage.

Table 7.7

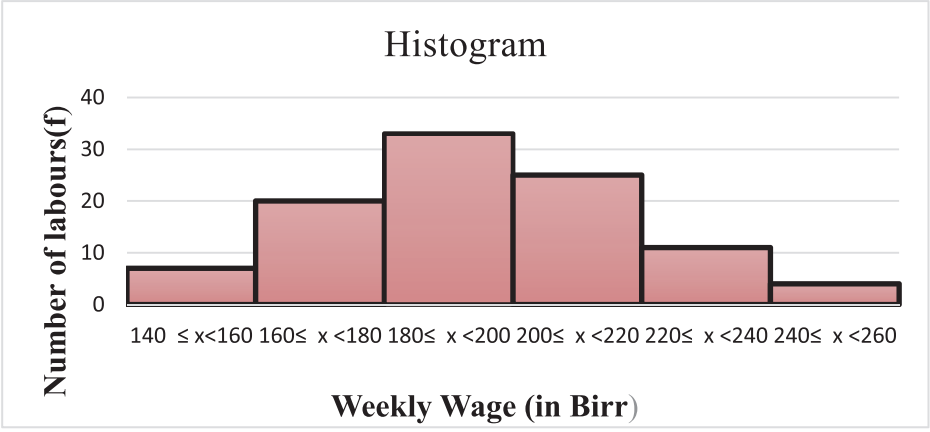
Weekly wage (in Birr)	Number of labourers *(f)*
$140 \leq x < 160$	7
$160 \leq x < 180$	20
$180 \leq x < 200$	33
$200 \leq x < 220$	25
$220 \leq x < 240$	11
$240 \leq x < 260$	4

Solution

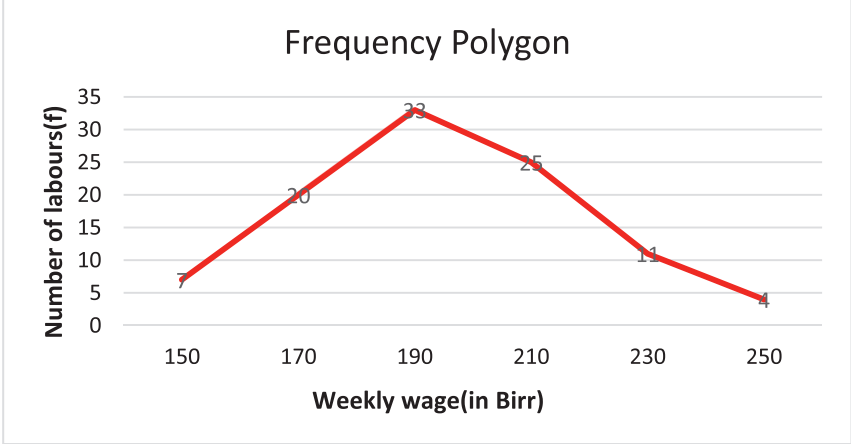
To construct histogram, and Frequency Polygon for the above data, you can apply the following steps;

1. Draw and label the x- and y- axis.
2. Choose a suitable scale for the frequencies and label it on the y- axis.
3. Represent the class limits for the histogram or the mid points for the frequency polygon on the x- axis.
4. Plot the points.
5. Draw the bars or lines to connect the points.

i) Histogram



ii) Frequency polygon



Exercise 7.5

1. Draw
   a. the histograms b. the Frequency polygons of the previous Exercise 7.4
2. The Table below shows weight of grade 11 section A male students.
   a. Draw the histogram b. Draw frequency polygon. Let the weight be w.

Table 7.8

Weight (w) (in kg)	No. of student
$40 \leq w < 45$	4
$45 \leq w < 50$	8
$50 \leq w < 55$	12
$55 \leq w < 60$	9
$60 \leq w < 65$	3

7.4 Measures of Central Tendency and Their Interpretation

Revision of mean for ungrouped data

Activity 7.6

1. Do you know how to calculate mean of ungrouped data?
2. Discuss how to calculate mean of the following ungrouped data.
   a. 8, 6, 8, 10 and 13
   b. 6, 7, 7, 5, 6, 2, 4 and 6

From Activity 7.6, you may recall how to calculate the mean for ungrouped data that you have learnt in grade 9.

Definition 7.9

The mean x of a set of data, denoted by $\bar{X}$ is equal to the sum of the data items divided by the number of items contained in the data set.

$$\bar{X} = \frac{x_1 + x_2 + x_3 + \ldots + x_n}{n} \text{ or}$$

If $x_1, x_2, \ldots, x_n$ is a set of data items, with frequencies $f_1, f_2, \ldots, f_n$ respectively, then their mean is given by

$$\bar{X} = \frac{f_1x_1 + f_2x_2 + \ldots + f_nx_n}{f_1 + f_2 + \ldots + f_n}$$

Example

Consider the following values which show the number of radios sold by an electronics shop for 26 days.

7	7	2	6	7	10	8	10	2	7
0	1	7	2	7	6	10	6	7	8
7	6	7	10	6	10				

a. Construct a frequency distribution table.

b. Find the mean from the raw data.

Solution

a. From the above data, you may have found the following frequency distribution table which shows the number of radios sold by the shop for 26 days;

Number of radios (x)	0	1	2	6	7	8	10
f	1	1	3	5	9	2	5

b. You use the above formula to find the mean from the frequency distribution table.

$$\bar{X} = \frac{f_1x_1 + f_2x_2 + f_3x_3 + f_4x_4 + f_5x_5 + f_6x_6 + f_7x_7}{f_1 + f_2 + f_3 + f_4 + f_5 + f_6 + f_7} = \frac{1 \times 0 + 1 \times 1 + 3 \times 2 + 5 \times 6 + 9 \times 7 + 2 \times 8 + 5 \times 10}{1 + 1 + 3 + 5 + 9 + 2 + 5}$$

$$= \frac{166}{26} = 6.38$$

Thus, the typical number of radios sold daily by the shop is about 6.4 or 62 % of the radios sold daily is above the average.

Exercise 7.6

1. Find the mean of the following data.

   3, 5, 7, 1, 8, 6

2. The following frequency distribution tables represent the score and weight of grade 11 students in mathematics. Find the mean for each of them.

   a.

Marks (out of 20)	5	9	10	12	13	19
f	2	10	5	5	7	1

   b.

Weight (kg)	41	43	52	54	60
f	1	14	15	21	14

7.4.1 Mean for Grouped Data

Activity 7.7

1. Do you know how to calculate mean of grouped data?
2. Suppose the first group of 8 students have mean score 7 in mathematics exam. A second and third group of 10 students and 12 students have mean score 8 and 6 in the same exam. What is the mean score of all students?

Form Activity 7.7, you have observed that the procedure for finding the mean for grouped data is similar to that of ungrouped data, except that the mid points of the classes are used for the x values.

Example 1

Calculate the mean of this grouped frequency table for students' test scores.

Table 7.9

Class interval	Frequency(f)
6 – 10	35
11 – 15	23
16 – 20	15
21 – 25	12
26 – 30	9
31 – 35	6

Solution

First, find the class marks (class midpoint).

Second, find the product of frequency class mark.

Third, find mean using the formula.

Table 7.10

Class interval	f_i	Midpoint (m_i)	$f_i m_i$
6 – 10	35	8	280
11 – 15	23	13	299
16 – 20	15	18	270
21 – 25	12	23	276
26 – 30	9	28	252
31 – 35	6	33	198
Total	100		1575

Now, total number of students = 100; Total marks(approximate) = 1575

$$\bar{x} = \frac{f_1m_1 + f_2m_2 + f_3m_3 + f_4m_4 + f_5m_5 + f_6m_6}{f_1 + f_2 + f_3 + f_4 + f_5 + f_6} = \frac{1575}{100} = 15.75$$

Hence, on average, the students' test score is approximately 15.75

Note

Steps to find the mean from a grouped frequency distribution

1. Find the class mark (midpoint)$m_i$of each class.
2. Multiply m_i by its corresponding frequency and add.
3. Divide the sum obtained in step 2 by the sum of the frequencies

$\bar{x} = \frac{f_1 m_1 + f_2 m_2 + f_3 m_3 + \ldots + f_i m_i}{f_1 + f_2 + f_3 + \ldots + f_i}$, where m_i is the class mark (midpoint) of a class and calculated by

$$m_i = \frac{\text{lower class boundary + upper class bundary}}{2}$$

Example 2

Table 7.11 indicates the age distribution of 30 students in a class of extension students attending evening classes. Find the mean age of these students.

Table 7.11

Age (in years)	Class midpoint (m_i)	Number of students(f_i)	$f_i m_i$
16 – 20	18	4	72
21 – 25	23	11	253
26 – 30	28	9	252
31 – 35	33	6	198
Total		30	775

Solution

$$\bar{x} = \frac{f_1 m_1 + f_2 m_2 + f_3 m_3 + f_4 m_4}{f_1 + f_2 + f_3 + f_4} = \frac{775}{30} = 25.83 \text{ years}$$

Thus, the average age of students in the class is about 25.83.

Example 3

Table 7.12 gives data on the height of 60 students in a class. Find the mean height of these students. Let the height be h.

Table 7.12

Height (in cm)	Class midpoint (m_i)	Number of students(f_i)	$f_i m_i$
$130 \leq h < 140$	135	5	675
$140 \leq h < 150$	145	6	870
$150 \leq h < 160$	155	18	2,790
$160 \leq h < 170$	165	22	3,630
$170 \leq h < 180$	175	9	1,575
Total		60	9540

Solution

$$\bar{x} = \frac{f_1m_1 + f_2m_2 + f_3m_3 + f_4m_4 + f_5m_5}{f_1 + f_2 + f_3 + f_4 + f_5} = \frac{9540}{60} = 159 \text{ cm.}$$

Thus, the average height of students in the class is about 159cm.

Exercise 7.7

1. The following frequency distribution tables represent the mid- year mark of grade 11 students in mathematics and their weight in kg. Find the mean for each of them.

   a.

Marks (out of 50)	41–45	36–40	31–35	26–30	21–25	16–20
f	1	8	8	14	7	2

b.

Weight (kg)	f
$40 \leq w < 45$	1
$45 \leq w < 50$	**7**
$50 \leq w < 55$	8
$55 \leq w < 60$	**7**
$60 \leq w < 65$	**5**
$65 \leq w < 70$	5
$70 \leq w < 75$	**6**
$75 \leq w < 80$	1

2. The following table gives the frequency distribution of the number of people attacked by COVID-19 Pandemic per day for 50 consecutive days reported by Ministry of Health of Ethiopia. Calculate the mean.

No. of People attacked	10–12	13–15	16 –18	19 – 21	Total
f	4	12	20	14	50

Revision of Median for ungrouped Data

Activity 7.8

1. Do you know how to calculate the median of ungrouped data?
2. Discuss how to calculate median of the following ungrouped data.
   a. 9, 7, 9, 11 and 14 b. 7, 8, 8, 6, 7, 3, 5 and 7

From Activity 7.8, you may have observed that median of ungrouped data is half way point in data, when the data is arranged in order. The median will be a value in the data or will fall between two values.

Note: Median is denoted by $(\tilde{X})$

Note

Steps in computing the median of ungrouped data

1. Arrange the data in increasing order.
2. Select the middle number.
3. 3. If n is odd, median = $\frac{(n+1)^{th}}{2}$ observation.
4. If n is even, median = $\frac{\left(\frac{n}{2}\right)^{th} \text{ observation} + \left(\frac{n}{2}+1\right)^{th} \text{ observation}}{2}$

Example 4

The following data shows the number of car accident for 9 days.

What will be the median of this data distribution?

7, 9, 6, 7, 11, 8, 4, 11, 7

Solution

Step 1. Arranging the data in an increasing order gives 4, 6, 7, 7, 7, 8, 9, 11,11

Step 2. Select the middle value

Since the number of observations is 9 and this number is odd;

therefore,

$\widetilde{X} = \frac{(n+1)^{th}}{2}$ item

$= \left(\frac{9+1}{2}\right)^{th}$ item = 5^{th} item shows = 7

Thus, the median is 7.

Therefore, the median value of car accident is 7.

Example 5

Find the median from the following data; 70, 73, 69, 82, 60, 59.

Solution

Step 1: Arranging the data in an increasing order gives 59, 60, 69, 70, 73, 82.

Step 2: Select the middle value.

Since the number of observations is 6 and this number is even; therefore,

$$\tilde{X} = \frac{\left(\frac{n}{2}\right)^{th}\text{ item} + \left(\frac{n}{2}+1\right)^{th}\text{ item}}{2}.$$

$$= \frac{\left(\frac{6}{2}\right)^{th}\text{ item} + \left(\frac{6}{2}+1\right)^{th}\text{ item}}{2}.$$

$$= \frac{3^{rd}\text{ item} + 4^{th}\text{ item}}{2}.$$

$$= \frac{69+70}{2} = \frac{139}{2} = 69.5$$

Hence, the median is 69.5. That is, the median value of the data is 69.5.

Exercise 7.8

1. Consider the following data which shows grade 11 students mathematics score out of 20.

10	11	12	11	13	15	15	13	12	11
10	9	13	12	11	10	9	13	13	12
11	10	11	12	13	15	13	12	11	10
20	20								

Find the median from the raw data.

2. The Bills paid (in Birr) for water consumption by Ato Girma in the last 8 months is as follows.

62, 78, 57, 66, 88, 58, 113, 92.

a. Find the median of Bills paid for the water consumption.

b. Calculate the mean and compare it with the median.

7.4.2 The Median of Grouped Data

Activity 7.9

1. Do you know how to calculate the median of grouped data?
2. Discuss how to calculate the median of the following data on mathematics test score of 20 students out of 10 (full mark).

Math test Score	1 – 2	3 – 4	5 – 6	7 – 8	9 – 10
f	2	4	8	5	1

From Activity 7.9, you can observe that the procedure for finding the median from grouped data is similar to that of ungrouped data; except preparing the cumulative frequency distribution.

Note

Steps to find the median of a grouped frequency distribution

1. Prepare a cumulative frequency distribution.
2. Find the class where the median is located. It is the lowest class for which the cumulative frequency equals or exceeds $\frac{n}{2}$.
3. Determine the median by the formula

$$\widetilde{X} = \mathrm{B_L} + \left(\frac{\frac{n}{2} - cf_B}{fc}\right)\mathrm{i}$$

where,

B_L = Lower limit of the class containing the median (median class)

n = total number of observations.

cf_B = the cumulative frequency in the class preceding ("coming before") the class containing the median.

fc = the number of observations (frequency) in the class containing the median.

i = the size of the class interval. (i.e. width of the median class)

Example

The following is the time 50 students take to travel to school. Find the median time to travel to school. Let t be the time.

Time travel to school(min)	$1 \leq t < 10$	$10 \leq t < 20$	$20 \leq t < 30$	$30 \leq t < 40$	$40 \leq t < 50$
f	8	14	12	9	7

Solution

From this, you can prepare the cumulative frequency column as follows;

Table 7.12

Time to travel to school(min)	**f**	**Cumulative frequency (Cf)**
$1 \leq t < 10$	8	8
$10 \leq t < 20$	14	8 + 14 = 22
$20 \leq t < 30$	**12**	**22 + 12=34**
$30 \leq t < 40$	9	34 + 9 = 43
$40 \leq t < 50$	7	43 + 7 =50
Total	50	

The median class is the class containing the $\left(\frac{50}{2}\right)^{th}$ item = 25^{th} item. In this case, it is in the 3^{rd} class.

Therefore, the median class is **$20 \leq t < 30$**

Thus, $B_L = 20$, $\frac{n}{2} = 25$, $fc = 12$, i = 40 – 30 = 10, $cf_B = 22$

Therefore,

$$\widetilde{X} = B_L + \left(\frac{\frac{n}{2} - cf_B}{fc}\right)i$$

$$= 20 + \left(\frac{25-22}{12}\right)10 = 20 + 2.5 \quad = 22.5$$

Thus, the median time to travel to school is 22.5 minutes.

Exercise 7.9

1. The following data shows the number of people attacked by COVID-19 epidemic within 40 days.

57	59	54	57	58	56	59	53	59	57
53	54	56	53	54	57	54	56	58	55
56	53	55	52	54	53	54	57	58	55
58	56	57	60	56	57	61	61	57	56

   a. Find the median from the raw data.

   b. Construct a grouped frequency distribution with 5 classes.

2. Calculate the median of the following data about weight of students in a class. Let w be the weight.

Weight (in kg)	Number of students
$35 \leq w < 40$	8
$40 \leq w < 45$	9
$45 \leq w < 50$	14
$50 \leq w < 55$	15
$55 \leq w < 60$	10
$60 \leq w < 65$	4

3. The following set of data shows about mark of students in mathematics exam:

Marks	41–47	48–54	55–61	62–68	69–75	76–82	83–89	90–96
No of students	4	7	9	14	9	8	5	4

Find the median marks of the students.

7.4.3 Quartiles, Deciles and Percentiles

Measures of central tendency like median is a measure of location that divides the data into two equal parts. In addition, there are other measures of position or location such as percentiles, deciles, and quartiles. These measures are used to locate the relative position of a data value in the data set. For example, if a value is located at

the 80th percentile, it means that 80% of the values fall below it in the distribution and 20% of the values fall above it. The *median* is the value that corresponds to the 50th percentile, since one-half of the values fall below it and the remaining half of the values fall above it. Before discussing these measures of position for grouped data, let us see how they work for ungrouped data and then move to grouped data.

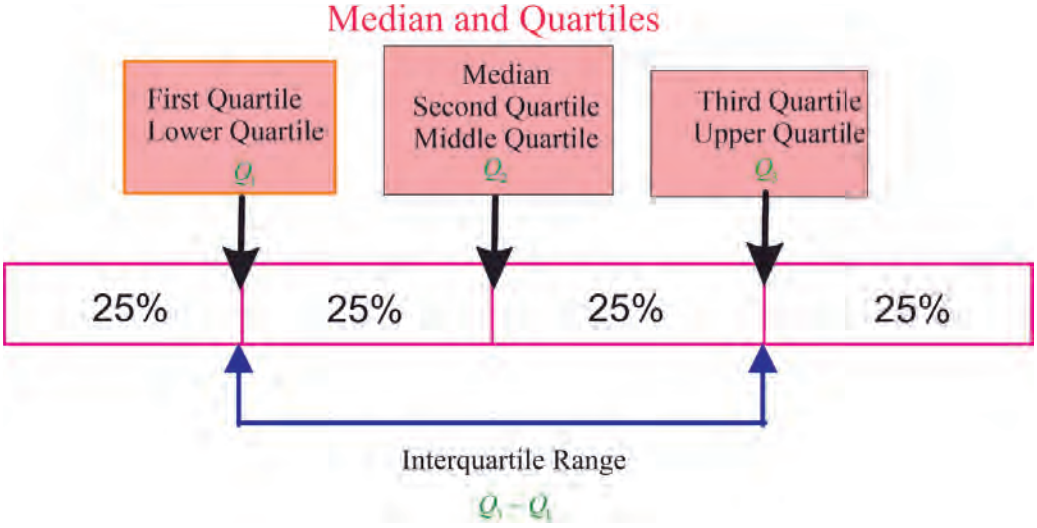
7.4.3.1 Quartiles, Deciles and Percentiles for Ungrouped Data

1. Quartiles

Activity 7.10

1. Do you know how to calculate the quartiles of ungrouped data?
2. Explain meaning and steps to calculate quartiles for ungrouped data?

A quartile divides data into three points, namely **a lower quartile, median**, and **upper quartile** to form four groups of the dataset. The lower quartile, or first quartile, is denoted as Q_1 and is the middle number that falls between the smallest value of the dataset and the median. The second quartile, Q_2, is also the median. The upper or third quartile, denoted as Q_3, is the central point that lies between the median and the highest number of the distribution.



Now, the four groups formed from the quartiles are the first group of values contains **the smallest number up to Q_1**; the second group includes **Q_1 to the median**; the third set is **the median to Q_3;** the fourth category comprises **Q_3 to the highest data point of the entire** set.

Definition 7.10

The difference of the upper quartile and the lower quartile is **the interquartile range (IQR); i.e., $IQR = Q_3 - Q_1$**

Example 1

Suppose the distribution of math scores in a class of 20 students in ascending order is:

59	60	65	65	68	69	70	72	75	75
76	77	81	82	84	87	90	93	95	98

Then, find Q_1, Q_2 and Q_3 and interpret what the value of each shows?

Solution

First, mark down the median, Q_2, which in this case is the average of the 10^{th} and 11^{th} score: 75.5.

Q_1 is the central point between the smallest score and the median. In this case, Q_1 lies between the 5^{th} and 6^{th} score: 68.5 (Note that the median can also be included when calculating Q_1 or Q_3 for an even score). If you were to include the median on either side of the middle point, then Q_1 will be the middle value between the 5^{th} and 6^{th} scores. It is the average of the 5^{th} and 6^{th} scores.

i.e., $\frac{(5^{th} + 6^{th})}{2} = \frac{(68 + 69)}{2} = 68.5.$

Q_3 is the middle value between Q_2 and the highest score: 98. $Q_3 = \frac{(84 + 87)}{2} = 85.5$

This means, Q_1 tells us that 25% of the scores are less than 68.5 and 75 % of the class scores are greater than 68.5. Q_2 (the median) shows that 50% of the scores are less

than 75.5, and 50% of the scores are above 75.5. Finally, Q_3 reveals that 25% of the scores are greater than 85.5 and 75% are less than 85.5.

Note

Steps to calculate quartiles for ungrouped data

1. Arrange the data in increasing order of magnitude.
2. If the number of observations is:

   i) odd, $Q_k = \left(\frac{k(n+1)}{4}\right)^{th}$ item

   ii) Even, $Q_k = \left(\frac{\left(\frac{kn}{4}\right)+\left(\frac{kn}{4}+1\right)}{2}\right)^{th}$ item

Example 2

Find Q_1, Q_2, Q_3 and IQR for the following data;

30, 43, 47, 51, 36, 34, 26, 14, 11, 5, 7, 54, 57

Solution

Arranging the data in increasing order of magnitude, you get,

5, 7, 11, 14, 26, 30, 34, 36,43, 47, 51, 54, 57

$Q_1 = \left(\frac{1(13+1)}{4}\right) = (3.5)^{th}$ item. What does this mean?

Q_1 lies between the 3^{rd} and 4^{th} items.

Therefore, $Q_1 = 3^{rd}$ item + 0.5(4^{th} item – 3^{rd} item)

$= x_3 + 0.5\,(x_4 - x_3)$

$= 11 + 0.5(14-11) = 11 + 1.5 = 12.5$

Thus, $Q_1 = 12.5$.

$Q_2 = \left(\frac{2(13+1)}{4}\right) = 7^{th}$ item

Thus, Q_2 is the 7^{th} item

Therefore, $Q_2 = 34$.

$Q_3 = \left(\frac{3(13+1)}{4}\right) = 10.5^{th}$ item

It is half of the way between the 10^{th} (x_{10}) and 11^{th} (x_{11}) items

Thus, $Q_3 = x_{11} + 0.5\,(x_{11} - x_{10})$

$= 47 + 0.5(47 - 43) = 47 + 2 = 49$

$IQR = Q_3 - Q_1 = 49 - 12.5 = 36.5.$

Exercise 7.10

1. For each of the following data sets:
   a. 12, 5, 22, 30, 7, 36, 14, 42, 15, 53, 25, 58, 62,63, 55
   b. 10, 13, 15, 12, 18, 15, 16, 12, 13, 20, 17, 14, 19, 18, 23,28
   c.

Score	15	19	20	22	24	25	31
f	11	16	15	7	5	5	2

   Find Q_1, Q_2, and Q_3

2. The marks obtained by 19 students of a class are given below:
   27, 36, 22, 31, 25, 26, 33, 24, 37, 32, 29, 28, 36, 35, 27, 26, 32, 35 and 28.
   Find:
   a. median b. lower quartile c. upper quartile d. interquartile range

2. Deciles

Activity 7.11

1. Do you know how to calculate the deciles of ungrouped data?
2. Explain meaning and steps to calculate deciles for ungrouped data?

Deciles divide a set of data into ten equal number of parts. There are nine deciles. Like the quartile, a decile consists of nine data points that divide a data set into 10 equal parts and it is usually used to assign decile ranks to a data set. A decile rank arranges the data in order from lowest to highest and is done on a scale of one to 10

where each successive number corresponds to an increase of 10 percentage points. In other words, there are nine decile points. Namely, D_1, D_2, D_3, D_4, D_5, D_6, D_7, D_8, and D_9. The 1st decile, or D_1, is the point that has 10% of the observations lies below it, D_2 has 20% of the observations lies below it, D_3 has 30% of the observations lies below it, and so on. The example below clarifies the point.

Example 3

The table below shows the ungrouped scores (out of 100) for 29 exam takers:

45	52	55	57	58	60	61	64	65	66
69	72	73	75	76	78	81	82	84	87
88	90	91	92	93	94	95	96	97	

Using the information presented in the table, find D_1, D_3 and D_5 and interpret what the value of each shows.

Solution

Using the information presented in the table, the 1st decile can be calculated as:

$$D_1 = \text{Value of } \left(\frac{29+1}{10}\right)^{th} \text{ data}$$

$$= \text{Value of } 3^{rd} \text{ data}$$

$$= 55$$

Thus, D_1 means that 10% of the scores of exam takers falls below 55.

Let's calculate the 3rd decile:

$$D_3 = \frac{\text{value of } 3(29+1)^{th}}{10}$$

$$= \text{Value of } 9^{th} \text{ position.}$$

$$= 65$$

This means, 30% of the scores of exam takers lies below 65.

What would you get if we were to calculate the 5th decile?

$$D_5 = \frac{\text{value of } 5(29+1)^{th}}{10}$$

$$= \text{Value of } 15^{th} \text{ position}$$

$$= 76$$

This means, 50% of the scores of exam takers fall below 76.

The 5th decile is also the median of the scores of exam takers. Looking at the data in the table, the median can be calculated as 76 = median = D_5. At this point, half of the scores of exam takers lie above and below the distribution.

In general, follow these steps, to calculate deciles;

Note

Steps to calculate deciles for ungrouped data

1. Arrange the data in increasing order of magnitude.
2. If the number of observations is:

i) odd, $D_j = \left(\frac{j(n+1)}{10}\right)^{th}$ item. ii) even, $D_j = \left(\frac{\left(\frac{jn}{10}\right) + \left(\frac{jn}{10} + 1\right)}{2}\right)^{th}$ item.

Example 4

Find D_3 and D_8 for the following data: 27, 40, 44, 48, 52, 33, 31, 23, 11, 7.

Solution

Arranging in increasing order of magnitude; you get,

7, 11, 23, 27, 31, 33, 40, 44, 48, 52.

$$D_3 = \left(\frac{\left(\frac{3(10)}{10}\right) + \left(\frac{3(10)}{10} + 1\right)}{2}\right)^{th} \text{ item}$$

$$= \left(\frac{3+4}{2}\right)^{th} \text{ item} = (3.5)^{\text{th}} \text{ item} = \frac{(23+27)}{2} = 25$$

This means, 30% of the data lies below 25.

$$D_8 = \left(\frac{\left(\frac{8(10)}{10}\right) + \left(\frac{8(10)}{10} + 1\right)}{2}\right)^{th} \text{ item}$$

$$= \left(\frac{8+9}{2}\right)^{th} \text{ item} = (8.5)^{\text{th}} \text{ item} = \frac{(44+48)}{2} = 46$$

Thus, 80% of the data lies below 46.

Exercise 7.11

Find D_2, D_3, D_5, D_8 for each of the following data sets:

a. 12, 5, 22, 30, 7, 36, 14, 42, 15, 53, 25, 58, 62,63, 55

b. 10, 13, 15, 12, 18, 15, 16, 12, 13, 20, 17, 14, 19, 18, 23, 2

c.

Score	15	19	20	22	24	25	31
f	11	16	15	7	5	5	2

3. Percentiles

Activity 7.12

1. Do you know how to calculate the Percentiles of ungrouped data?
2. Explain meaning and steps to calculate Percentiles for ungrouped data?

Percentiles divide the data set into 100 equal groups. There are 99 percentiles, namely, P_1, P_2, … P_{99}.

Percentiles are also used to compare an individual's test score with the national norm. For example, tests such as the National examination are taken by students in 8^{th} grade. A student's scores are compared with those of other students locally and nationally by using percentile ranks.

Percentiles and percentages are two different concepts. Thus, the percentage is a mathematical quantity which is written out of a total of 100 whereas a percentile is the percentage of values found under the specific values. Percentiles are mostly used for the ranking system. For instance, if you get 72 correct answers out of a possible 100. You obtain a percentage score of 72. There is no indication of your position with respect to the rest of the class. You could have scored the highest, the lowest, or somewhere in between. On the other hand, if a raw score of 72 corresponds to the 64^{th} percentile, then you did better than 64% of the students in your class.

Note

Steps to calculate Percentiles for ungrouped data

1. Arrange the data in increasing order of magnitude.
2. If the number of observations is:

i) odd, $P_j = \left(\frac{j\,(n+1)}{100}\right)^{th}$ item. ii) even, $P_j = \left(\frac{\left(\frac{jn}{100}\right) + \left(\frac{jn}{100} + 1\right)}{2}\right)^{th}$ item.

Example 5

Find P_{47} and P_{83} for the following data:

27	40	57	48	52	33	31	23	11	7
35	34	24	34	59	35	10	23	54	18
29	43	45	47	49	37	42	34	45	47
23	39	55	37	58	48	26	37	17	49

Solution

Arranging in increasing order of magnitude; you get,

7,10, 11, 18, 23, 23, 23, 24, 26, 27, 29, 31, 33, 34, 34, 34, 35, 35, 37, 37, 37, 39, 40, 42, 43, 45, 45, 47, 47, 48, 48, 49, 49, 53, 54, 55, 57, 58, and 59.

$$P_{47} = \left(\frac{\left(\frac{47(40)}{100}\right) + \left(\frac{47(40)}{100} + 1\right)}{2}\right)^{th} \text{ item}$$

$$= \left(\frac{18.8+19.8}{2}\right)^{th} \text{ item} = (19.3)^{\text{th}} \text{ item} = 37 + 0.3(37 - 37) = 37$$

$$P_{83} = \left(\frac{\left(\frac{83(40)}{100}\right) + \left(\frac{83(40)}{100} + 1\right)}{2}\right)^{th} \text{ item}$$

$$= \left(\frac{33.2 + 34.2}{2}\right)^{\text{th}} \text{ item} = (33.7)^{\text{th}} \text{ item} = 49 + 0.7(53 - 49) = 49 + 0.7(4)$$

$$= 51.8$$

Note that P_{83} = 51.8, that means 83% of the data values are less than 51.8 and the rest are above it.

Exercise 7.12

Find P_{50}, P_{24}, and P_{87} for each of the following data sets:

a. 12, 5, 22, 30, 7, 36, 14, 42, 15, 53, 25, 58, 62,63, 55

b. 10, 13, 15, 12, 18, 15, 16, 12, 13, 20, 17, 14, 19, 18, 23, 2

c.

Score	15	19	20	22	24	25	31
f	11	16	15	7	5	5	2

7.4.3.2 Quartiles, Deciles and Percentiles for Grouped data

When you have a very large raw data, grouping the data in a frequency distribution will make it easier.

1. Quartiles for grouped data

Activity 7.13

1. Do you know how to calculate the Quartiles for grouped data?
2. Explain steps to calculate quartiles for grouped data by giving your own examples.

Just like quartiles for ungrouped data, when the observation is arranged in increasing order then the values that divide the whole data into four (4) equal parts are called quartiles for grouped frequency distributions. These values are denoted by Q_1, Q_2 and Q_3. It is to be noted that 25% of the data falls below Q_1, 50% of the data falls below Q_2 and 75% of the data falls below Q_3.

Example 6

Find the quartiles of the following grouped data

Score	41 – 50	51– 60	61– 70	71– 80	81– 90	91– 100	101– 110
f	17	29	20	12	9	25	8

Solution

Table 7.13

Score	f	cf
41 – 50	17	17
51 – 60	29	46
61 – 70	20	66
71 – 80	12	78
81 – 90	9	87
91 – 100	25	112
101 – 110	8	120

Q_1 is the 30^{th}item in the distribution. By assuming that the items are equally spread through each class, we calculate the value of the required item by means of proportions.

Now, since the first 17 items lie in earlier classes, Q_1 is the $30 - 17 = 13^{th}$item in a class of 29 items. This means it lies $\left(\frac{13}{29}\right)^{th}$ of the way into the class. Since this class has an interval length of 10, this means that $\frac{13}{29} \times 10 = 4.48$ is to be added to the lower end. Now, the quartile class starts at 51, so that the first quartile is $51 + 4.48 = 55.48$.

Similarly, $Q_3 = 91 + \frac{90 - 87}{25} \times 10 = 91 + 1.2 = 92.2$.

Thus, let us summarize the above example in the following formula:

The k^{th} **quartile** for a grouped frequency distribution is:

$$Q_K \ (k^{th} \text{ quartile}) = B_L + \left(\frac{\frac{kn}{4} - cf_B}{fc}\right)i$$

$k = 1, 2, 3$ and

B_L = lower class limit of the k^{th} quartile class

n = number of observations

cf_B = the cumulative frequency before the k^{th} quartile class

f_c = the number of observations (frequency) in the k^{th} quartile class

i = the size of the class interval

Note

Steps to find quartiles for grouped data

1. Prepare a cumulative frequency distribution.
2. Find the class where the kth quartile belongs: the $\left(\frac{kn}{4}\right)^{th}$ item.
3. Use the formula above.

Example 7

Find Q_1, Q_2, Q_3 and IQR of the following distribution;

Table 7.14

Score	f	cf
141 – 150	21	21
151 – 160	19	40
161 – 170	15	55
171 – 180	17	72
181 – 190	8	80
191 – 200	5	85
201 – 210	4	89
211 – 220	3	92
221 – 230	2	94
231 – 240	2	96
241 – 250	4	100

Solution

n = 100

Q_1 is $\left(\frac{100}{4}\right)^{th}$ item; that is, 25th item which falls in the 2nd class. cf_B = 21, f_c = 19 and i = 10

$$Q_1 = B_L + \left(\frac{\frac{1n}{4} - cf_B}{fc}\right)i$$

$$= 151 + \left(\frac{\frac{1 \times 100}{4} - 21}{19}\right)10$$

$= 151 + \left(\frac{25-21}{19}\right)10$

$= 151 + \left(\frac{4}{19}\right)10$

$= 151 + 2.11$

$= 153.11$

Q_2 is $\left(\frac{2 \times 100}{4}\right)^{th}$ item; that is, 50th item, it is found in the 3nd class. $cf_B = 40$, $f_c = 15$ and $i = 10$

$Q_2 = B_L + \left(\frac{\frac{2n}{4} - cf_B}{fc}\right)i$

$= 161 + \left(\frac{\frac{2 \times 100}{4} - 40}{15}\right)10$

$= 161 + \left(\frac{50-40}{15}\right)10$

$= 161 + \left(\frac{10}{15}\right)10 = 161 + 6.7 = 167.7$

Q_3 is $\left(\frac{3 \times 100}{4}\right)^{th}$ item; that is, 75th item, and Q_3 found in the 5th class. $cf_B = 72$, $f_c = 8$ and $i = 10$

$Q_3 = B_L + \left(\frac{\frac{3 \times n}{4} - cf_B}{fc}\right)i$

$= 181 + \left(\frac{\frac{3 \times 100}{4} - 72}{8}\right)10$

$= 181 + \left(\frac{75-72}{8}\right)10$

$= 181 + \left(\frac{3}{8}\right)10 = 181 + 3.7 = 184.7$

$IQR = Q_3 - Q_1 = 184.7 - 153.11 = 31.59$

Note

Q_2 = median i.e., the 2nd quartile is the same as the median.

Exercise 7.13

1. The following table gives the amount of time (in minutes) spent on the internet each evening by a group of 56 students. Let t be the time.

Time spent on internet	$10 \leq t < 15$	$15 \leq t < 20$	$20 \leq t < 25$	$25 \leq t < 30$	$30 \leq t < 35$
No of students	9	10	18	10	9

Find Q_1, Q_2, and Q_3

2. Find Q_1, Q_2 and Q_3 of the following data. It is a distribution of marks obtained in a mathematics exam (out of 50).

Score	30–32	33–35	36–38	39– 41	42–44	45– 47
Number of students	12	18	16	12	8	6

   a. From the above data, if students in the top 25% are to be awarded a certificate, what is the minimum mark for a certificate?

   b. If students whose scores are in the bottom 25% of the marks are considered as failures, then what is the maximum failing mark?

2. Deciles for grouped data

Activity 7.14

1. Do you know how to calculate the deciles for grouped data?
2. Explain steps and formulas to calculate deciles for grouped data by giving your own examples.

Just like deciles for ungrouped data, when the observation is arranged in increasing order, then the values that divide the whole data into ten (10) equal parts are called deciles for grouped frequency distributions. These values are denoted by D_1, D_2, ...,

D_9. It is to be noted that 10% of the data falls below D_1, 20% of the data falls below D_2, ..., and 90% of the data falls below D_9.

The **j^{th} decile** for grouped frequency distributions is calculated in a similar way as follows;

$$D_j(jth\ decile) = B_L + \left(\frac{\frac{jn}{10} - cf_B}{fc}\right)i$$

$j = 1, 2, 3, \ldots, 9$ and

B_L = lower class limit of the j^{th} decile class.

n = number of observations.

cf_B = the cumulative frequency before the j^{th} decile class.

f_c = the number of observations (frequency) in the j^{th} decile class, and

i = the size of the class interval

Note

Steps to find deciles for grouped data

1. Prepare a cumulative frequency distribution.
2. Find the class where the j^{th} decile belongs: the $\left(\frac{jn}{10}\right)^{th}$ item
3. Use the formula above

Example 8

Find D_6 and D_8 of the following data.

Table 7.15

Score	f	cf
141 – 150	21	21
151 –160	19	40
161 – 170	15	55
171 – 180	17	72
181 – 190	8	80
191 – 200	5	85
201 – 210	4	89
211 – 220	3	92
221 – 230	2	94
231 – 240	2	96
241 – 250	4	100

Solution

i) D_6 is $\left(\frac{6\times100}{10}\right)^{th}$ item $= (60)^{th}$ item, it is found in the 4th class. $cf_B = 55$, $f_c = 17$ and $i = 10$

Thus,

$$D_6 = B_L + \left(\frac{\frac{6n}{10} - cf_B}{fc}\right)i$$

$$= 171 + \left(\frac{\frac{6\times100}{10} - 55}{17}\right)10$$

$$= 171 + \left(\frac{60 - 55}{17}\right)10$$

$$= 171 + \left(\frac{5}{17}\right)10 = 171 + 2.94 = 173.94$$

ii) D_8 is $\left(\frac{8\times100}{10}\right)^{th}$ item $= (80)^{th}$ item, it is found in the 5th class. $cf_B = 72$, $f_c = 8$ and $i = 10$

Thus, $D_8 = B_L + \left(\frac{\frac{8n}{10} - cf_B}{fc}\right)i$

$$= 181 + \left(\frac{\frac{8\times100}{10} - 72}{8}\right)10$$

$$= 181 + \left(\frac{80 - 72}{8}\right)10$$

$$= 181 + \left(\frac{8}{8}\right)10 = 181 + 10 = 191$$

Exercise 7.14

1. The following Table gives the amount of time (in minutes) spent on the internet each evening by a group of 41 students. Let t be the time.

   Find D_3, D_5, and D_7.

Time spent on internet	$10 \le t < 15$	$15 \le t < 20$	$20 \le t < 25$	$25 \le t < 30$	$30 \le t < 35$
No of students	3	9	10	12	7

2. The following Table shows the frequency distribution of monthly income of workers in a factory. Let x be the income in thousands.

Find, D_2, D_5 and D_8

Income (in thousands)	Number of workers
$1 \leq x < 5$	10
$5 \leq x < 9$	12
$9 \leq x < 13$	8
$13 \leq x < 17$	7
$17 \leq x < 21$	5
$21 \leq x < 25$	8
$25 \leq x < 29$	4
$29 \leq x < 33$	6

3. Percentiles for grouped data

Activity 7.15

1. Do you know how to calculate the percentiles for grouped data?
2. Explain steps and formulas to calculate percentiles for grouped data by giving your own examples.

Just like percentiles for grouped data, when the observation is arranged in increasing order then the values that divide the whole data into hundred (100) equal parts are called percentiles for grouped frequency distribution. These values are denoted by P_1, P_2, ..., P_{99}. It is to be noted that 1% of the data falls below P_1, 2% of the data falls below P_2, ..., and 99% of the data falls below P_{99}.

$$P_j(jth \text{ percentile}) = B_L + \left(\frac{\frac{jn}{100} - cf_B}{fc}\right)i$$

Where $j = 1, 2, 3, \ldots, 99$ and

B_L = lower class limit of the j^{th} percentile class;

n = number of observations;

cf_B = the cumulative frequency before the j^{th} percentile class;

f_c = the number of observations (frequency) in the j^{th} *percentile* class; and

i = the size of the class interval.

Note

Steps to find percentile for grouped data

1. Prepare a cumulative frequency distribution.
2. Find the class where the jth percentile belongs: the $\left(\frac{jn}{100}\right)^{th}$ item
3. Use the formula $P_j(jth \text{ percentile}) = B_L + \left(\frac{\frac{jn}{100} - cf_B}{fc}\right)i$

Example 9

Find P_{35}, and P_{91} of the following data.

Table 7.16

Score	f	Cumulative frequency (Cf)
141 – 150	21	21
151 – 160	19	40
161 – 170	15	55
171 – 180	17	72
181 – 190	8	80
191 – 200	5	85
201 – 210	4	89
211 – 220	3	92
221 – 230	2	94
231 – 240	2	96
241 – 250	4	100

Solution

i) P_{35} is $\left(\frac{35 \times 100}{100}\right)^{th}$ item $= (35)^{th}$ item, and it is found in the 2nd class.

$cf_B = 21$, $f_c = 19$ and $i = 10$

Thus, $p_{35} = B_L + \left(\frac{\frac{35n}{100} - cf_B}{fc}\right)i$

$$= 151 + \left(\frac{\frac{35 \times 100}{100} - 21}{19}\right)10 = 151 + \left(\frac{35 - 21}{19}\right)10 = 151 + \left(\frac{14}{19}\right)10$$

$$= 151 + 7.37 = 158.37$$

This means, 35% of the data falls below 158.37.

ii). P_{91} is $\left(\frac{91\times100}{100}\right)^{th}$ item $= (91)^{th}$ item, it is found in the 8th class. $cf_B = 89$, $f_c = 3$ and i = 10

Thus, $p_{91} = B_L + \left(\frac{\frac{91n}{100} - cf_B}{fc}\right)i$

$$= 211 + \left(\frac{\frac{91\times100}{100} - 89}{3}\right)10 = 211 + \left(\frac{91-89}{3}\right)10$$

$$= 211 + \left(\frac{2}{3}\right)10$$

$$= 217.67$$

This means, 91% of the data falls below 217.67.

Exercise 7.15

1. The following Table shows mathematics test scores of 100 students.

   Find P_{25}, P_{61}, P_{80}

Scores	21 – 30	31– 40	41 –50	51 – 60	61–70	71 – 80	81– 90	91– 100
No of Students	4	18	9	13	14	10	24	8

2. The following Table gives the amount of time spent (in minutes) spend studying for a mathematics test by a group of 41 students. Let t be the time.

   Find P_{33}, P_{50} and P_{75}.

Time spent on studying(min)	$30 \le t < 35$	$35 \le t < 40$	$40 \le t < 45$	$45 \le t < 50$	$50 \le t < 55$
No of students	3	12	20	2	4

3. The following Table gives a frequency distribution of weight (in kilogram) of 57 children at a day care centre. Let w be the weight.

Weight(kg)	No children
$10 \leq w < 20$	5
$20 \leq w < 30$	19
$30 \leq w < 40$	10
$40 \leq w < 50$	13
$50 \leq w < 60$	4
$60 \leq w < 70$	4
$70 \leq w < 80$	2

Calculate

a. The maximum weight of lower 30% of the children

b. The minimum weight of upper 30% of the children.

7.4.4 The Mode for a Grouped Data

Activity 7.16

1. Do you know how to calculate the mode of ungrouped data as you have seen in grade 9?
2. Consider the following sets of data and discuss the most frequently occurring value in the data.
   a. 5, 3, 5, 8 and 9 b. 8, 9, 9, 7, 8, 2 and 5

From Activity 7.16, you have observed the most frequently occurring value for a data.

Definition 7.11

The mode is the value that occurs most often in a data. Mode is denoted by $(\hat{X})$

A data set that has only one value that occurs with the greatest frequency is said to be **unimodal.**

If a data set has two values that occur with the same greatest frequency, both values are considered to be the mode and the data set is said to be **bimodal.**

If a data set has more than two values that occur with the same greatest frequency, each value is used as the mode, and the data set is said to be **multimodal.**

When no data value occurs more than once, the data set is said to have ***no mode***. A data set can have more than one mode or no mode at all. These situations will be shown in some of the examples that follow.

Example 1

Find the mode of the bonuses of students for one semester. The bonuses are:

a. 18, 14, 15, 10, 11, 3, 10, 12, 10 marks.

b. 6, 10, 9, 6, 10, 4 marks.

c. 5, 6, 7, 11, 12 marks.

d.

x	7	12	15	17	19	23
f	6	4	6	5	6	5

Solution

a. In this observation, the most frequent value is 10. Therefore, the mode is $\widehat{X}$ = 10 since it appears three times. This data has only one mode and is called unimodal.

b. Here both 6 and 10 appear twice but the rest appear only once. Hence the modes are 6 and 10. This distribution has two modes. Such distributions are said to be **bimodal**.

c. Every number appeared only once. Hence there is no mode for this distribution.

d. Three values 7, 15 and 19 all appear 6 times. Hence the modes are 7, 15 and 19. Distributions that have more than two modes are called **multimodal.**

Activity 7.17

1. Do you know how to calculate the mode of grouped data?
2. Explain steps to find mode from grouped data and calculate mode of grouped data by giving your own examples.

For Activity 7.17, just like ungrouped data, you have to see the frequency of each number or variable, and the variable that has the greater frequency is the mode. This changes when you work with grouped data, because when you work with grouped data there are no numbers to count how many times each number is repeated. Instead the data is organized in intervals and the way the mode is found changes.

Now, before you find any mode(s) that might exist, check the class interval of all classes should be equal (uniform class interval).

You can use the following steps to compute the mode of grouped or continuous frequency distribution with equal class intervals:

Step 1: Prepare the frequency distribution table in such a way that its first column consists of the observations and the second column the respective frequency.

Step 2: Determine the class of maximum frequency by inspection. This class is called the modal class.

Step 3: To calculate mode, use the following formula:

$$\text{Mode} = \widehat{X} = B_L + \left(\frac{d_1}{d_1 + d_2}\right)i$$

Where,

B_L = lower class limit of the modal class.

d_1= the difference between the frequency of the modal class(f_{mo}) and the frequency of the preceding class (pre-modal class) = $(f_{mo}) - f_1$

d_2= the difference between the frequency of the modal class and the frequency of the subsequent class (next class) = $(f_{mo}) - f_2$

i = size of the class interval.

Example 2

Following is the distribution of the lifespans, in days, of the bees in a colony. Calculate the mode of the distribution. **Table 7.17**

Lifespan(days)	Number of bees
16 – 35	10
36 – 55	12
56 – 75	19
76 – 95	31
96 – 115	33
116 – 135	7
136 – 155	5

Solution

96 – 115 is the modal class since it is a class with the highest frequency.

$B_L = 96$ $d_1 = 33 - 31 = 2$ $d_2 = 33 - 7 = 26$ $i = 115 - 96 = 20$

$$\hat{X} = B_L + \left(\frac{d_1}{d_1 + d_2}\right)i$$

$$= 96 + \left(\frac{2}{28}\right)20 = 96 + 1.36 = 97.43$$

Exercise 7.16

1. The following is a distribution of the number of drivers who violated traffic-safety in Addis Ababa. Find mode of the distribution.

Violated traffic safety	1–3	4–7	8–11	12–15	16 –19	20 – 23
No of drivers	6	10	20	22	6	2

2. The following represents the number of minutes that students spent studying for a math test. Find the mode of distribution. Let t be the time.

Studying time (in minutes)	$0 \leq t < 10$	$10 \leq t < 20$	$20 \leq t < 30$	$30 \leq t < 40$	$40 \leq t < 50$
No of students	2	10	6	4	3

7.5 Real-life Application of Statistics

There are numerous applications of statistics in different professions. For instance, statistics can be applied in clinical trial and design, corporate sectors, weather forecasting, sports and financial markets. Moreover, it can be used in a variety of domains including business, industry, agriculture, computer science, science, health science and other disciplines. Now, you will discuss further applications involving statistics.

Example

The number of people who paid a tax were registered for 80 days and the data are as follows:

74	96	101	83	105	83	88	84	89	74
100	101	93	89	89	94	76	94	98	92
110	82	81	109	75	70	83	74	104	64
85	73	74	84	111	94	98	98	95	102
64	64	91	115	76	63	61	100	84	94
70	89	75	92	83	96	84	73	96	109
65	100	95	60	98	98	73	64	76	81
84	113	99	78	73	73	94	90	89	75

a. Construct a grouped frequency distribution with 10 classes.

b. Find the mean, mode and median from the grouped frequency distribution.

c. Find Q_2, D_5, D_7, P_{50}, and P_{75}.

Solution

a. **Table 7.19**

Class interval	No of days(f)	m_i	fm_i	Cumulative frequency (Cf)
60 – 65	8	62.5	500	8
66 – 71	2	68.5	137	10
72 – 77	15	74.5	1117.5	25
78 – 83	8	80.5	644	33
84 – 89	12	86.5	1038	45
90 – 95	12	92.5	1110	57
96 – 101	14	98.5	1379	71
102 –107	3	104.5	313.5	74
108 – 113	5	110.5	552.5	79
114 – 119	1	116.5	116.5	80

b.

$$\bar{x} = \frac{f_1m_1 + f_2m_2 + f_3m_3 + f_4m_4 + f_5m_5 + f_6m_6 + f_7m_7 + f_8m_8 + f_9m_9 + f_{10}m_{10}}{f_1 + f_2 + f_3 + f_4 + f_5 + f_6 + f_7 + f_8 + f_9 + f_{10}} = \frac{6908}{80} = 86.35$$

This means, the mean number of people who paid a tax is 86.35.

The median class is that class containing the $\left(\frac{80}{2}\right)$ item = 40^{th} item.

It is in the 5^{th} class.

Therefore, the median class is **84 – 89.**

Thus, B_L = **84**, $\frac{n}{2} = 40$, $fc = 12$, i = 6, $cf_B = 33$, Therefore,

$$\tilde{X} = B_L + \left(\frac{\frac{n}{2} - cf_B}{fc}\right)i$$

$$= 84 + \left(\frac{40 - 33}{12}\right)6 = 84 + 3.5 \quad = 87.5$$

Thus, the median of the number of the people who paid a tax is 87.5.

72 – 77 is the modal class, since it is a class with the highest frequency.

$B_L = 72$ $\quad d_1 = 15 - 2 = 13$ $\quad d_2 = 15 - 8 = 7$ and $i = 6$

$$\hat{X} = B_L + \left(\frac{d_1}{d_1 + d_2}\right)i$$

$$= 72 + \left(\frac{13}{20}\right)6 = 75.9$$

Hence, the mode of the number of people who paid a tax is 75.9

c. $Q_2 = D_5 = P_{50} = 87.5$

This means, in 50% of the days, the number of people who paid a Tax fall below 87.

The 5th decile and the 50th percentile is also the median of the number of people who paid a tax a day. Looking at the data in the Table 7.19, the median can be calculated as 87.5 = median = $D_5 = P_{50}$. At this point, in half of the days, the number of the people who paid a tax lies above and below 87.5.

D_7 is $\left(\frac{7 \times 80}{10}\right)^{th}$ item, that is, 56th item which lies in the 6th class. $cf_B = 45$, $f_C = 12$ and i = 6

$$D_7 = B_L + \left(\frac{\frac{7n}{10} - cf_B}{fC}\right)i = 90 + \left(\frac{\frac{7 \times 80}{10} - 45}{12}\right)6 = 95.5$$

This means, in 70% of the days, the number of the people who paid a tax lies below 95.5.

P_{75} is $\left(\frac{75 \times 80}{100}\right)^{th}$ item, that is, 60th item which lies in the 7th class. $cf_B = 57$, $f_C = 14$ and i = 6

$$P_{75} = B_L + \left(\frac{\frac{75n}{100} - cf_B}{fC}\right)i = 96 + \left(\frac{\frac{75 \times 80}{100} - 57}{14}\right)6 = 97.29$$

This means, in 75% of the days, the number of the people who paid a tax lies below 97.29.

Exercise 7.17

1. The lifespans, in days, of the bees in a colony are recorded.

Lifespan(days)	1–20	21 – 40	41 – 60	61 – 80	81 – 100
Number of bees	6	11	22	32	29

   a. Calculate the mean for lifespan of the bees.

   b. State the modal number of days.

   c. Find Q_2, D_6, P_{25}

2. The table below shows the frequency distribution of the mass of 200 rebars that have to be transported from the warehouse to the construction site. Let m be the mass of the rebar.

Mass(kg)	Number of rebars
$10 \leq m < 20$	32
$20 \leq m < 30$	38
$30 \leq m < 40$	64
$40 \leq m < 50$	35
$50 \leq m < 60$	22
$60 \leq m < 70$	9

Calculate the mean, median and mode mass of the steel bars.

Problem Solving

1. A house building company wanting to find out what type of houses they should build most often in a region carried out a survey in that region to find out the number of people in a family. Will they use mean, median or mode to decide what type of houses should be built most?
2. A car battery factory wants to give a warranty to their customers as to the lifetime of their batteries. i.e., they want to tell the customer if you have a problem with the battery in the next months, we will replace your battery with new one. They checked the lifetime of 100 batteries. Will they use mean, median or mode to decide on the number of months to guarantee their batteries?
3. In a mathematics test, Tolosa got 62%. For the whole class, the mean mark was 64% and the median mark was 59%. Which average tells Tolosa whether he is in the top half or the bottom half of the class?
4. The mean age of three people is 22 and their median age is 20. The range of their ages is 16. How old is each person?

Summary

1. Quantitative data can be quantified in terms of number value.
2. Qualitative data cannot be quantified as number value and expressed as quality.
3. The characteristics of a unit being observed that may assume more than one of a set of values to which a numerical measure can be assigned is variable.
4. Discrete data is a numerical type of data that includes whole, concrete numbers with specific and fixed data values determined by counting.
5. Continuous data is one which takes any fractional point along a specified interval of values.
6. Frequency means the number of times a certain value of a variable is repeated in the given data.
7. A grouped frequency distribution is constructed to summarize a large sample of data.
8. The appropriate class width is given by

$$\text{Class width} = \frac{\text{Largest Value in ungrouped data} - \text{Smallest Value in ungrouped data}}{\text{number of classes required}}.$$

9. A measure of location is a single value that is used to represent a mass of data. The common measures of location are mean, median, mode, quartiles, deciles and percentiles.

$$\text{Mean} = \bar{X} = \frac{x_1 + x_2 + x_3 + \ldots + x_n}{n} \text{ for raw data.}$$

$$\bar{x} = \frac{f_1x_1 + f_2x_2 + f_3x_3 + \ldots + f_nx_n}{f_1 + f_2 + f_3 + \ldots + f_n} \text{ for ungrouped data.}$$

$$\bar{x} = \frac{f_1m_1 + f_2m_2 + f_3m_3 + \ldots + f_im_i}{f_1 + f_2 + f_3 + \ldots + f_i} \text{ for grouped data.}$$

10. Mode is the value with the highest frequency.
11. If a distribution has a single mode, it is "unimodal".
    If it has two modes, it is "bimodal".
    If it has more than two modes, it is called "multimodal".

12. Median of ungrouped data is given by

$\widetilde{X} = \left(\frac{n+1}{2}\right)^{th}$ item, if n is odd. $\widetilde{X} = \frac{\left(\frac{n}{2}\right)^{th}\text{ item} + \left(\frac{n}{2}+1\right)^{th}\text{ item}}{2}$, if n is even.

13. Median of grouped data is given by $\widetilde{X} = B_L + \left(\frac{\frac{n}{2} - cf_B}{fc}\right)i$

14. Quartiles for ungrouped frequency distributions are given by

i) $Q_k = \left(\frac{k(n+1)}{4}\right)^{th}$ item; if number of observations is odd.

ii) $Q_k = \left(\frac{\left(\frac{kn}{4}\right) + \left(\frac{kn}{4}+1\right)}{2}\right)^{th}$ item; if number of observations is even.

15. Decile for ungrouped frequency distributions are given by

i) $D_j = \left(\frac{j(n+1)}{10}\right)^{th}$ item; if number of observations is odd.

ii) $D_j = \left(\frac{\left(\frac{jn}{10}\right) + \left(\frac{jn}{10}+1\right)}{2}\right)^{th}$ item; if number of observations is even.

16. Percentile for ungrouped frequency distributions are given by

i) $P_j = \left(\frac{j(n+1)}{100}\right)^{th}$ item; if number of observations is odd.

ii) $P_j = \left(\frac{\left(\frac{jn}{100}\right) + \left(\frac{jn}{100}+1\right)}{2}\right)^{th}$ item; if number of observations is even.

17. Quartiles for grouped frequency distributions are given by

$Q_K = B_L + \left(\frac{\frac{kn}{4} - cf_B}{fk}\right)i.$

18. Decile for grouped frequency distributions are given by

$D_j = B_L + \left(\frac{\frac{jn}{10} - cf_B}{fc}\right)i.$

19. Percentile for grouped frequency distributions are given by

$P_j = B_L + \left(\frac{\frac{jn}{100} - cf_B}{fc}\right)i.$

20. The mode is given by

Mode $= \hat{X} = B_L + \left(\frac{d_1}{d_1 + d_2}\right)$ i, for grouped frequency distributions.

Review Exercise

1. Construct a grouped frequency distribution table and histogram for the following data using 6 classes:

61	83	92	87	74	60	71	57	61	69
85	40	85	59	80	52	56	81	77	79
78	82	43	64	67	48	81	68	37	43
65	85	49	69	61	54	56	69	68	78
74	59	76	65	69	49	57	38	73	81

2. Find mean, mode(s) and median of each of the following scores:

   a. 12, 13, 26, 15, 25, 38, 28, 27, 32

   b. 8, 7, 11, 13, 17, 9, 8, 7, 6, 11

   c.

Scores(x)	22	23	24	26	29	36
f	6	13	30	23	17	9

3. The following table gives the daily income of 100 workers of a factory: Let x be the daily income.

Daily income(birr)	Number of Workers
$200 \leq x < 220$	18
$220 \leq x < 240$	28
$240 \leq x < 260$	21
$260 \leq x < 280$	21
$280 \leq x < 300$	12

   Find the mean, median and mode of the above data.

4. In a small business 2 cleaners earn Birr 340 each, 6 persons handling the machinery earn Birr 600 each, the manager earns Birr 1500 and the director Birr 3500 per month. Which measure, mean, median or mode best represents these data?

5. The number of people who paid a tax were registered for 80 days and the data are as follows:

74	96	101	83	105	83	88	84	89	74
100	101	93	89	89	94	76	94	98	92
110	82	81	109	75	70	83	74	104	64
85	73	74	84	111	94	98	98	95	102
64	64	91	115	76	63	61	100	84	94
70	89	75	92	83	96	84	73	96	109
65	100	95	60	98	98	73	64	76	81
84	113	99	78	73	73	94	90	89	75

a. Construct a grouped frequency distribution, with 10 classes.

b. Find the mean, mode and median from the data.

c. Find Q_2, D_5, D_7, P_{50}, and P_{75}.

UNIT 8

PROBABILITY

Unit Outcomes

By the end of this unit, you will be able to:

- Distinguish certain and uncertain events.
- Know principles of counting.
- Explain the concept of probability.
- Understand Binomial Theorem.
- Calculate the probability of an event.
- Calculate the probability of a compound event.
- Apply facts and principles in computation of probability.
- Represent probabilities as fractions, decimals and percentages.
- Interpret probabilities as fractions, decimals and percentages.
- Represent the probability of an event as fraction or decimal between 0 and 1 or as a percentage.
- Use probability concept to solve real life problems.

Unit Contents

8.1 Introduction
8.2 Fundamental Principle of Counting
8.3 Permutations and Combinations
8.4 Binomial Theorem
8.5 Random Experiments and Their Outcomes
8.6 Events
8.7 Probability of an Event
8.8 Real-life Application of Probability
Summary
Review Exercise

- **combination**
- **dependent events**
- **exhaustive events**
- **fundamental counting principles**
- **independent events**
- **permutation**
- **probability of an event**

8.1 Introduction

From the time you awake until you go to bed, you make decisions regarding the possible events that are governed at least in part by chance. For example, should you carry an umbrella to school today? Should you accept that new home work? Thus, probability can be defined as the chance of an event occurring. Many people are familiar with probability from observing or playing games of chance, such as card games, slot machines, or lotteries. In addition to being used in games of chance, it is used in the fields of insurance, investments, and weather forecasting and in various other areas. Finally, probability is the basis of inferential statistics. For example, predictions are based on probability and hypotheses are tested by using probability.

Blaise Pascal (1623 – 1662) suffered the most appalling ill-health throughout his short life. He is best known with Fermat on probability. This followed correspondence with a gentleman gambler who was puzzled as to why he lost so much in betting on the basis of appearance of a throw of dice. Pascal's work on probability become of enormous importance and showed for the first time that absolute certainty is not a necessity in mathematics and science. He also studied physics, but his last years were spent in religious meditation and illness.

Source: Cambridge IGCSE Mathematics Fifth Edition.

Revision about Probability

You will revise some terms before you proceed to the next section as discussed in grade 9.

1. An **experiment** is an activity (measurement or observation) that generates well-defined results (Outcomes).
2. An **outcome (Sample point)** is the result of a single trial of an experiment.
3. A **sample space (S)** is a set that contains all possible outcomes of an experiment.
4. An **Event** is any subset of a sample space.

Example 1

If a fair die is rolled once the possible results are either 1, 2, 3, 4, 5 or 6, then from this:

a. Give the sample space.
b. Give the event of obtaining odd numbers.
c. Give the event of obtaining even numbers.
d. Give the event of obtaining number seven.

Solution

a. sample space(S) is {1, 2, 3, 4, 5, 6}.

b. the event of obtaining odd numbers is {1, 3, 5}.

c. the event of obtaining even numbers is {2, 4, 6}.

d. the event of obtaining number seven is $\emptyset$; which is an impossible event.

Example 2

When a "fair" coin is tossed, the possible results are either head (H) or tail (T). Consider an experiment of tossing a fair coin three times.

a. What are the possible outcomes?

b. Give the sample space.

c. Give the event of H appearing on the second throw.

d. Give the event of at least one T appearing.

Solution

a. HHH, HHT, HTH, THH, TTH, THT, HTT, TTT.

b. S = {HHH, HHT, HTH, THH, TTH, THT, HTT, TTT}.

c. A = {HHH, HHT, THH, THT}.

d. B = {HHT, HTH, THH, TTH, THT, HTT, TTT}.

Note

Events which have the same chance of occurring are **equally likely events.**

Exercise 8.1

1. A fair die is rolled once:

   a. Give the event of obtaining a number equal to or greater than 4.

   b. Give the event of obtaining number zero.

2. A fair coin is tossed twice:
   a. What are possible outcomes?
   b. Give the sample space.
   c. Give the event of at least one H appearing.

8.2 Fundamental Principle of Counting

Activity 8.1

1. Consider that six students leave the class; re-enter one by one and take a seat. In how many ways can you seat these students?
2. Consider students observing car plate numbers. The first digit is from 26 English alphabet, the other five digits are a number from 0 to 9. Discuss in how many different ways a car plate number is written.

From Activity 8.1, in order to calculate probabilities, you have to know the number of elements of an event and the number of elements of the sample space. Thus, if the experiment is leaving and re-entering of 6 students into the class, what is the total number of possible outcomes? If an event E is defined by number of ways the students can seat, similarly if the experiment is car plate number, what is the total number of possible outcomes? If an event E is defined by "6 English alphabet and 6-digit number", then how do you find n(E)? From this, you can observe that counting plays a very important role in finding probabilities of events.

Thus, when the number of possible outcomes is very large, it will be difficult to find the number of possible outcomes by listing. One may use fundamental principle of counting which helps us to find the number of ways of occurrence (selections) of events in a given order such as the multiplication and the addition principle.

8.2.1 Multiplication Principle

Consider the following example before stating the multiplication principle.

Example 1

Consider that Girma goes to the nearest snack to have breakfast. He can take tea, coffee, or milk with bread, cake, or sandwich. How many possibilities does he have?

Solution

As shown below, use a tree diagram to show the number of events.

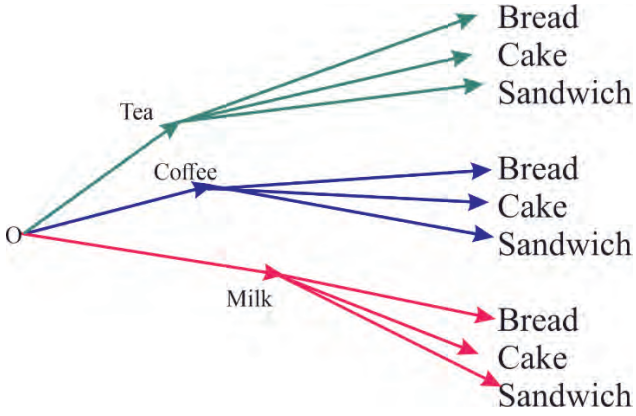


Figure 8.1

There are $3 \times 3 = 9$ possibilities as shown in Figure 8.1

From these possible choices, Girma can take tea with bread, tea with cake, tea with sandwich or coffee with bread, coffee with cake, coffee with sandwich or milk with bread, milk with cake, milk with sandwich.

The above example, therefore, illustrates the multiplication principle of counting.

Multiplication principle

If an event can occur in **m** different ways, and for every such choice another event can occur in **n** different ways, then both the events can occur in the given order in **m** × **n** different ways. That is, the number of ways in which a series of successive things can occur is found by multiplying the number of ways each thing can occur.

Example 2

Suppose there are 6 seats arranged in a row. In how many different ways can six people be seated on them?

Solution

The first man has 6 choices, the 2^{nd} man has 5 choices, the 3^{rd} man has 4 choices, the 4^{th} has 3 choices, the 5^{th} man has 2 choices and the sixth man only one choice.

Therefore, the total number of possible seating arrangements is

$6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$.

Example 3

A coin is tossed and a die is rolled. Find the number of outcomes for the sequence of events.

Solution

For a coin, 2 possible outcomes: H and T.

For a die, 6 possible outcomes :1, 2, 3, 4, 5 and 6.

Now when coin is tossed and a die is rolled simultaneously, $(2 \times 6) = 12$ possible outcomes.

Exercise 8.2

1. Abebech has got only clean clothes of 3 t-shirts and 5 pairs of jeans. How many different combinations can Abebech choose?
2. Suppose that a man has 5 coats, 10 shirts and 8 different trousers. In how many different ways can a man dress?
3. A paint manufacturer wishes to manufacture several different paints. The categories include:

Table 8.1

Category	Painting materials
Colour	Red, blue, white, black, green, brown, yellow
Type	Latex, oil
Texture	Flat, semigloss, high gloss
Use	Outdoor, indoor

How many different kinds of paint can be made if you can select one colour, one type, one texture, and one use?

4. There are 16 microcomputers in a computer center. Each microcomputer has 21 ports. How many different ports to a microcomputer in the center are there?

8.2.2 Addition Principle

Before stating the addition principle, let us see the following examples:

Example 1

Suppose that you want to buy a computer from one of two makes A_1 and A_2. Suppose also that those makes have 12 and 18 different models, respectively. How many models are there altogether to choose from?

Solution

Since you can choose one of 12 models of make A_1 or one of 18 of A_2, there are altogether 12 + 18 = 30 models to choose from a given model.

Choosing one from given models of the either make is an **event** and the choices for either event is the **outcomes** of the event. Thus, the event "selecting one from make A_1", for example, has 12 outcomes. This shows the Addition Principle of Counting.

Addition principle

If A and B are mutually exclusive events (i.e., the occurrence of one excludes that of the other) where an event A can occur in **m** ways and another event B can happen in **n** ways, respectively, then the total number of outcomes for the event A or B is **n** + **m** ways.

Note

Two events are said to be mutually exclusive, if both cannot occur simultaneously.

Example 2

If 4 red and 6 green marbles are placed in a bag. How many marbles are there to choose from?

Solution

Since the red and green marbles cannot be chosen together or at the same time, the number of outcomes is added together. So, there are 4 + 6 = 10 marbles to choose from.

Example 3

A question paper has two parts where one part contains 10 questions and the other 6 questions. If a student has to choose only one question from either part, in how many ways can the student do it?

Solution

The student can choose one question in 10 + 6 = 16 ways.

Example 4

Suppose that either a member of the Mathematics faculty or a student who is Mathematics major is chosen as a representative to a university committee. How many different choices are there for this representative if there are 37 members of the Mathematics faculty and 83 Mathematics majors and no one is both a faculty member and a student?

Solution

There are 37 ways to choose a member of the Mathematics faculty and there are 83 ways to choose a student who is Mathematics major. Choosing a member of the Mathematics faculty is never the same as choosing a student who is Mathematics major because no one is both a faculty member and a student. By addition principle it follows that there are $37 + 83 = 120$ possible ways to pick this representative.

Example 5

Tigist is selecting an outfit. She has 5 different pairs of pants, 4 different skirts and 12 different T-shirts. How many outfits are possible?

Solution

Tigist cannot wear a pair of pants and a skirt at the same time. So, she can wear either "pants and T-shirt" or "skirt and T-shirt."

When she wears a pair of pants and a T-shirt, possible outfits are

$$5 \times 12 = 60$$

When she wears a skirt and a T-shirt, possible outfits are

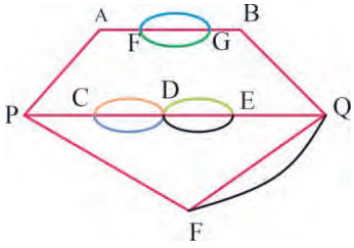
$$4 \times 12 = 48$$

Therefore, total possible outfits are

$$5 \times 12 + 4 \times 12 = (5 + 4) \times 12 = 108.$$

Exercise 8.3

1. Suppose there are 5 chicken dishes and 8 beef dishes. How many selections does a customer have?
2. There are 5 vegetarian entrée options and 7 meat entrée options on a dinner menu. What is the total number of entrée options?
3. Consider the following road system from P to Q.
   In how many different ways could one travel from P to Q?



4. Aster will draw one card from a standard deck of playing cards. How many ways can she choose:
   a. an even number?
   b. a king or a queen?
   c. a heart, a diamond or a club?
   d. a king or a black?

8.3 Permutations and Combinations

Factorial

Activity 8.2

1. What knowledge do you have about factorial of a number?
2. Compute a. 5! b. 9!

Definition 8.1

Factorial of a number denoted by n! is the product of all positive integers less than or equal to n:

$n! = n \times (n - 1) \times (n - 2) \times (n-3) \ldots 3 \times 2 \times 1$ and

$0! = 1 \quad 1! = 1$

Example 1

Compute

a. 4! b. 7! c. $\frac{10!}{8!}$ d. $\frac{45!}{42!}$ e. $\frac{8!}{5!\,2!}$

Solution

According to the definition 7.1 above:

a. $4! = 4 \times 3 \times 2 \times 1 = 24.$

b. $7! = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 5040.$

c. $\frac{10!}{8!} = \frac{10 \times 9 \times 8!}{8!} = 90.$

d. $\frac{45!}{42!} = \frac{45 \times 44 \times 43 \times 42!}{42!} = 85, 140.$

e. $\frac{8!}{5!\,2!} = \frac{8 \times 7 \times 6 \times 5!}{5!\,2!} = 168.$

8.3.1 Permutation

Permutations (1)

Activity 8.3

1. Compute the following:

   a. $\frac{7!}{(7-5)!}$ b. $\frac{11!}{(11-3)!}$ c. $\frac{9!}{(9-2)!}$

2. Consider a business owner who has a choice of 5 locations in which to establish her business. She decides to rank each location according to certain criteria, such as price of the store and parking facilities. Discuss the number of different ways she can rank the 5 locations.

From Activity 8.3, you can define permutation formally as follows.

Definition 8.2

The arrangement of n objects in a specific order using r objects at a time is called a permutation of n objects taking r objects at a time. It is written as P (n, r) or nPr, and the formula is

$$nPr = \frac{n!}{(n-r)!}, \text{ where } 0 < r \leq n.$$

This formula can be derived this way. Suppose there are five balls of different colours. You want to pick up three of them and line them up. How many different ways will you have to arrange the three balls? This answer is denoted as ${}_5P_3$. Let's calculate this. You pick up the first ball. You will have 5 choices. Then for second ball, you will have 4 balls to choose from. Similarly, for the third ball, you will have 3 remaining balls to choose from. Then the number of all possible ways is calculated as

$${}_5P_3 = 5 \times 4 \times 3 = 60$$

This is in fact the same as

$${}_5P_3 = \frac{5 \times 4 \times 3 \times 2 \times 1}{2 \times 1} = \frac{5!}{2!} = \frac{5!}{(5-3)!}$$

This was the case where n = 5 and r = 3. From this you can generalize that

$$nPr = \frac{n!}{(n-r)!}.$$

Note

The number of permutations of a set of n objects taken all together is denoted by P (n, n) or nPn and is equal to n!.

Thus, P (n , n) = n!

Example 2

Compute the following permutation:

a. P (7, 4) b. P (9, 6) c. P (14, 4)

Solution

a. $P(7, 4) = \frac{7!}{(7-4)!} = 840.$

b. $P(9, 6) = \frac{9!}{(9-6)!} = 60{,}480.$

c. $P(14, 4) = \frac{14!}{(14-4)!} = 24{,}024.$

Exercise 8.4

1. Find the factorial of each of the following numbers:
   a. 5! b. 8! c. 13!
2. Compute the following permutation:
   a. P (5, 1) b. P (5, 2) c. P (5, 3) d. P (5, 5)

Permutation (2)

Example 3

Five students are contesting an election for 5 places in the executive committee of environmental protection club in their school. In how many ways can their names be listed on the ballot paper?

Solution

You have to arrange 5 names in 5 places. Therefore, the number of ways of listing their names on the ballot paper, $P(5, 5) = 5! = 120$.

Example 4

Suppose you have letters A, B, C, and D:

a) How many permutations are there taking all the four?

b) How many permutations are there taking two letters at a time?

Solution

a. Here, n = 4, i.e., there are four distinct objects.

⇒ There are 4! = 24 permutations.

b. Here, n = 4, r = 2

⇒There are ${}_4P_2 = \frac{4!}{(4-2)!}$ = 12 permutations.

Exercise 8.5

1. A mathematics debating team consists of 4 speakers:
   a. In how many ways can all 4 speakers be arranged in a row for a photo?
   b. How many ways can the leader and vice - leader be chosen?
2. 3 different statistics books, 5 different mathematics books, and 3 different physics books are arranged on a shelf. How many different arrangements are possible if;
   a. The books in each particular subject must all stand together
   b. Only the statistics books must stand together
3. R, S, T, U, V and W are six students. In how many ways can they be seated in a row if:
   a. there are no restrictions on the seating.
   b. R and S must sit beside each other.
   c. U, V and W must sit beside each other.
   d. R and W must sit at the end of each row.

Permutations with Repetition

When you arrange four letters, A, B, C, and D, you will have ${}_4P_4 = 4 \times 3 \times 2 \times 1 =$ 24 different ways of arrangements. However, if you have four letters of A, B, C and C, how many different ways of arrangements will you have?

Suppose you can differentiate two C's as C_1 and C_2. Then, the total number of ways of arrangements is 24. In this calculation, ABC_1C_2 and ABC_2C_1 are different

arrangements, but in fact you cannot differentiate those two permutations: ABCC and ABCC are the same arrangement. Similarly, BC_1AC_2 and BC_2AC_1 are the same arrangement, BCAC. Thus, when there are two C's, the total number of permutations, 24, indicates duplications. For instance, you have counted ABCC twice as ABC_1C_2 and ABC_2C_1. Since C_1 and C_2 can be arranged in 2! = 2 ways (C_1C_2 and C_2C_1), the total number of 24 should be divided by 2 to obtain the correct total number of arrangements, which is 12.

When there are some same objects included in the objects, the total number of arrangements of the objects should be counted avoiding duplications.

Definition 8.3

The number of arrangements of n objects in which n_1 are alike, n_2 are alike, …, n_r alike objects of the r^{th} type is $\frac{n!}{n_1! \times n_2! \times \ldots \times n_r!}$, where $n_1 + n_2 + \ldots + n_r = n$ and $0 < r \leq n$.

Example 5

How many different permutations can be made from the letters in the word "ADDIS" ?

Solution

Here n = 5 of which there are two D's.

So, 1 is A, 2 are D, 1 is I and 1 is S.

$\Rightarrow n_1 = 1,\ n_2 = 2,\ n_3 = 1,\ n_4 = 1$

Using definition 8.3, there are

$$\frac{5!}{n_1! \times n_2! \times n_3 \times n_4!} = \frac{5!}{1! \times 2! \times 1! \times 1!} = \frac{5 \times 4 \times 3 \times 2 \times 1}{2 \times 1} = 60 \text{ permutations.}$$

Example 6

How many different permutations can be made from the letters in the word "ADDIS" if the rearrange letters which start from S?

Solution

If you fix the first letter to be S, then you have four remaining letters, A, D, D, and I, to arrange. You have to think of the permutations of them. Since there are 2 D's, using definition 8.3, you have

$\frac{4!}{1! \times 2! \times 1!} = \frac{4 \times 3 \times 2 \times 1}{2 \times 1} = 12$ permutations:

They are:

SADDI	SDADI	SIADD
SADID	SDAID	SIDAD
SAIDD	SDDAI	SIDDA
	SDDIA	
	SDIAD	
	SDIDA	

Exercise 8.6

1. Determine how many different permutations are possible using the word ABABA.
2. Find how many ways you can rearrange the letters of BANANA; if
   a. there is no restriction.
   b. the rearranged letters start with N.
3. How many different permutations can be made from the letters in the word "MATHEMATICS"?
4. How many different permutations can be formed from the letters of the word MATHEMATICS which start from C?

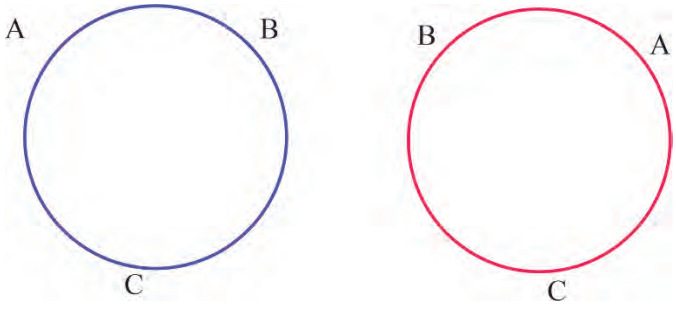
5. How many permutations can be made from the letters in the word MATHEMATICS which start with M?

Circular Permutations

Activity 8.4

Discuss the difference between arrangements of objects in a straight line and around a circle.

Suppose you have three students named A, B, and C. You have already determined that they can be seated in a straight line in 3! or 6 ways. your next problem is to see in how many ways these students can be seated in a circle. You draw a diagram.



It happens that there are only two ways we can seat three students in a circle, relative to each other's positions. This kind of permutation is called a circular permutation which depends on the relative positions of the students after you fix the position of one student. In such cases, no matter where the first student sits, the permutation is not affected. Each student can shift as many places as they like, and the permutation will not be changed. You are interested in the position of each student in relation to the others. Thus, in circular permutations, the first student is considered a place holder, and where he/she sits does not matter.

Definition 8.4

The number of permutations of n elements in a circle is $\frac{n!}{n} = (n-1)!$

Example 7

In how many different ways can five people be seated at a round table?

Solution

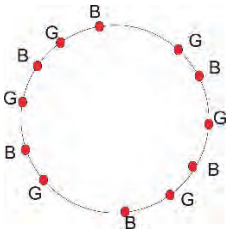
The number of ways will be (5 – 1)! or 24.

Example 8

In how many ways 6 boys and 5 girls dine at a round circular table, if no two girls are to sit together.

Solution

First let us allot the seats to boys. Now 6 boys can have (6−1)! circular permutation, i.e., the number of permutations in which boys can take their seats is 5! = 120. Next the 5 girls occupy seats marked(G). There are 6 spaces between the boys, which can be occupied by 5 girls in ${}_6P_5 = 720$ ways. Hence total number of ways is



$5! \times {}_6P_5 = 120 \times 720 = 86{,}400.$

Example 9

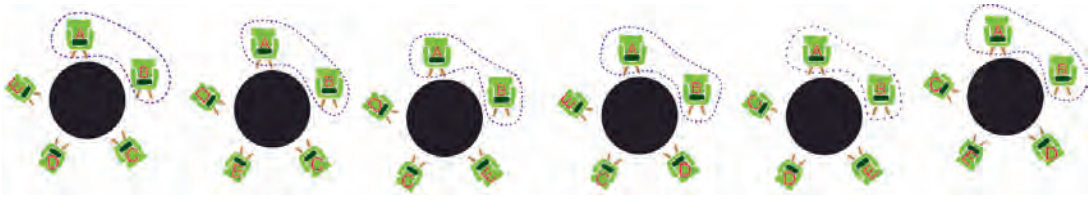
Find the number of ways in which 5 people A, B, C, D, and E can be seated at a round table, such that

a. A and B always sit together.

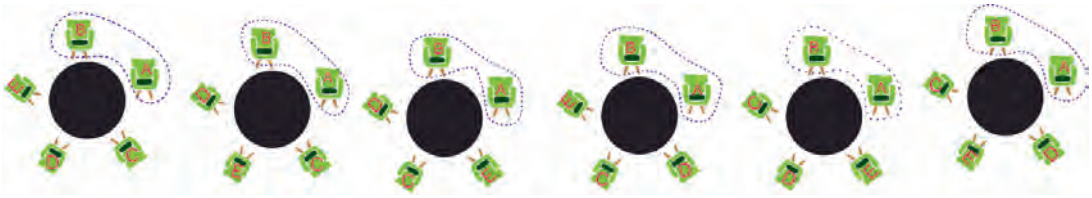
b. C and D never sit together.

Solution

a. If you wish to make seat A and B together in all arrangements, you can consider these two as one unit, along with 3 others. So, effectively you have to arrange 4 people in a circle, the number of ways being (4 – 1)! or 6. Let's take a look at these arrangements:



But in each of these arrangements, A and B can themselves interchange places in 2 ways as shown in the figures below.



Therefore, the total number of ways will be 6 × 2 or 12.

b. The number of ways in this case would be obtained by removing all those cases (from the total possible) in which C and D are together. The total number of ways will be (5 – 1)! or 24. Similar to (**a**) above, the number of cases in which C and D are seated together, will be 12. Therefore, the required number of ways will be 24 – 12 or 12.

Exercise 8.7

1. Calculate circular permutation of 4 persons sitting around a round table.
2. In how many ways can four couples be seated at a round table if the men and women want to sit alternately?
3. In how many ways 8 boys and 3 girls can sit around a circular table, so that no two girls sit together?

8.3.2 Combination

Combinations (1)

Activity 8.5

1. Compute the following:

   a. $\frac{P(7,3)}{3!}$ b. $\frac{P(14,7)}{7!}$ c. $\frac{P(n,r)}{r!}$ d. $\frac{P(n,n)}{n!}$

2. Suppose a dress designer wishes to select two colours of material to design a new dress, and he has on hand four colours. Discuss how many different possibilities can there be in this situation?

From Activity 8.5, you have observed that combination can be obtained by dividing permutation by factorial of r objects selected from n objects.

Example 1

Given the letters A, B, C, and D, list the permutations and combinations for selecting two letters.

Solution

The permutations are:

AB	BA	CA	DA
AC	BC	CB	DB
AD	BD	CD	DC

In permutations, AB is different from BA. But in combinations, AB is the same as BA since the order of the objects does not matter in combinations. Therefore, if duplicates are removed from a list of permutations, what is left is a list of combinations, as shown:

AB	~~BA~~	~~CA~~	~~DA~~
AC	BC	~~CB~~	~~DB~~
AD	BD	CD	~~DC~~

Hence, the combinations of A, B, C, and D are AB, AC, AD, BC, BD, and CD. (Alternatively, BA could be listed and AB crossed out, etc.). This example leads us to the definition of combinations.

Definition 8.5

The number of combinations of r objects selected from n objects without considering the order of selection, is denoted by $C(n, r) = {}_nC_r = \binom{n}{r} = C_r^{\,n}$ and defined by

$$C(n, r) = \frac{n!}{(n-r)!\, r!} \quad \text{where } 0 < r \leq n.$$

To arrive at a formula for nCr, observe that the *r* objects in nPr can be arranged among themselves in *r*! ways.

Thus, $C(n, r) = \frac{nPr}{r!} = \frac{\frac{n!}{(n-r)!}}{r!} = \frac{n!}{(n-r)!\, r!}$

Hence, the number of possible combinations of *n* objects taken *r* at a time is given by the formula

$$C(n, r) = \frac{n!}{(n-r)!\, r!} \quad \text{where } 0 < r \leq n.$$

From this, you can see that the number of ways that a committee of three members can be selected from four individuals is given by

$C(4, 3) = \frac{4!}{1!\, 3!} = 4$ ways.

Example 2

Compute the following:

a. C (7, 3) b. C (11, 5) c. C (13, 10)

Solution

a. $C(7, 3) = \frac{7!}{(7-3)!\,3!} = \frac{7!}{4!3!} = \frac{7 \times 6 \times 5 \times 4!}{4! \times 3 \times 2 \times 1} = 35.$

b. $C(11, 5) = \frac{11!}{(11-5)! \times 5!} = \frac{11!}{6! \times 5!} = \frac{11 \times 10 \times 9 \times 8 \times 7 \times 6!}{6! \times 5 \times 4 \times 3 \times 2 \times 1} = 462.$

c. $C(13, 10) = \frac{13!}{(13-10)!\,10!} = \frac{13 \times 12 \times 11 \times 10!}{3 \times 2 \times 1 \times 10!} = 286.$

Exercise 8.8

1. Compute each of the following.
   a. C (5, 2) b. C (5, 3) c. C (9, 1) d. C (11, 9) e. C (n, n)
2. C (5, 2) = C (5, 3). Explain in words why this happens.
3. If C (n, 3) = C (n, 4), find n.

Combinations (2)

Example 3

In an examination paper, there are 10 questions. In how many different ways can a student choose six questions in all?

Solution

The student is to choose 6 questions from the 10 questions. Their order does not matter.

Hence, he/she can do it in C (10, 6) ways. This means, $C(10, 6) = \frac{10!}{(10-6)!\,6!} = 210.$

Example 4

In HIV/AIDS club there are 7 women and 5 men. A committee of 3 women and 2 men is to be chosen. How many different possibilities are there?

Solution

Here, you must select 3 women from 7 women, which can be done in ${}_7C_3$, or 35, ways. Next, 2 men must be selected from 5 men, which can be done in ${}_5C_2$, or 10 ways.

Finally, by the fundamental counting rule, the total number of different ways is $35 \times 10 = 350$, since you are choosing both men and women separately. Using the formula gives

$${}_7C_3 \times {}_5C_2 = \frac{7!}{(7-3)!\,3!} \times \frac{5!}{(5-2)!\,2!} = 350.$$

Example 5

In how many ways can Bekele invite at least one of his friends out of 6 friends to an art exhibition?

Solution

At least one means that he can invite either one, two, three, four, five or all 6.

Therefore, the total number of ways in which he can invite at least one of his friends is given by (Addition principle)

$$C(6, 1) + C(6, 2) + C(6, 3) + C(6, 4) + C(6, 5) + C(6, 6)$$
$$= 6 + 15 + 20 + 15 + 6 + 1 = 63.$$

Exercise 8.9

1. Select 5 students from a class of 25 to write solutions to a homework problem on the board. If it doesn't matter who does which question, how many ways can these 5 students be picked?
2. A newspaper editor has received 8 books to review. He decides that he can use 3 reviews in his newspaper. How many different ways can these 3 reviews be selected?

3. A committee of 5 students has to be formed from 8 boys and 4 girls. In how many ways can this be done when the committee consists of:
   a. exactly 3 girls? b. at least 3 boys? c. at most 3 girls?
4. There are 7 women and 5 men in mathematics department:
   a. In how many ways can a committee of 4 people be selected?
   b. In how many ways can this committee be selected if there must be 2 men and 2 women on the committee?
   c. In how many ways can this committee be selected if there must be at least 2 women on the committee?

8.4 Binomial Theorem

Pascal's Triangle

Activity 8.6

1. Expand $(a + b)^0$, $(a + b)^1$ and $(a + b)^2$. Discuss the pattern in the expansions and its corresponding coefficients?
2. What is the coefficient of a^3b^2 after expanding $(2a + b)^5$?

From Activity 8.6, imagine the pattern in the expansions and its corresponding coefficients. Again, the expansion form of $(a + b)^n$ where n = 0, 1, 2 is

$(a + b)^0 = 1$

$(a + b)^1 = a + b$

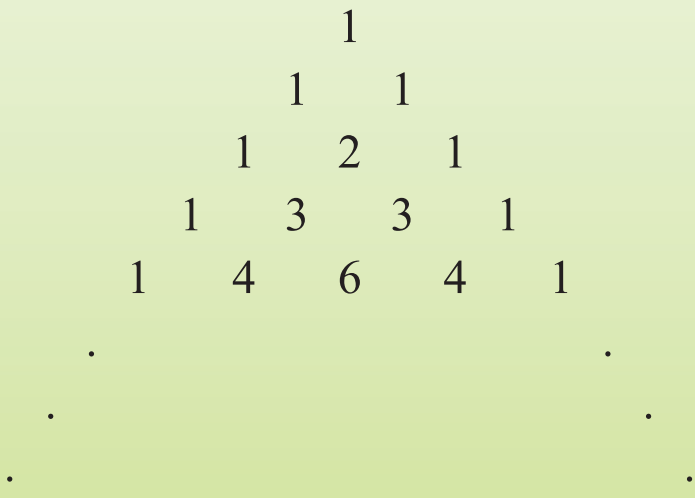
$(a + b)^2 = a^2 + 2ab + b^2$

In these expansions, you observe that the total number of terms in the expansion is one more than the index. For example, in the expansion of $(a + b)^2$, number of terms is 3 whereas the index of $(a + b)^2$ is 2. Powers of the first quantity 'a' go on decreasing by 1 whereas the powers of the second quantity 'b' increases by 1, in the successive terms. In each term of the expansion, the sum of the indices of a and b is the same and is equal to the index of a + b.

Thus, arranging the coefficients in these expansions in the form of triangle defines Pascal's Triangle.

Definition 8.6

Pascal's triangle is a triangular arrangement of numbers that gives the coefficients in the expansion of any binomial expression, such as $(a + b)^n$ for $n = 0, 1, \ldots, n$



Note

Pascal's triangle can be used to visualize many properties of the binomial coefficient and the binomial theorem.

Exercise 8.10

Using the Pascal's triangle, find the coefficients in the expansion of $(a + b)^n$

a. when $n = 5$. b. when $n = 6$.

Binomial theorem

For any positive integer n, the binomial expansion of $(a + b)^n$ is given by

$$(a+b)^n = \binom{n}{0}a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \binom{n}{r}a^{n-r}b^r + \ldots + \binom{n}{n}b^n$$

Example 1

Expand $(a+b)^4$

$(a+b)^4 = \binom{4}{0}a^4 + \binom{4}{1}a^3b + \binom{4}{2}a^2b^2 + \binom{4}{3}ab^3 + \binom{4}{4}b^4$

$= a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4.$

Example 2

Find the coefficient of a^3b^2 in the expansion of $(a+b)^5$

$(a+b)^5 = \binom{5}{0}a^5 + \binom{5}{1}a^4b + \binom{5}{2}a^3b^2 + \binom{5}{3}a^2b^3 + \binom{5}{4}ab^4 + \binom{5}{5}b^5$

Thus, the coefficient of $a^3b^2 = \binom{5}{2} = \frac{5!}{2!\,3!} = 10.$

Example 3

Expand $(a-2b)^4$ using Pascal's triangle.

Solution

Pascal's triangle

1

1 1

1 2 1

1 3 3 1

1 4 6 4 1

Hence, $(a-2b)^4 = a^4 + 4(a^3)(-2b) + 6a^2(-2b)^2 + 4(a)(-2b)^3 + (-2b)^4$

$= a^4 - 8a^3b + 24a^2b^2 - 32ab^3 + 16b^4$

Exercise 8.11

1. Expand each of the following using the Binomial Theorem:
   a. $(a+b)^7$ b. $(a+b)^8$ c. $(a-3b)^9$
2. Without writing all the expanded terms, answer the following:
   a. What is the coefficient of a^5b^3 in the expansion of $(a+b)^8$?
   b. What is the coefficient of a^2b^4 in the expansion of $(a+b)^6$?
   c. What is the coefficient of the term containing a^4b^2 in $(2a+b)^6$?
3. Expand $(x-3)^4$ using pascal's triangle.

8.5 Random Experiments and Their Outcomes

Random Experiments (1)

Activity 8.7

Find the sample spaces for each of the following random experiments:

a. Tossing a coin
b. Tossing a pair of coins
c. Rolling a die

From Activity 8.7, you have observed that if the experiment is repeated under identical conditions, it does not necessarily produce the same results every time but the outcome in a trial is one of the several possible outcomes which can be defined as random experiment or probability experiment.

Definition 8.7

A **random experiment** is any well-defined procedure that produces an observable outcome that could not be perfectly predicted in advance.

Example 1

1. Tossing a fair coin is a random experiment.
2. Rolling of a die is a random experiment.
3. Drawing a card from a pack of cards is a random experiment.

Definition 8.8

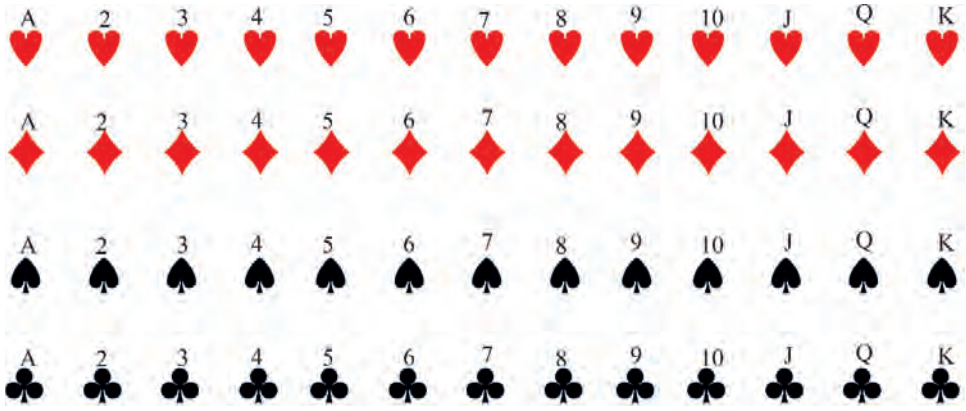
Sample space is the set of all possible outcomes of a random experiment associated with it and denoted by S.

Example 2

Find the sample spaces for drawing one card from an ordinary deck of cards.

Solution

There are 4 suits (hearts, clubs, diamonds, and spades) and 13 cards for each suit (ace through King). Thus, there are 52 outcomes in the sample space as shown below.



Example 3

Find the possible outcomes of selecting a 4-digit PIN from 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9.

Solution

There are 10 possible values for each digit of the PIN.

So, $10 \times 10 \times 10 \times 10 = 10^4 = 10{,}000$.

Example 4

A box contains 24 different balls. Find the number of possible outcomes when you select three balls from the box.

Solution

There are C (24, 3) = 2,024 number of possible outcomes.

Exercise 8.12

1. Find the sample space for the gender of the children if a family has four children.
2. A fair die is thrown. How many favourable outcomes are there for getting an even number.
3. Find the sample space for drawing one card from an ordinary deck of cards.
4. Find the possible outcomes of the football match between St. Gorge and Ethiopia Coffee (for each team, draw, loss or win).

Random Experiments (2)

Example 5

Each of 5 cards has one of the letters A, B, C, D and E on them. The cards are shuffled. Find number of different arrangements (possible outcomes).

Solution

Number of different arrangements = $5! = 5 \times 4 \times 3 \times 2 \times 1$.

Note

Outcomes of a random experiment are said to be **equally likely** when each element has **equal chance of being chosen**.

Example 6

In tossing coin, any one of the outcomes H or T, has an equal chance of appearing at the top. Thus, they are considered as **equally likely.**

Note

In a random experiment, the outcomes which insure the occurrence of a particular result is called **favourable outcomes** to that particular result.

Example 7

a. There are four blue marbles and one red marble in a jar. You pick up one marble from the jar. What is the number of favourable outcomes to get a red marble only.

b. If you draw one slip from a box that contains 12 slips of paper numbered 1 to 12, how many favourable outcomes are there for choosing a slip with an even number on it?

c. In picking a playing card from a pack of 52 cards, what is the number of favourable outcomes to get a picture card?

Solution

a. There is 1 favourable outcome. This is red marble.

b. There are 6 favourable outcomes. These are 2, 4, 6, 8,10 and 12.

c. There are 12 favourable outcomes. These are 4 Jacks, 4 Queens and 4 Kings.

Exercise 8.13

1. What is meant by equally likely?
2. Two balls are to be selected with replacement from a bag that contains one red, one blue, one green and one orange ball. Use the counting principle to determine the number of favourable outcomes to get blue.
3. Each of 5 cards has one of the letters A, B, C, D, and D on them. The cards are shuffled. What is the number of favourable outcomes that the letters A and B are together?
4. In a certain state's lottery, 48 balls numbered 1 through 48 are placed in a machine and six of them are drawn at random. If the six numbers drawn match the numbers that a player had chosen, the player wins 1,000,000 birrs. In this lottery, the order the numbers are drawn in doesn't matter. Find the number of favorable outcomes of the lottery drawn that you win one-million-birr prize if you purchase a single lottery ticket.

8.6 Events

Events and Sample Space

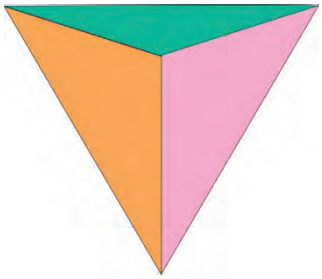
Activity 8.8

1. What does an event mean?
2. List some events of the following experiments:
   a. Tossing a coin three times.
   b. Selecting a number at random from rolling a die.
   c. Drawing a ball from a bag containing 4 red and 6 white balls.

For Activity 8.8, remember that a subset of the sample space associated with a random experiment is called an event. It is denoted by E or other uppercase letters.

Example 1

The four faces of a regular tetrahedron are numbered 1, 2, 3 and 4. If it is thrown and the number on the bottom face (on which it stands) is registered, then list all the possible events of this experiment.



Solution

The sample space = {1, 2, 3, 4}.

Example 2

Suppose our experiment is tossing a fair coin. The sample space for this experiment is S = {H, T}. This sample space has a total of four possible events: a head is thrown ({H}), a tail is thrown ({T}), a head and a tail is thrown ({H}∩{T}= {} = ∅ because this event is impossible) and a head **or** a tail is thrown ({H}∪{T} = {H, T} = S because this always happens).

Thus, the list of the possible events is ∅, {H}, {T}, and {H, T}.

Note

We can determine the possible number of events that can be associated with an experiment whose sample space is S. As events are subsets of a sample space, and any set with m elements has 2^m subsets, the number of events associated with a sample space with m elements is 2^m.

Example 3

Suppose our experiment is recording the gender of three children of three families. Where B and G standing for boy and girl. List events of the first result is a boy.

Solution

The sample space is S = {BBB, BBG, BGB, BGG, GBB, GBG, GGB, GGG}.

E = {BBB, BBG, BGB, BGG} is an event.

Example 4

Consider the random experiment of throwing a die. List an event of getting a number(s) which is(are) less than 3.

Solution

The sample space = {1, 2, 3, 4, 5, 6}.

The possible events = {1, 2}.

Example 5

A committee of 5 people is to be selected from a group of 5 men and 6 women. What is the number of an event that the committee contains 2 men and 3 women?

Solution

Number of possible committee compositions = C (11, 5) = 462

Number of an event that the committee contains 2 men and 3 women is

C (5, 2) × C (6, 3) = 200.

Exercise 8.14

1. Suppose you throw a fair die once. List an event of getting an odd number.
2. Suppose you throw a fair die twice and record the numbers. List an event of getting an odd number twice.
3. Three students are to be chosen from a class of 8 girls and 10 boys. What is the number of an event that the three selected students are two girls and one boy?

8.6.1 Types of Events

Various Events (1)

Activity 8.9

What do you know about types of events?

Definition 8.9

1. **Simple Event (Elementary Event)** is an event containing only one sample element.

Example 1

Suppose you randomly select one student from your class and observe whether the student selected each time is a man or a woman. The occurrence of man is a simple event.

2. **Compound Event**

An event that contains more than one sample element is called a compound event.

Example 2

When a die is rolled, if you are interested in the event of "getting an odd number", then the event will be a compound event, i.e., {1, 3, 5}.

3. **Impossible event**

An event that cannot happen is called an impossible event.

Example 3

If a die is thrown, then S = {1, 2, 3, 4, 5, 6}. Let E be the event of getting number 8, then E is an impossible event and denoted by E = ∅.

4. Certain or sure Event:

You know that every set is subset of itself. Thus, sample space (S) is a subset of itself and hence S is an event. This is called a Sure or certain event.

Example 4

If a die is thrown, then S = {1, 2, 3, 4, 5, 6}. Let E be the event of getting a number ≤ 6, then E = 1, 2, 3, 4, 5 or 6 is a sure or certain event.

5. Occurrence or Non-occurrence of an event

An event is said to occur if the outcome is associated to the event's sample space. Otherwise, it is a non-occurrence event.

Example 5

If a die is thrown, then S = {1, 2, 3, 4, 5, 6}. Let E be the event of getting an even number, then E = {2, 4, 6}. When you throw the die, if the outcome is 4, as $4 \in E$, then you say that E has occurred. If in another trial, the outcome is 3, then as $3 \notin E$, you say that E has not occurred (not E).

Exercise 8.15

You draw a card from a pack of 52 playing cards:

a. How many possible events are there?

b. Is the event that you draw 9 of spade a simple event or a compound event?

c. Is the event that you draw any card of spade a simple event or a compound event?

d. How do you call the event of drawing the card of 0 of heart?

Various Events (2)

6. Algebra of events

In a random experiment, let S be sample space, and let E_1 and E_2 be the events in S.

a. **Complement of an Event E_1**, denoted by E_1' (not E_1) consists of all events in the sample space that are not in E_1.

Example 6

In drawing a marble from a jar containing 6 red, 4 blue and 2 green marbles, let E be the event of green marble. Give the complement of the event.

Solution

Not E = E ' = 6 red and 4 blue marbles.

Example 7

Let a die be rolled once. Let E be the event of a prime number appearing at the top; i.e., E = {2, 3, 5}. Give the complement of the event.

Solution

E ' = {1, 4, 6}.

Note: E ' = S – E = {a: a ∈S and a ∉ E}.

a. Event E_1or E_2 $= E_1 \cup E_2$ is an event that occurs when either one of E_1 or E_2 or both occur.

Example 8

In tossing 3 coins, let E_1 = having exactly one Head; i.e., {TTH, THT, HTT} and E_2 = having exactly one tail; i.e., {HHT, HTH, THH}, list events of $E_1 \cup E_2$.

Solution

$E_1 \cup E_2$ = {exactly one Head or exactly one tail}.

= {TTH, THT, HTT, HHT, HTH, THH}.

Event E_1 and $E_2 = E_1 \cap E_2$ is an event that occurs only when both E_1 and E_2 occurs.

Example 9

In throwing two dice,

Let E_1 = first die has 5 = {(5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6)}.

E_2 = sum on both dies is 6 = {(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)}.

List events of $E_1 \cap E_2$

Solution

Events of $E_1 \cap E_2$ = {(5, 1)}.

7. Exhaustive Events

A set of events is said to be exhaustive events if the performance of the experiment always results in the occurrence of at least one of them. Thus, if a set of events E_1, E_2, …, En are subsets of a sample space, S, they are said to be exhaustive,

if $E_1 \cup E_2 \cup \ldots, \cup E_n = S$.

Example 10

If a die is thrown, give instances of exhaustive events.

Solution

The sample space is S = {1, 2, 3, 4, 5, 6}. From this, the events E_1 = {2, 4},

E_2= {2, 4, 6} E_3 = {1, 3, 5} are exhaustive events.

Since $E_1 \cup E_2 \cup E_3$ = {2, 4} ∪ {2, 4, 6} ∪ {1, 3, 5} = {1, 2, 3, 4, 5, 6} = S.

8. Mutually Exclusive Events

Two events E_1 and E_2 are said to be mutually exclusive events if the occurrence of any one of them excludes the occurrence of the other event. Thus, they cannot occur simultaneously.

Example 11

Say whether or not the following are mutually exclusive events.

a. When a coin is tossed once, the events {H} and {T}.

b. When a die is thrown, E_1 = getting an even number.

E_2 = getting 1 and 3.

c. When a die is rolled, E_1 = getting an odd number.

E_2 = getting a prime number.

d. When a card is drawn, the events of Kings and Aces.

Solution

a. Either you get head or tail but we cannot get both at the same time. Thus, {H} and {T} are mutually exclusive events. Since $E_1 \cap E_2 = \emptyset$.

b. E_1 and E_2 are mutually exclusive because $E_1 \cap E_2 = \{2, 4, 6\} \cap \{1, 3\} = \emptyset$.

c. E_1 and E_2 are not mutually exclusive because 5 is odd and prime at the same time.

d. Kings and Aces are mutually exclusive events.

9. Exhaustive and Mutually Exclusive Events

If S is a sample space associated with a random experiment and if $E_1, E_2, ..., E_n$ are subsets of S such that

a. $E_i \cap E_i = \emptyset$ for $i \neq j$ and,

b. $E_1 \cup E_2 \cup \ldots \cup E_n = S$ then the collection of the events $E_1, E_2, \ldots, E_n$ forms a mutually exclusive and exhaustive set of events.

Example 12

If a die is thrown, the events {2, 4, 6}, and {1, 3, 5} are mutually exclusive and exhaustive events. But, the events {1, 2}, {2, 3}, {4, 5, 6} are not because $\{1, 2\} \cap \{2, 3\} = \{2\} \neq \emptyset$.

10. Independent Events

Events are said to be independent, if the occurrence or non-occurrence of one does not affect the occurrence or non-occurrence of the other.

Example 13

In a simultaneous throw of two coins, the event of getting a tail on the first coin and the event of getting a tail on the second coin are independent.

Example 14

a. Landing on heads after tossing a coin and rolling a 5 on a single 6-sided die.
b. Choosing a marble from a jar and landing on heads after tossing a coin.
c. Choosing a 3 from a deck of cards, replacing it, and then choosing an ace as the second card.
d. Rolling a 4 on a single 6-sided die, and then rolling a 1 on a second roll of the die.

11. Dependent Events

Events are said to be dependent, if the occurrence or non- occurrence of one event affects the occurrence or non-occurrence of the other.

Example 15

If a card is drawn from a well shuffled a pack of cards and the card is not replaced, then the result of drawing a second card is dependent on the first draw.

Exercise 8.16

When two dice thrown, consider following events:

E_1 = getting a prime number and E_2 = getting an even number.

a. List E_1 and E_2.

b. Are E_1 and E_2 exhaustive events?

c. Are E_1 and E_2 mutually exclusive events?

8.7 Probability of an Event

8.7.1 Revision on Probability

In this unit, you will revise the classical and the experimental approach of an event before you proceed to the next section as discussed in grade 9 such as:

1. Classical (mathematical) approach, and
2. Empirical (relative frequency) approach before you proceed to the next section.

1. Classical (Mathematical) Approach (1): Simple cases

Definition 8.10

In the classical approach to probability, the probability of an event occurring E is defined as the number of elements of the sample space included in the event, divided by the total number of elements in the sample space, when all outcomes of a random experiment are equally likely and mutually exclusive.

i.e., $P(E) = \frac{\text{Number of outcomes favourable to E}}{\text{Total number of outcomes}} = \frac{n(E)}{n(S)}$.

Example 1

A fair die is tossed once. Find the probability of getting:

a. Number 5.

b. a number greater than or equal to 4.

c. number 7.

Solution

First identify the sample space, say S

$S = \{1, 2, 3, 4, 5, 6\}$

$\Rightarrow n(S) = 6$

a. Let E be the event of number 5

$E = \{5\}$

$\Rightarrow n(E) = 1$

Thus, $P(E) = \frac{n(E)}{n(S)} = \frac{1}{6}$.

b. Let E be the events of a number greater than or equal to 4

$E = \{4, 5, 6\}$

$\Rightarrow n(E) = 3$

$P(E) = \frac{n(E)}{n(S)} = \frac{3}{6} = 0.5$.

c. Let E be the events of number 7

$E = \emptyset$

$\Rightarrow n(E) = 0$

$P(E) = \frac{n(E)}{n(S)} = \frac{0}{6} = 0$.

Exercise 8.17

1. If a die is rolled once, find the probability of getting an even number.
2. If a die is rolled twice, then find the following probabilities.
   a. The probability of getting the sum 2.
   b. The probability of getting the sum greater than 6.
   c. The probability of getting the sum greater than 9 or an odd.

Classical (mathematical) Approach (2): Cases using Permutations and Combinations

There are cases in which permutations and combinations are needed to calculate the number of outcomes.

Example 2

There are 4 blue balls and 3 red balls in a bag. You pick up 3 balls at random. Find the probability of getting 2 blue balls and 1 red ball.

Solution

First, calculate the total number of possible outcomes. You pick up 3 balls from 7 balls (their order does not matter), so the number of possible outcomes is given by

$${}_7C_3 = \frac{7!}{(7-3)!\times 3!} = \frac{7\times 6\times 5\times 4\times 3\times 2\times 1}{(4\times 3\times 2\times 1)\times(3\times 2\times 1)} = 7\times 5 = 35.$$

Second, calculate the number of possible outcomes where 2 blue balls and 1 red ball are chosen. Since you select 2 blue balls from 4 (their order does not matter), there are ${}_4C_2$ possible outcomes for blue balls. For red balls, there are ${}_3C_1$ possible outcomes. Since picking up blue balls and picking up red balls are independent, the number of possible outcomes of picking up 2 blue balls and 1 red ball is given by

$${}_4C_2 \times {}_3C_1 = \frac{4!}{2!2!} \times \frac{3!}{2!1!} = 6\times 3 = 18.$$

Therefore, the probability is given by $\frac{{}_4C_2 \times {}_3C_1}{{}_7C_3} = \frac{18}{35}$.

Exercise 8.18

1. A box of 10 candles consists of 3 defective and 7 non-defective candles. If 5 of these candles are selected at random, what is the probability in which:
   a. 3 will be defective?
   b. 4 will be non-defective?
   c. all will be non-defective?

2. If 3 books are picked at random from a shelf containing 4 math books, 3 books of chemistry, and a dictionary, what is the probability that:
   a. the dictionary is selected?
   b. 2 math and 1 book of chemistry are selected?
3. Three cars are chosen at random from a certain car station containing 8 defective and 12 non-defective cars. What is the probability that?
   a. all are defective?
   b. all are non-defective?
   c. two are defective and the other is non- defective?

2. Empirical (relative frequency) approach

This approach is based on the observations obtained from random experiment. The empirical frequency of an event E is the relative frequency of event E.

Thus, $P(E) = \frac{\text{frequency of E}}{\text{total number of observations}} = \frac{f_E}{n}$.

Example 3

A travel agent determines that in every 70 reservations she makes 14 will be for a cruise. What is the probability that the next reservation she makes will be for a cruise?

Solution

Let E be the event of cruise

$P(E) = \frac{f_E}{n} = \frac{14}{70} = 0.20$.

Example 4

If records show that 30 out of 50,000 bulbs produced are defective, find the probability of a newly produced bulb to be defective.

Solution

Let E be the event that the newly produced bulb is defective.

$P(E) = \frac{f_E}{n} = \frac{30}{50{,}000} = 0.0006.$

Exercise 8.19

1. In a sample of 50 people, 21 had type O blood, 22 had type A blood, 5 had type B blood, and 2 had type AB blood. Find the following probabilities:
   a. A person has type O blood.
   b. A person has type A or type B blood.
   c. A person has neither type A nor type O blood.
   d. A person does not have type AB blood.
2. Ten of the 500 randomly selected cars manufactured at a certain auto factory are found to be lemons (defective). Assuming that the lemons are manufactured randomly, what is the probability that the next car manufactured at this auto factory is a lemon?

8.7.2 The Axiomatic Approach of Probability

This approach includes both the classical and empirical definitions of probability.

Let **A** be a random experiment and S be a sample space associated with **A**. With each event **E** a real number called the probability of E, denoted by P(E), that satisfies the following properties called axioms of probability or postulates of probability:

1. $P(E) \geq 0$.
2. $P(S) = 1$, if E = S (the sure or certain event).
3. $P(E_1 \cup E_2) = P(E_1) + P(E_2)$ If E_1 and E_2 are mutually exclusive events.
4. $0 \leq P(E) \leq 1$; i.e., the probability of an event is always between 0 and 1.
5. $P(\emptyset) = 0$ if $E = \emptyset$ (the impossible event)

6. If E ∪ E ' = S then P (E ∪ E ') = P(S) = 1, and P (E ') = 1 – P(E),

   where E ' = S – E. (not E); i.e., the sum of the probability of occurrence event (E) and non-occurrence event (E ') is 1, (P(E) $+ P(E') = 1$).

Note

Probability (P) is a function whose domain is the set of subsets of S (Sample space) and whose range is the set of real numbers between 0 and 1(both inclusive).

Example 1

A box contains 8 white balls. One ball is drawn at random. Find the probability of getting:

a. white ball b. red ball

Solution

a. The box contains all white balls. Hence, we are sure that white will occur. Then, the probability of getting a white ball is one.

   Thus, $P(w) = \frac{n(w)}{n(s)} = \frac{8}{8} = 1$.

b. The box contains no red balls. It is impossible to get a red ball and the probability is zero.

Hence, $P(R) = \frac{n(R)}{n(S)} = \frac{0}{8} = 0$.

Example 2

A bag contains 5 red, 7 black, and 6 white marbles. One marble is drawn at random. What is the probability that the marble is

a. black b. not black

Solution

a. P(black) = $\frac{7}{18}$.

b. P (not black) = 1 − P(black)

$= 1 - \frac{7}{18} = \frac{11}{18}$

Thus, P(black) + P(not black) = $\frac{7}{18} + \frac{11}{18} = 1$.

Example 3

If one card is drawn from a deck, find the probability of picking these results:

a. King b. Not king c. King and Queen d. King or Queen

Solution

Number of ways king can happen, n(king) = 4 (there are 4 kings)

Number of ways Queen can happen, n(Queen) = 4 (there are 4 Queens)

Total number of outcomes, n(S) = 52 (there are 52 cards in total)

a. P(king) = $\frac{\text{n(king)}}{\text{n(S)}} = \frac{4}{52} = \frac{1}{13}$.

b. P (Not king) = $1 - \frac{4}{52} = 1 - \frac{1}{13} = \frac{12}{13}$.

c. A card cannot be a king and a Queen at the same time. The chance of picking a King and a Queen at the same time is zero (or none).

Hence, P (a king and a Queen) = $\frac{0}{52} = 0$

d. P (a king or a Queen) = P (a king) + P (a Queen)

$= \frac{1}{13} + \frac{1}{13} = \frac{2}{13}$.

Example 4

Which of the following cannot be valid assignments of probabilities for outcomes of sample space $S = \{ a_1, a_2, a_3, a_4, a_5, a_6, a_7 \}$

	a_1	a_2	a_3	a_4	a_5	a_6	a_7
a	0.2	0.001	0.09	0.03	0.01	0.008	0.3
b	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$
c	0.1	0.2	0.3	0.4	0.5	0.6	0.7
d	-0.7	0.007	0.3	0.4	-0.2	0.1	0.3
e	$\frac{1}{13}$	$\frac{2}{13}$	$\frac{3}{13}$	$\frac{4}{13}$	$\frac{5}{13}$	$\frac{6}{13}$	$\frac{7}{13}$

Solution

a. Valid because all properties are satisfied.

b. not valid because the sum of the properties is $\frac{7}{6}$ which is greater than 1.

c. not valid, because the sum of all the probabilities is 2.8 which is greater than 1 i.e., $0 \leq P(E) \leq 1$ is not satisfied.

d. not valid, because probabilities of w_1 and w_5 are negative and hence $0 \leq P(E) \leq 1$ is violated.

e. not valid, because the sum of all the probabilities, $\frac{28}{13}$ is greater than 1.

Exercise 8.20

1. If one card is drawn from a deck, find the probability of getting these results:
   a. A queen b. A club c. not a club
   d. A 6 or a spade e. A red card and a 7
2. If two dice are rolled one time, find the probability of getting these results:
   a. A sum of 9 b. A sum of 7 or 11
   c. A sum less than 9 d. A sum greater than or equal to 10

8.7.3 Odds in Favour of and Odds Against an Event

Activity 8.10

1. Find the smallest share if you divide the number below by the ratio 2: 3
   a. 20 b. 35 c. 42
2. Find the largest share if you divide the number below by the ratio 2: 3
   a. 20 b. 35 c. 42

From Activity 8.10, you can formally define odds in favor of and odds against an event as follows:

Definition 8.11

If **P** and **P'** are probability of the occurrence and non- occurrence of an events respectively, then the ratio of **P: P '** is called **the odds in favour of the event** and the ratio of **P ': P** is called **the odds against the event**.

Note

If the probability of an event occurring is P, then the probability of the event not occurring (P ') is 1– P.

Example 1

If a race horse runs 100 races and wins 25 times and loses the other 75 times, what are the probability of winning and the odds of the horse winning?

Solution

The probability of winning is $\frac{25}{100} = 0.25$ and the probability of losing is $\frac{75}{100} = 0.75$.

The odds in favor of winning is $0.25 : 0.75 = \frac{25}{75} = 0.333$.

Example 2

The odds against certain events are 6: 8. Find the probability of its occurrence.

Solution

Let E be the event. Then we are given that n (E ') = 6 and n(E) = 8

Thus, n(S) = n (E') + n(E) = 6 + 8 = 14

Thus, $P(E) = \frac{n(E)}{n(S)} = \frac{8}{14} = \frac{4}{7}$.

Example 3

In throwing a die,

a. Find the odds in favor of getting 3 dots?

b. Find odds against getting 3 dots?

Solution

Let S be possible outcomes in throwing a die,

Let E be event of getting 3 dots.

Let E ' be event not getting 3 dots.

You are given that n(S) = 6, n(E) = 1 and n (*E* ') = (6 - 1) = 5

a. Odds in favor $= \frac{n(E)}{n(E')} = \frac{1}{5} = 1:5$

Therefore, odds in favor of getting "3 dots" is $\frac{1}{5}$ or $1:5$.

b. the Odds against $= \frac{n(E')}{n(E)} = 5:1 = \frac{5}{1}$

Therefore, the odds against getting "3 dots" is 5: 1 or $\frac{5}{1}$.

Exercise 8.21

Find the odds in favor of and against each event:

a. Rolling a die and getting a number less than 3.

b. Drawing a card and getting a red card.

c. Tossing two coins and getting two tails.

8.7.4 The Rules of Addition of Probability

Activity 8.11

Consider two events E_1 and E_2, what condition should be applied to the rule: $P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$ and $P(E_1 \cup E_2) = P(E_1) + P(E_2)$. Discuss and verify using venn diagram?

From Activity 8.11, you have observed that, if $E_1, E_2, \ldots, E_n$ form a set of exhaustive events of a sample space S, then $E_1 \cup E_2 \cup \ldots \cup E_n = S$. Moreover, the probability of an event E, i.e., P(E) is given by

$$P(E) = \frac{\text{number of outcomes favoring E}}{\text{total of outcomes in the sampling sapce}} = \frac{n(E)}{n(S)}.$$

Having this concept, we can easily calculate probabilities of two events by making the use of the addition rule stated below.

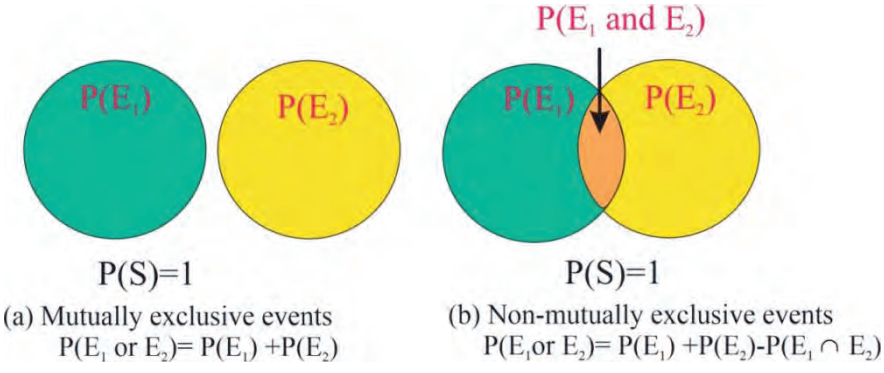
Rule 1: If E_1 and E_2 are any two events, then,

$$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2) \text{ and}$$

Rule 2: If two events are mutually exclusive, (i.e., $E_1 \cap E_2 = \emptyset$) then

$$P(E_1 \cap E_2) = 0 \text{ so that } P(E_1 \cup E_2) = P(E_1) + P(E_2).$$

Moreover, you can verify this rule using venn diagram as follows



(a) Mutually exclusive events
$P(E_1 \text{ or } E_2) = P(E_1) + P(E_2)$

(b) Non-mutually exclusive events
$P(E_1 \text{or } E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$

Example 1

a. Find the probability of getting an odd number or 4 in one roll of a die.

b. Find the probability of getting Head or Tail in tossing a coin once.

c. A die is rolled once. Find the probability that it is even or it is divisible by 3.

Solution

a. $S = \{1, 2, 3, 4, 5, 6\}$,

Let E_1 be event of getting an odd number,

E_2 be event of getting 4;

then E_1 and E_2 are mutually exclusive events.

Thus, $P(E_1 \cup E_2) = P(E_1) + P(E_2) = \frac{3}{6} + \frac{1}{6} = \frac{4}{6} = \frac{2}{3}$.

b. Let E_1 = getting Head.

E_2 = getting Tail.

The events are mutually exclusive

Thus, $P(E_1 \text{ or } E_2) = P(E_1) + P(E_2) = \frac{1}{2} + \frac{1}{2} = 1$.

c. $S = \{1, 2, 3, 4, 5, 6\}$

Let E_1 = getting an even number = $\{2, 4, 6\}$.

E_2 = getting a number divisible by 3 = $\{3, 6\}$;

then E_1 and E_2 are not mutually exclusive, because $E_1 \cap E_2 = \{6\}$

Therefore, $P(E_1 \text{ or } E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$

$$= \frac{3}{6} + \frac{2}{6} - \frac{1}{6} = \frac{2}{3}.$$

Example 2

Two dice are rolled. Find the probability of getting:

a. A sum of 8, 9, or 10

b. Doubles or a sum of 7

c. A sum greater than 9 or a sum of 12

Solution

Die 1	Die2					
	1	**2**	**3**	**4**	**5**	**6**
1	(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)
2	(2, 1)	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
3	(3, 1)	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
4	(4, 1)	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
5	(5, 1)	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
6	(6, 1)	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

From the above table

a. $n(S) = 36$

Let E_1 = getting a sum of 8 = {(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)}.

E_2 = getting a sum of 9 = {(3, 6), (4, 5), (5, 4), (6, 3)}.

E_3 = getting a sum of 10 = {(5, 5), (6, 4), (4, 6)};

then E_1, E_2 and E_3 are mutually exclusive because $E_1 \cap E_2 = E_1 \cap E_3 = E_2 \cap E_3 = \emptyset$

Therefore, $P(E_1 \text{ or } E_2 \text{ or } E_3) = P(E_1) + P(E_2) + P(E_3)$

$$= \frac{5}{36} + \frac{4}{36} + \frac{3}{36} = \frac{12}{36} = \frac{1}{3}.$$

b. $n(S) = 36$

Let E_1 = getting Doubles = {(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)}.

E_2 = getting a sum of 7 = {(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)};

then E_1 and E_2 are mutually exclusive because $E_1 \cap E_2 = \emptyset$

Therefore, $P(E_1 \text{ or } E_2) = P(E_1) + P(E_2)$

$$= \frac{6}{36} + \frac{6}{36} = \frac{12}{36} = \frac{1}{3}.$$

c. $n(S) = 36$

Let E_1 = getting a sum greater than 9 = {(4,6), (5,5), (5,6), (6,4). (6,5), (6,6)}

E_2 = getting a sum of 12 = {(6, 6)};

then E_1 and E_2 are not mutually exclusive because $E_1 \cap E_2$ = {(6, 6)}

Therefore, $P(E_1 \text{ or } E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$

$$= \frac{6}{36} + \frac{1}{6} - \frac{1}{6} = \frac{1}{6}.$$

Note

The probability rules can be extended to three or more events.

1. For three mutually exclusive events E_1, E_2, and E_3,

$$P(E_1 \cup E_2 \cup E_3) = P(E_1) + P(E_2) + P(E_3)$$

2. For three events that are not mutually exclusive,

$$P(E_1 \cup E_2 \cup E_3) = P(E_1) + P(E_2) + P(E_3) - P(E_1 \cap E_2) - P(E_2 \cap E_3) - P(E_1 \cap E_3) + P(E_1 \cap E_2 \cap E_3).$$

Exercise 8.22

1. Two dice are rolled. Find the probability of getting:
   a. a sum of 3, 6, or 11
   b. doubles or a sum of 8
   c. a sum greater than 8 or a sum divisible by 4
2. At a convention there are 7 mathematics instructors, 5 computer science instructors, 3 statistics instructors, and 4 science instructors. If an instructor is selected, then find the probability of getting a science instructor or a math instructor.
3. A single card is drawn at random from an ordinary deck of cards. Find the probability that it is either an ace or a black card.
4. In throwing a die, consider the following events.

   E_1 = the number that shows up is odd

   E_2 = the number that shows up is prime

   E_3 = the number that shows up is less than 4

   a. Determine the event $E_1 \cap E_2$
   b. Determine the number of elements in $E_2 \cap E_3$
   c. Determine the number of elements in $E_1 \cap E_2 \cap E_3$
   d. Determine $P(E_1 \cap E_3)$
   e. Determine $P(E_1 \cup E_2 \cup E_3)$

8.7.5 The Rule of Multiplication of Probability

Independent and Dependent Events

Activity 8.12

1. What knowledge do you have about rules of addition of probability?
2. Consider two events E_1 and E_2. Discuss what condition should be applied to the rule:

$P(E_1 \text{ and } E_2) = P(E_1 \cap E_2) = P(E_1) \cdot P(E_2)$ and $P(E_1 \cap E_2) = P(E_1) \cdot P(E_2 \setminus E_1)$, whenever $P(E_1) \neq 0$.

The multiplication rule can be used to find the probability of two or more events in sequence. It is based on the concepts of independence or dependence of events discussed earlier. Let us take a brief revision of independent and dependent events.

When the occurrence of the first event does not affect the occurrence of the second event in such a way that the probability is not changed, the events are called independent whereas when the occurrence of the first event affects the occurrence of the second event in such a way that the probability is changed, the events are called dependent.

Example 1

A jar contains 4 black and 3 white balls. You draw two balls one after the other with replacement (the second is drawn after the first is replaced). Find the probability that the first ball is black and the second ball is also black.

Solution

Let event E_1 be the first ball is black.

Let event E_2 be the second ball is black;

then, $P(E_1) = \frac{4}{7}$ and $P(E_2) = \frac{4}{7}$

$P(E_1 \cap E_2) = P(E_1) \times P(E_2) = \frac{4}{7} \times \frac{4}{7} = \frac{16}{49}$.

Example 2

Suppose you repeat the above experiment without replacement (the second ball is drawn without the first ball being replaced).

Solution

$P(E_1) = P$ (The first ball is black.) $= \frac{4}{7}$

If the first ball is black, then $P(E_2) = \frac{3}{6}$ (One black ball has been removed.)

If the first ball is not black, then $P(E_2) = \frac{4}{6} = \frac{2}{3}$.

Example 3

A box contains 3 red balls, 2 blue balls, and 5 white balls. A ball is selected and its colour is noted. Then it is replaced. A second ball is selected and its colour is noted. Find the probability of each of the following:

a. Selecting 2 blue balls.

b. Selecting 1 blue ball and then 1 white ball.

c. Selecting 1 red ball and then 1 blue ball.

Solution

a. Let E_1 = Selecting blue in the first draw.

E_2 = Selecting blue in the second draw.

Since, the balls are replaced after each draw, the events are independent.

$P(E_1 \cap E_2) = P(E_1) \times P(E_2) = \frac{2}{10} \times \frac{2}{10} = \frac{1}{25}$.

b. Let E_1 = Selecting blue in the first draw.

E_2 = Selecting white in the second draw.

Since, the balls are replaced after each draw, the events are independent.

$P(E_1 \cap E_2) = P(E_1) \times P(E_2) = \frac{2}{10} \times \frac{5}{10} = \frac{1}{10}$.

c. Let E_1 = Selecting red in the first draw.

E_2 = Selecting blue in the second draw.

Since, the balls are replaced after each draw, the events are independent.

$P(E_1 \cap E_2) = P(E_1) \times P(E_2) = \frac{3}{10} \times \frac{2}{10} = \frac{3}{50}$.

Exercise 8.23

1. A jar contains 6 red balls, 3 green balls, 5 white balls and 7 yellow balls. Two balls are chosen from the jar, with replacement. What is the probability that both balls chosen are green?
2. A die is rolled and a coin is tossed. Find the probability of getting a prime number on the die and a head in the coin.

Conditional Probability

Identifying dependence or independence is most important in using the multiplication rule of probability. When occurrence of one event depends on the occurrence of another event, you say the second event is conditioned by the first event. This leads into what is called conditional probability.

The **conditional probability** of an event E_2 in relationship to an event E_1 is the probability that event E_2 occurs after event E_1 has already occurred. The notation for conditional probability is $P(E_2 \backslash E_1)$. This notation does not mean that E_2 is divided by E_1; rather, it means the probability that event E_2 occurs given that event E_1 has already occurred. If the occurrence or non -occurrence of E_1 does not affect the probability of E_2, or if $\mathbf{E_1}$ and $\mathbf{E_2}$ are independent, then

$\mathbf{P(E_2 \backslash E_1) = P(E_2)}$. This defines **multiplication rule of probability**. These are:

Rule 1: When two events are independent, the probability of both occurring, denoted by P (E_1 and E_2) or P ($E_1 \cap E_2$) or P ($E_1 E_2$) is given by:

$$P(E_1 \cap E_2) = P(E_1) \times P(E_2)$$

Rule 2: When two events are dependent, the probability of both occurring is given by:

$$P(E_1 \cap E_2) = P(E_1) \times P(E_2 \setminus E_1), \text{ whenever } P(E_1) \neq 0$$
$$= P(E_2) \times P(E_1 \setminus E_2) \text{ whenever } P(E_2) \neq 0$$

Example 4

A bag contains 6 red, 5 blue, and 4 yellow balls. Two balls are drawn, but the first ball is drawn without replacement. Find the following:

a. P (red, then blue)

b. P (blue, then blue)

Solution

a. Let E_1 = getting red in the first draw.

E_2 = getting blue in the second draw

Since, the balls are not replaced, so events are dependent.

$$P(E_1 \cap E_2) = P(E_1) \times P(E_2 \setminus E_1) = \frac{6}{15} \times \frac{5}{14} = \frac{1}{7}.$$

b. Let E_1 = getting blue in the first draw.

E_2 = getting blue in the second draw

Since, the balls are not replaced, so events are dependent.

$$P(E_1 \cap E_2) = P(E_1) \times P(E_2 \setminus E_1) = \frac{5}{15} \times \frac{4}{14} = \frac{2}{21}.$$

Note

1. Multiplication rule 1 can be extended to three or more independent events by using the formula

$$P(E_1 \cap E_2 \cap E_3 \cap \ldots \cap E_k) = P(E_1) \times P(E_2) \times P(E_3) \times \ldots \times P(E_k).$$

2. Multiplication rule 2 can be extended to three or more dependent events by using the formula

$$P(E_1 \cap E_2 \cap E_3 \cap \ldots \cap E_k) = P(E_1) \times P(E_2 \setminus E_1) \times P(E_3 \setminus E_1 \cap E_2) \times \ldots \times P(E_k \setminus E_1 \cap E_2 \cap \ldots \cap E_{k-1}).$$

Example 5

A jar contains 3 red, 5 green, 2 blue and 6 yellow marbles. Three marbles are drawn one after the other. Find the probability of getting a green marble on the first draw, a yellow marble on the second draw and a red marble on the third draw; if

a. each marble is drawn, but then is replaced back before the next draw.

b. the marbles are drawn without replacement.

Solution

Let E_1 = getting green, in the first draw,

E_2 = getting yellow in the second draw,

E_3 = getting red in the third draw.

a. The marbles are replaced after each draw. The events are independent.

$$P(E_1 \cap E_2 \cap E_3) = P(E_1) \times P(E_2) \times P(E_3) = \frac{5}{16} \times \frac{6}{16} \times \frac{3}{16} = \frac{45}{2,048}.$$

b. The marbles are not replaced after each draw. The events are dependent.

$$P(E_1 \cap E_2 \cap E_3) = P(E_1) \times P(E_2 \setminus E_1) \times P(E_3 \setminus (E_1 \cap E_2)) = \frac{5}{16} \times \frac{6}{15} \times \frac{3}{14} = \frac{3}{112}.$$

Exercise 8.24

1. If 2 cards are selected from a standard deck of 52 cards without replacement, find these probabilities:
   a. Both are spades. b. Both are the same suit. c. Both are kings.
2. Three cards are drawn from an ordinary deck and not replaced. Find the probability of these events:
   a. Getting 3 jacks b. Getting an ace, a king, and a queen in order
   c. Getting a club, a spade, and a heart in order d. Getting 3 clubs

Sequential Events

In the previous section, you saw how to determine probability of independent or dependent events using multiplication rules of probability. It is also possible to show events that are sequential using tree diagrams and tables, and calculate probabilities from these.

Example 6

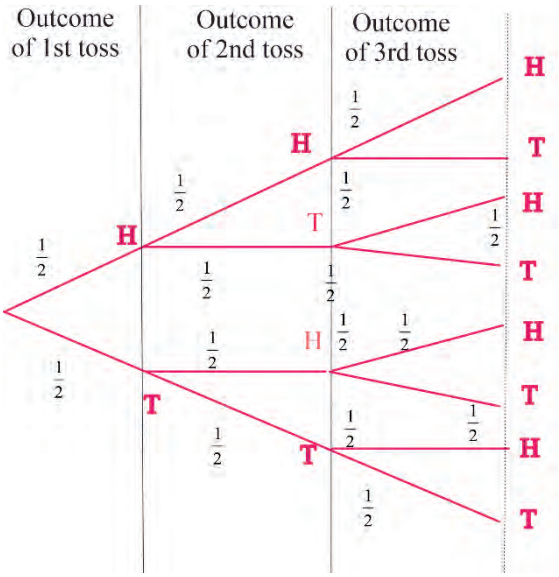
A fair coin is tossed three times. Find the probability that all outcomes will be heads.

Solution

Using the multiplication rule $(HHH) = P(H) \times P(H) \times P(H) = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$

You can use a tree diagram and/or table to show the possible outcomes.

Using tree diagram



Squential Events	Probability of Sequential Events
HHH	
HHT	
HTH	
HTT	
THH	
THT	
TTH	
TTT	

Therefore, the probability that all outcomes are head is $\frac{1}{8}$.

Exercise 8.25

1. A fair die is rolled three times. Find the probability that all outcomes will be 6 dots.
2. If 2 cards are selected from a standard deck of 52 cards without replacement, find these probabilities:
   a. both are Aces. b. both are hearts.

8.8 Real-life Application of Probability

Probability theory is widely used in the area of studies such as statistics, finance, insurance policy, traffic signals, medical decisions, and weather forecasting. Now, you will discuss real-life applications involving probability as follows.

Example 1

A traffic light at a certain road crossing starts green at 06: 30 hours and continues to be green till 06: 32 hours and again turns green at 06: 36 hours and continues green till 06: 38 hours. This cycle is repeated throughout the day. If a person's arrival time at this crossing is random and uniform over the interval 18: 20 to 18: 35 hours, then find the probability that he has to wait the signal.

Solution

In every 6 minutes, the light remains green for 2 minutes and red for 4 minutes. So, in the interval, 18: 20 hours-18: 35 hours; i.e., 15 minutes, the light will remain green for $\frac{2}{6} \times 15 = 5$ minutes and red for 10 minutes. Thus, the probability that he has to wait at the signal $= \frac{10}{15} = \frac{2}{3} \approx 67\ \%$.

Example 2

The source of federal government revenue for a specific year is 50% from individual income taxes, 32% from social insurance payroll taxes, 10% from corporate income taxes, 3% from excise taxes and 5% from other.

If a revenue source is selected at random, what is the probability that it comes from individual or corporate income taxes?

Solution

P (revenue comes from individual or corporate income taxes) = P (revenue comes from individual) + P (revenue comes from corporate income taxes)

$= 0.5 + 0.1$

$= 0.60.$

Example 3

On New Year's Eve, the probability of a person driving while intoxicated is 0.32, the probability of a person having a driving accident is 0.09, and the probability of a person having a driving accident while intoxicated is 0.06. What is the probability of a person driving while intoxicated or having a driving accident?

Solution

P (intoxicated or accident) = P(intoxicated) + P(accident) − P (intoxicated and accident) = 0.32 + 0.09 − 0.06 = 0.35.

Exercise 8.26

1. If the probability that a person lives in an industrialized country of the world is $\frac{1}{5}$, find the probability that a person does not live in an industrialized country.
2. The top-10 selling computer software titles last year consisted of 3 for doing taxes, 5 antivirus or security programs, and 2 "other." Choose one title at random:
   a. What is the probability that it is not used for doing taxes?
   b. What is the probability that it is used for taxes or is one of the "other" programs?
3. In a group of 40 people, 10 are healthy and every person of the remaining 30 has either high blood pressure, a high level of cholesterol or both.
   a. If 15 have high blood pressure and 25 have high level of cholesterol, how many people have blood pressure and a high level of cholesterol?
   b. If a person is selected randomly from this group, what is the probability that he\she
      i) has high blood pressure only? ii) has high level of cholesterol only?
      iii) has high blood pressure and high level of cholesterol?
      iv) has either high pressure or high level of cholesterol?

Problem Solving

1. A company screens job application for illegal drug use at a certain stage in their hiring process. The specific test they use has a false positive rate of 2% and a false negative rate of 1%. Suppose that 5% of all their applicants are actually using illegal drug and we randomly select an applicant. Given the applicant tests positive, what is the probability that they are actually on drugs?
2. A problem is given to three students D, E, F whose respective chances of solving it are $\frac{2}{7}$, $\frac{4}{7}$ and $\frac{4}{9}$ respectively. What is the probability that the problem is solved?
3. Three bags contain 3 red, 7 black; 8 red, 2 black; and 4 red and 6 black balls, respectively. One of the bags is selected at random and a ball is drawn from it. If the ball drawn is red, find the probability of that it is drawn from the third bag.
4. A bag contains 20 marbles of equal size of which 12 are red, x is blue and the rest are white.
   a. If the probability of selecting a blue marble is $\frac{1}{4}$, find x.
   b. A marble is drawn and then replaced. A second marble is drawn. Find the probability that neither marble is red.

Summary

1. **Addition Principle of counting**

   If an operation can be performed in **m** different ways and another operation can occur in **n** different ways and the two operations are mutually exclusive, (the performance of one excludes the other) then either of the two can be performed in **m + n** ways.

2. **Multiplication Principle of counting**

   If an event can occur in **m** different ways and for every such choice another event can occur in **n** different ways, then both events can occur in the given order in **m × n** different ways.

3. If n is a natural number, then n factorial, denoted by $n!$ is defined by

   $n! = n \times (n-1) \times (n-2) \times \ldots \times 2 \times 1$ $(0! = 1)$.

4. Permutations are the number of arrangements of n objects taking r of them at a time, and denoted by nPr or P (n, r) where $nPr = P(n, r) = \frac{n!}{(n-r)!}$,

   where $0 < r \leq n$.

5. The number of combinations of n things taking r at a time is given by

   $$n\,Cr = \binom{n}{r} = \frac{P(n,r)}{r!} = \frac{n!}{(n-r)!\,r!}.$$

6. The Binomial Theorem $(a+b)^n = \binom{n}{0}a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \binom{n}{r}a^{n-r}b^r + \ldots + \binom{n}{n}b^n$.

7. Probability of an event is defined as follows:

   If an experiment results in n equally likely outcomes and $m < n$ is the number of the ways favourable for event E, then $P(E) = \frac{m}{n}$.

8. Addition rule of probability

   Rule 1. If E_1 and E_2 are any two events, then,

   $$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2) \text{ and}$$

Rule 2. If two events are mutually exclusive; (i.e., $E_1 \cap E_2 = \emptyset$) then

$P(E_1 \cap E_2) = 0$ so that

$P(E_1 \cup E_2) = P(E_1) + P(E_2)$.

9. Multiplication rule of probability

Rule 1: When two events are independent, the probability of both occurring, denoted by P (E_1 and E_2) or P ($E_1 \cap E_2$) or P ($E_1 \bullet E_2$) is given by

$P(E_1 \cap E_2) = P(E_1) \bullet P(E_2)$.

Rule 2: When two events are dependent, the probability of both occurring is given by

$P(E_1 \cap E_2) = P(E_1) \bullet P(E_2 \setminus E_1)$, whenever $P(E_1) \neq 0$

$= P(E_2) \bullet P(E_1 \setminus E_2)$ whenever $P(E_2) \neq 0$.

Review Exercise

1. A student can choose a computer project from one of three lists. The three lists contain 23, 15 and 19 possible projects, respectively. No project is on more than one list. How many possible projects are there to choose from?
2. Compute each of the following:
   a. ${}_8P_2$ b. ${}_{12}P_{10}$
   c. ${}_8C_2$ d. ${}_{12}C_{10}$
3. A committee of 7 students has to be formed from 9 boys and 5 girls. In how many ways can this be done when the committee contains:
   a. exactly three girls?
   b. at least three girls?
   c. 2 girls and 5 boys?
4. 7 boys and 6 girls are to be seated around a table. Find the number of ways that this can be done in each of the following cases:
   a. there is no restriction
   b. no girls are adjacent
   c. all girls form a single block
   d. a particular girl G is adjacent to two particular B_1 and B_2
5. A bag contains 6 red, 5 blue, and 4 yellow balls. 2 balls are drawn, but the first ball is drawn without replacement. Find the following:
   a. P (blue, then red). b. P (red, then red). c. P (yellow, then red)
6. In a pack of 52 cards, a card is drawn at random without replacement. Find the probability of drawing a queen followed by a jack.
7. Suppose that a group of 15 students contain eight boys (B) and seven girls (G). If two students are chosen randomly without replacement, find the probability that the two students chosen are both boys using multiplication rule?